INT422

IT PROJECT MANAGEMENT

Dr. S.Markkandeyan
Senior Assistant Professor,
School of Computing,
SASTRA Deemed to be University,
Thanjavur - 613005

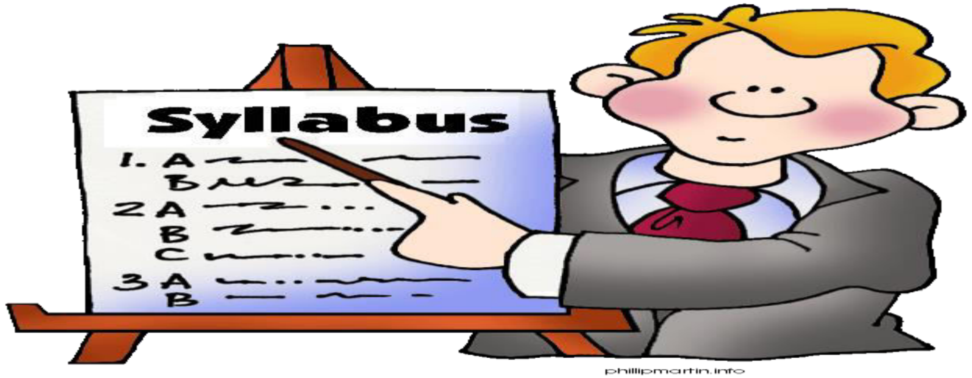
- COURSE OBJECTIVES
- SYLLABUS
- TEXT AND REFERENCE BOOKS
- ASSIGNMENT
- CIA

Course Objective

To help the learners

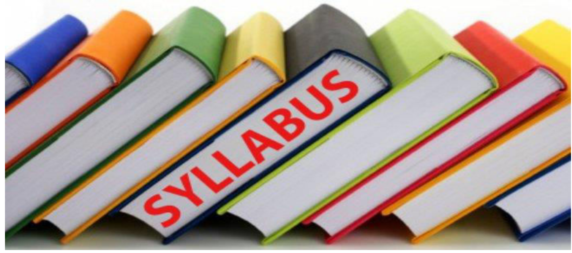
- This course will help the learner to understand the key aspects of software projects
- The strategies required for managing projects from their birth to completion
- Agile project management techniques such as Scrum and DevOps.

UNIT – I



UNIT - I 12 Periods

Project Overview and Feasibility Studies: Identification of Market and Demand Analysis, Project Cost Estimate, Financial Appraisal. **Project Scheduling**: Introduction to PERT and CPM - Critical Path Calculation - Precedence Relationship - Difference between PERT and CPM - Float Calculation and its importance - Cost reduction by Crashing of activity.

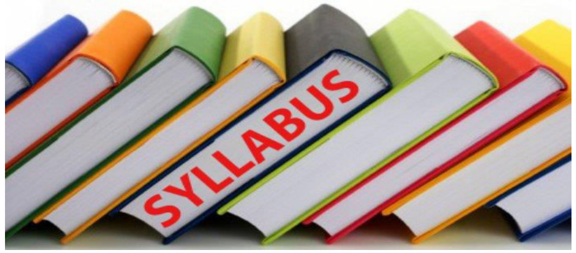


UNIT - II **11 Periods**

Cost Control and Scheduling: Project Cost Control (PERT/Cost) - Resource Scheduling & Resource Leveling. **Project Management Features**: Risk Analysis - Project Control – Project Audit and Project Termination.

UNIT – III 11 Periods

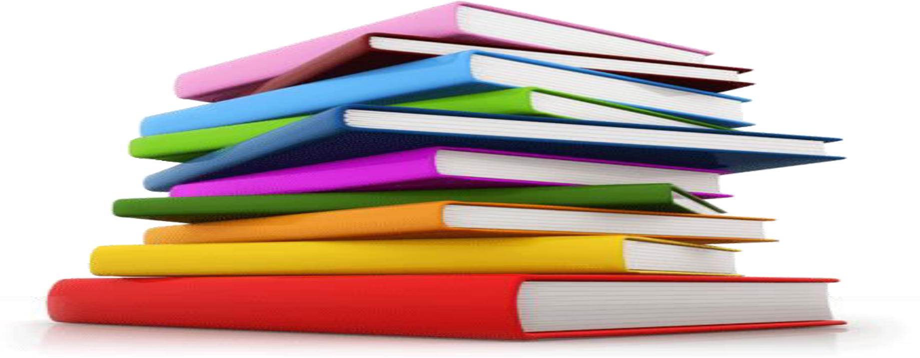
Agile Project Management: Introduction to Agile Principles and Agile methodologies - Relationship between Agile Scrum, Lean, DevOps and IT Service Management (ITIL). **Scrum**: Various terminologies used in Scrum (Sprint, product backlog, sprint backlog, sprint review, retro perspective) - Various Scrum roles - Best practices of Scrum.



UNIT - IV 11 Periods

DevOps: Overview and its Components - Containerization Using Docker - Managing Source Code and Automating Builds - Automated Testing and Test Driven Development – Continuous Integration and Configuration Management - Continuous Deployment – Automated Monitoring. **Other Agile Methodologies**: Introduction to XP, FDD,DSDM & Crystal.

REFERENCES



1. Roman Pichler, *Agile Product Management with Scrum,* Addison-Wesley Professional, 2010
2. Mark C. Layton, Steven J. Ostermiller, *Agile Project Management For Dummies*, Wiley, 2017
3. Rubin, Kenneth S. *Essential Scrum: A Practical Guide to the Most Popular Agile Process*. United Kingdom, Addison-Wesley, 2012.
4. Ken Schwaber, *Agile Project Management with Scrum (Microsoft Professional),* Microsoft Press, 2004

ASSIGNMENTS

Assignments - I (10 Marks)
Assignments - II (10 Marks)
Assignments - III (10 Marks)
Average – 10 Marks

CIA

shutterstock.com • 688292506

CIA- I (20 Marks)
CIA- II (20 Marks)
CIA- III (20 Marks)
Any Best Two - 40 Marks

DAY	Hour
MONDAY	2
WEDNESDAY	3
THURSDAY	3

Staff Room – LTC 410 (B)
markkandeyan@cse.sastra.edu

Unit 1 – Lecture 1

Project Overview and Feasibility Study in IT Project Management

Introduction

A **project** is defined as a "temporary endeavor with a beginning and an end and it must be used to create a unique product, service or result".

Examples

- Construction of Chennai Airport
- Computerizing Apollo Hospital

Project Management:

- A project in any organization is collaboration across departments to achieve a single well defined objective.
- The process of planning, organizing and managing resources to achieve the organizational objective is called project management.

Introduction

Management Activities

- Planning - Deciding what is to be done
- Organizing - Making arrangements
- Staffing - Selecting the right people for the job
- Directing - Giving instructions
- Monitoring - Checking on progress
- Controlling - Taking action to remedy hold-ups
- Innovating - Coming up with new solutions
- Representing - Liaising with users, etc

Product-driven project :

- A project will be to create a product.
- The details of the product is provided by the client.

Objective-driven project :

- A project is to meet an objective.
- The Client may have a problem and asks a specialist to recommend solutions.

Introduction

SMART Goal of Project Management:(Specific, Measurable, Achievable, Relevant, Time bound)

Example

Develop a web based appointment management system for poly clinics deliverable in March, 2012 as per ISO 9000 standards within the budget of ` 50 Lakhs.

Specific - application development

Measurable - ISO 9000 standards, Below ` 50 lakhs

Achievable - using available Web Technology

Relevant - Not something very great or not something never done by anyone before

Time bound - Before March, 2012

Unit 1 - Topic 1

Project Overview and Feasibility Studies

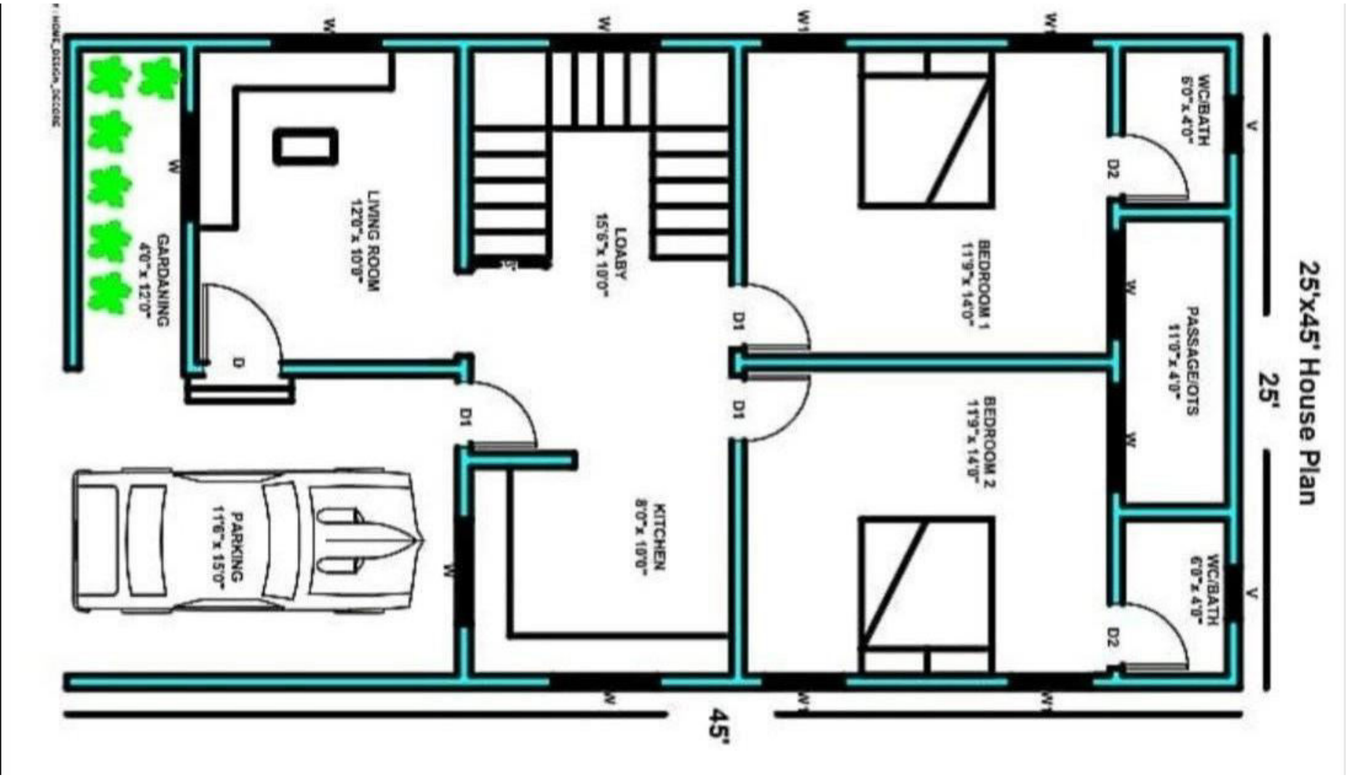
Introduction to Software Project

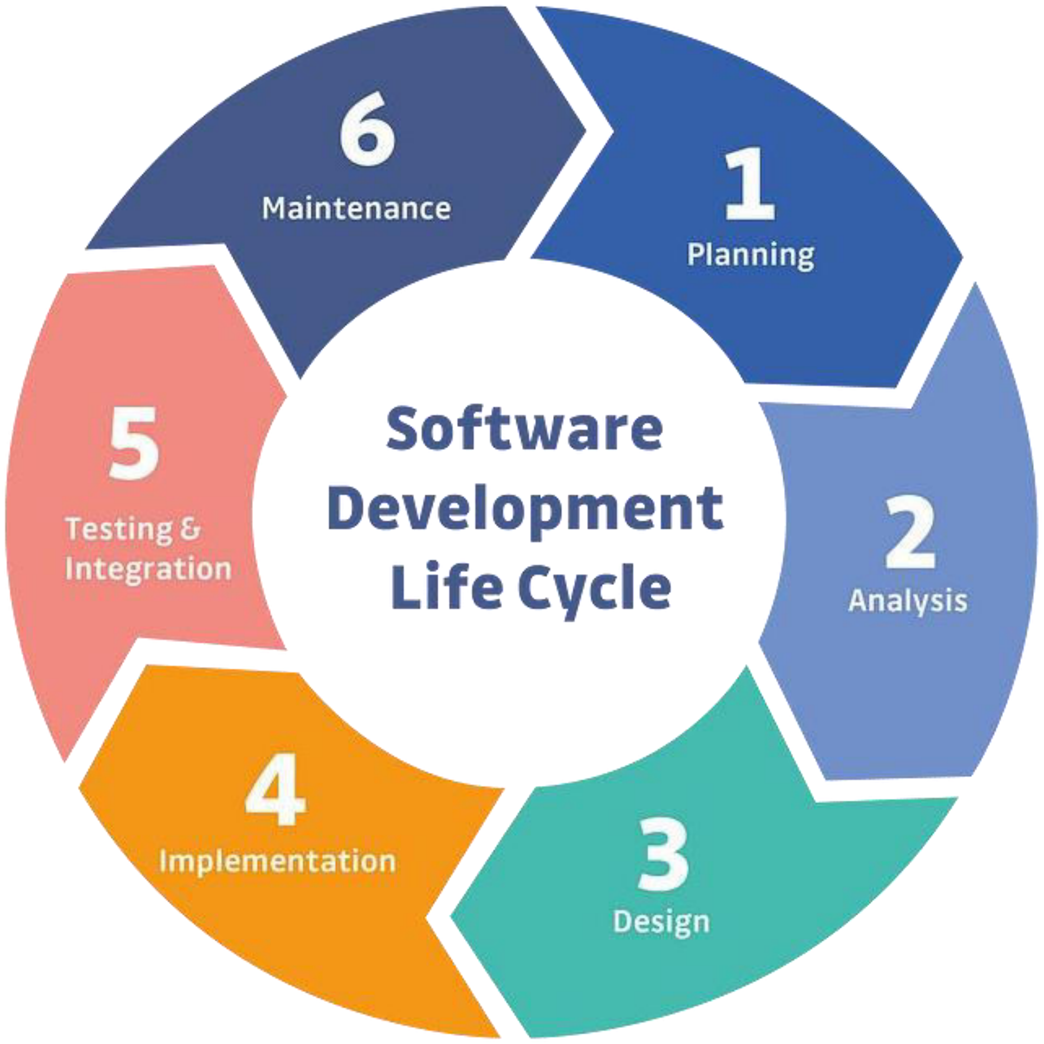
- A software project is like **building** a **digital tool** or **program**.
- It involves planning, designing, coding, testing, and launching software to solve a specific problem or perform a particular task.
- Just as a construction project has blueprints, a software project has plans and steps to follow, with a team of people working together to create the final product.

Software Project Overview

- A software project overview provides **a high-level summary of the key aspects and goals** of a software project.
- It typically includes information such as the project's **purpose, scope, objectives, stakeholders, major features, and the overall timeline**.
- This overview serves as a **roadmap** for the **project team and stakeholders**, helping everyone understand the project's direction and what to expect.
- **It's a concise** yet comprehensive introduction to the essential elements of the software development effort.

Software Development Life Cycle





1.1 Introduction to Project Management

Definition of Project Management:

- Project management involves planning, executing, and closing projects **to achieve specific goals within a specified timeframe and budget**.

1.1.1 Introduction to Software Project Management

- **S**oftware **P**roject **M**anagement **involves** planning, organizing, and overseeing the development of a software product from its initial concept to its completion.
- It **includes tasks** such as defining project goals, creating schedules, allocating resources, managing budgets, and coordinating the efforts of a team of developers.
- The **goal** of SPM is to ensure that the project is completed on time, within budget, and meets the specified requirements and quality standards.
- It involves various methodologies and practices to efficiently guide the software development process and achieve successful project outcomes.

1.1.2 Key Elements of Project Management

1. Scope: Defines the project's **objectives, deliverables**, and **constraints**.
2. Schedule: Outlines the project **timeline, milestones**, and **deadlines**.
3. Budget: **Allocate**s **resources** and **estimates costs** for the project.
4. Risk Management: **Identifies** and **manages** potential **risks** that may impact the project.

1.2 Project Overview

- Importance of Project Overview:
  - The project overview provides a **high-level understanding** of the project's **purpose, goals**, and **key components**.
- Components of Project Overview:

1. Project Title: Clearly **defines the project's name**.
2. Objectives: **Lists the specific, measurable goals** of the project.
3. Stakeholders**: Identifies individuals or groups with an interest** in the project.
4. Scope Statement: **Describes the boundaries and limitations** of the project.

1.3 Feasibility Studies

- Definition of Feasibility Studies:
  - Feasibility studies **assess the viability of a project** before committing **resources, considering technical, economic, legal, operational, and scheduling factors.**
- Types of Feasibility Studies:
  - Technical Feasibility: **Examine**s the project's **technical requirements and capabilities**.
  - Economic Feasibility: **Analyze**s the **financial aspects**, including **costs and benefits**.
  - Legal Feasibility: **Assess**es compliance with **laws and regulations**.
  - Operational Feasibility: **Evaluate**s the project's **impact on day-to-day operations**.
  - Scheduling Feasibility: **Determine**s **if** the **project can be completed within** the required **timeframe.**

1.3.1 Steps in Conducting Feasibility Studies

- **Preliminary Investigation:**
    a. Identify the Problem or Opportunity.
    b. Conduct Initial Project Scope Analysis.
- **Technical Feasibility:**
    a. Assess Technical Requirements.
    b. Evaluate Technological Capabilities.
- **Economic Feasibility:**
    a. Estimate Costs and Benefits.
    b. Calculate Return on Investment (ROI).

1.3.1 Steps in Conducting Feasibility Studies

- **Legal Feasibility:**
  a. Review Legal and Regulatory Requirements.
  b. Ensure Compliance with Laws.
- **Operational Feasibility:**
  a. Analyze Impact on Operations.
  b. Consider Employee and Stakeholder Acceptance.
- **Scheduling Feasibility:**
  a. Develop Project Timeline.
  b. Identify Critical Path and Milestones.

Summary

- Importance of Comprehensive Project Overview and Feasibility Studies:
  - Proper planning, clear objectives, and thorough feasibility assessments are crucial for the success of any project.
- Next Steps:
  - Once the project overview and feasibility studies are complete, the **project can proceed to the planning and execution phases**.
- Remember, **effective project management begins with a solid foundation of understanding** the project's purpose, scope, and feasibility.

Example for Software Project Overview

Online Shopping App

Purpose:

Develop a user-friendly mobile application for online shopping, providing a convenient and secure platform for users to browse, select, and purchase products.

Scope:

The project encompasses the design, development, testing, and launch of the online shopping app for iOS and Android platforms. It will include features such as product catalog, user authentication, shopping cart, and secure payment processing.

Example for Software Project Overview

Online Shopping App

Objectives:

1. Create an intuitive and visually appealing user interface.
2. Implement a robust backend system for product management and order processing.
3. Ensure data security and user privacy.
4. Test the application thoroughly to identify and fix bugs.
5. Launch the app on schedule and within the allocated budget.

Example for Software Project Overview

Online Shopping App

Stakeholders:

- Project Team: Developers, Designers, Testers
- Product Owner: Responsible for defining and prioritizing features
- End Users: Individuals who will use the online shopping app

Major Features:

1. User Registration and Authentication
2. Product Catalog with Search and Filter options
3. Shopping Cart for adding and managing selected items
4. Secure Checkout with multiple payment options
5. Order History and Tracking

Example for Software Project Overview

Online Shopping App

Timeline:

- Planning and Design: 4 weeks
- Development: 8 weeks
- Testing and Debugging: 3 weeks
- Launch: Week 16

Budget:

- $150,000 (including development, testing, and marketing costs)

This overview provides a snapshot of the key aspects of the project, outlining its purpose, scope, objectives, stakeholders, major features, timeline, and budget.

Example for Feasibility study

- Look at the NOTES

Unit 1 - Topic 2

Key Aspects of SPM & Identification and Formulation of Software Projects

Key Aspects in Software Project Management

- SPM involves Planning + Executing + Supervising the Development
- SPM requires the skills of technical expertise, leadership skills, and knowledge of project management principles .

Key Aspects in Software Project Management

1. Project Planning
2. Team Management
3. Resource Management
4. Timeline Management
5. Quality Assurance
6. Change Management

Key Aspects in Software Project Management

7. Communication
8. Risk Management
9. Documentation
10. Project Monitoring and Control
11. Post-Implementation Review

Introduction to Software Project Identification and Formulation

A. Definition of Software Project Identification and Formulation

- Software Project Identification: The process of **recognizing and defining a potential software project** based on organizational **needs, opportunities, or challenges.**
- Software Project Formulation: The **systematic planning and structuring of the identified project**, laying the foundation for successful execution.

Introduction to Software Project Identification and Formulation

B. Importance of Identification and Formulation

- Strategic Alignment: **Ensure** that software projects **align** with the overall organizational **strategy and goals**.
- Resource Optimization: **Proper identification and formulation** contribute to **efficient resource utilization**.
- Risk Mitigation: **Early identification** allows for the assessment and **mitigation** of potential risks.

Software Project Identification

A. Factors Influencing Identification

- Business Needs:
  - **Addressing** specific business **challenges**.
  - **Capitalizing** on market opportunities.
- Technological Advancements:
  - Leveraging **emerging technologies** for competitive **advantage**.
- Regulatory Requirements:
  - Ensuring **compliance with** industry **regulations**.

Software Project Identification

B. Techniques for Identification

- Stakeholder Interviews:
  - **Engage with key stakeholders** to **understand** their **needs** and **expectations**.
- Market Analysis:
  - **Evaluate** market **trends and demands**.
- SWOT Analysis:
  - Assess internal **strengths** and **weaknesses**, external **opportunities**, and **threats**.

Software Project Formulation

A. Key Steps in Formulation

- Define Project Scope:
  - Clearly **outline the boundaries and objectives** of the software project.
- Requirements Gathering:
  - **Identify and document** functional and non-functional requirements.
- Feasibility Study:
  - **Assess** technical, operational, and economic **feasibility.**
- Resource Planning:
  - **Allocate** human, financial, and technical **resources**.

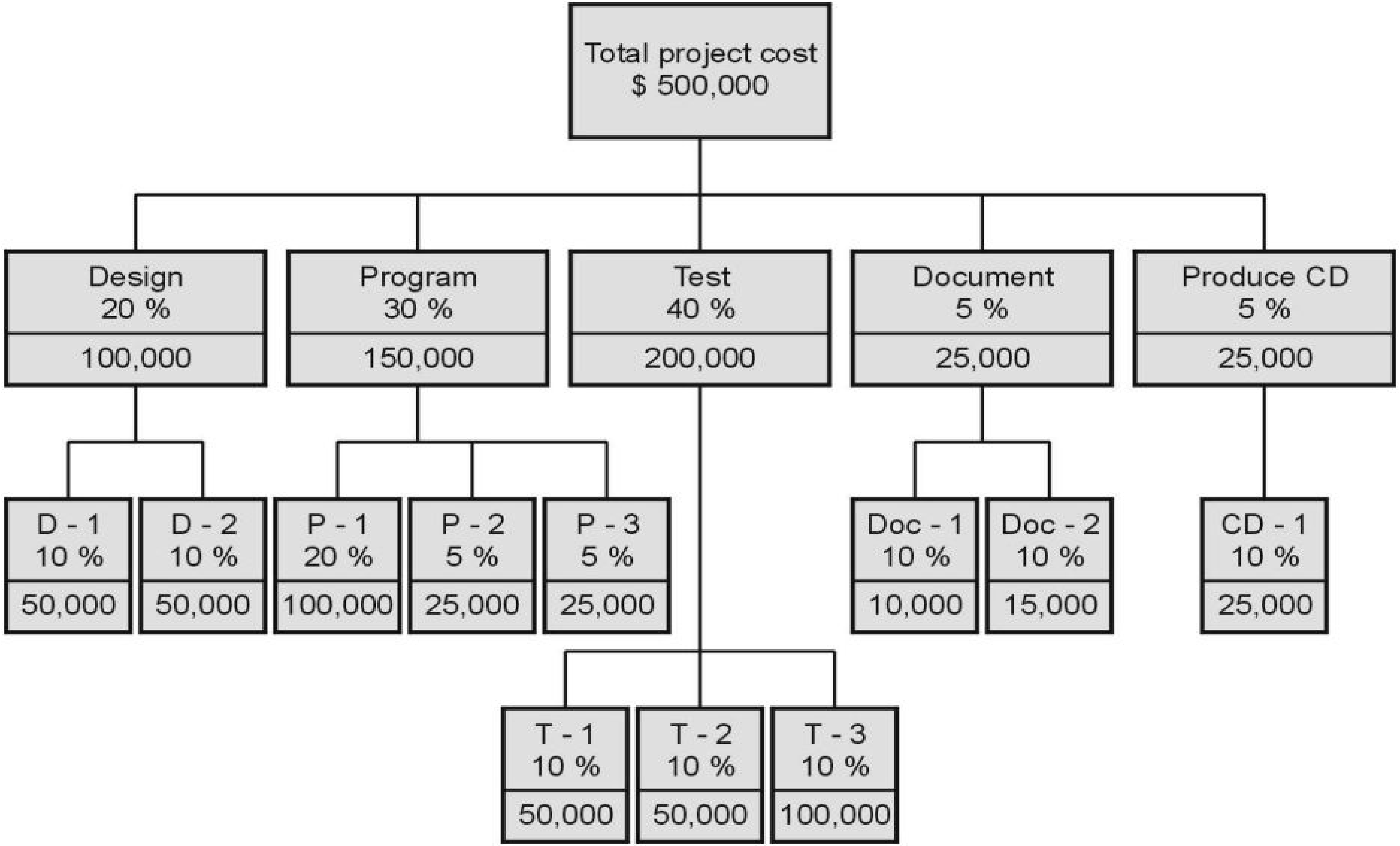
Software Project Formulation

B. Project Planning and Scheduling

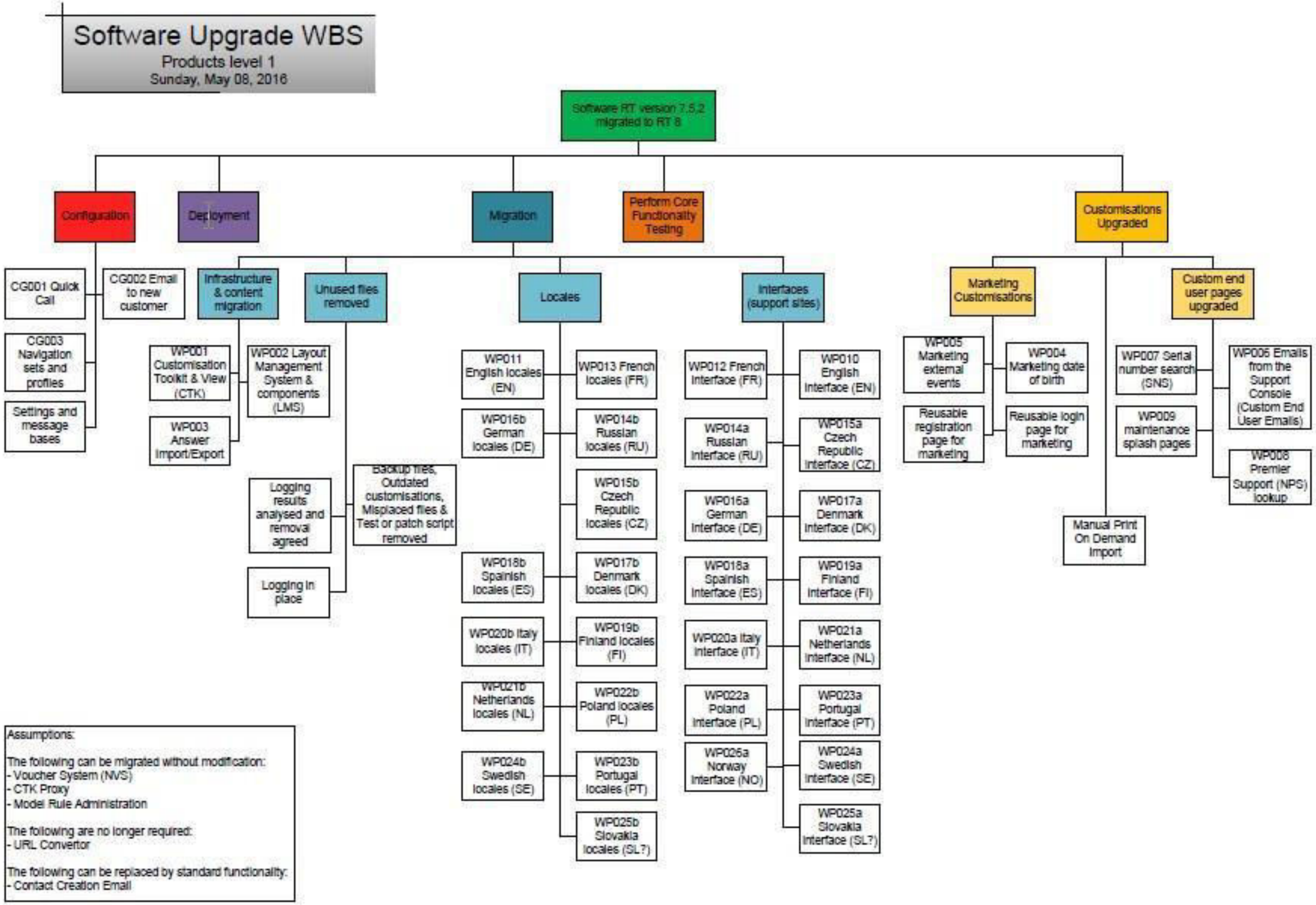
- Work Breakdown Structure (WBS):
  - **Break** down the **project into** manageable **tasks**.
- Gantt Charts:
  - **Visualize** project **timelines** and **dependencies**.
- Risk Management Plan:
  - **Identify** and **address** potential risks.

Work Breakdown Structure Example

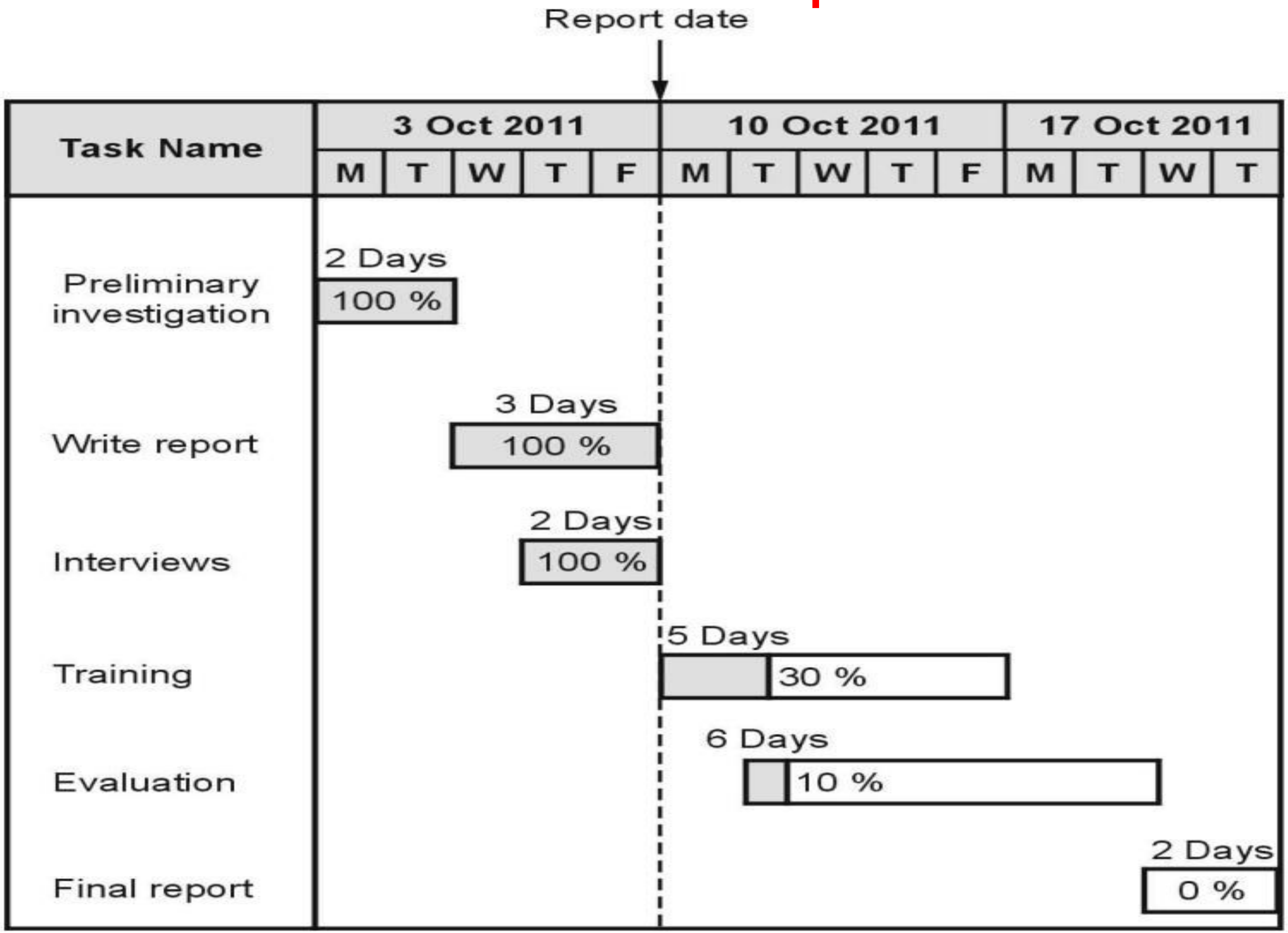
Apportion Method of Allocating Project Costs Using the Work Breakdown Structure



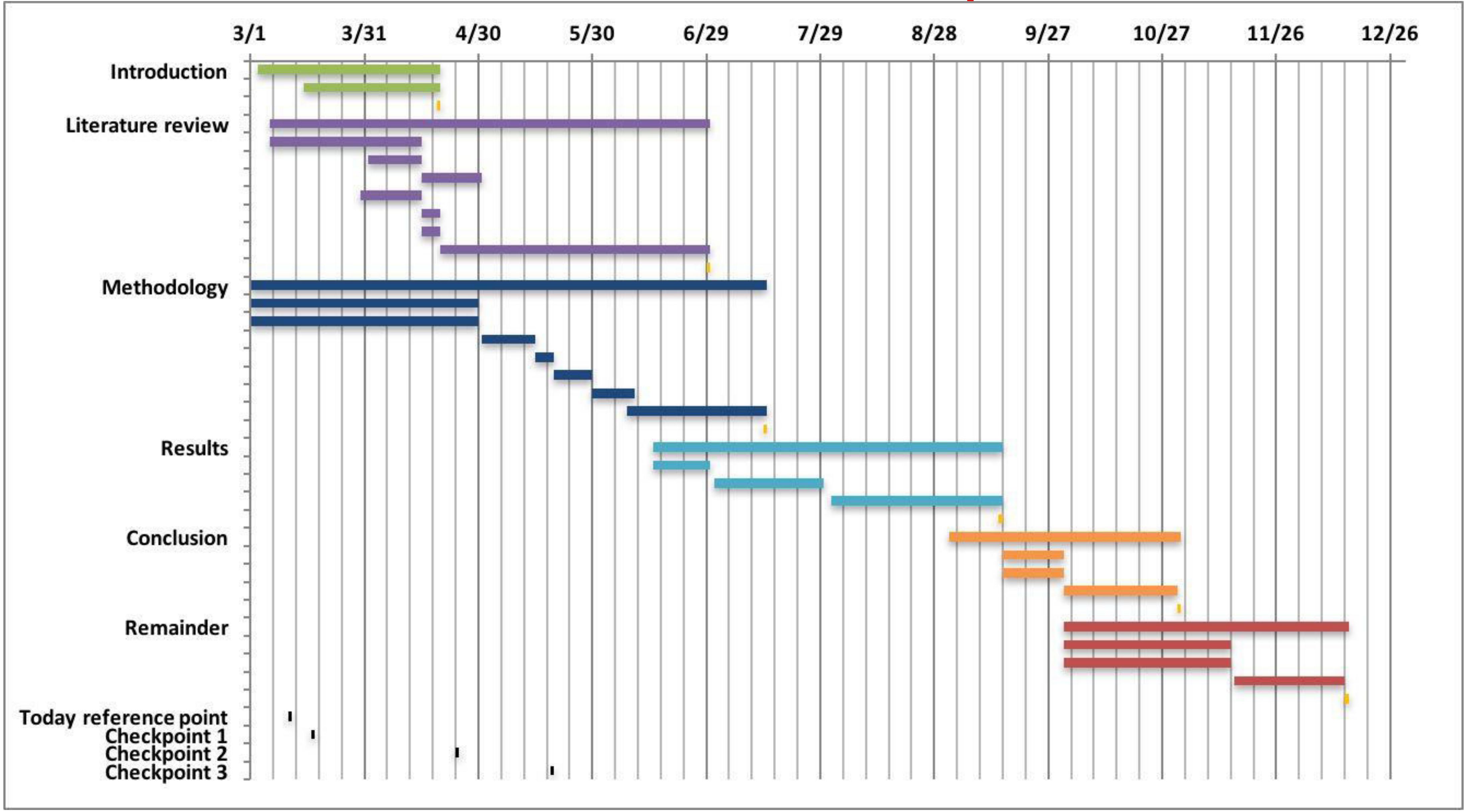
Work Breakdown Structure Example



Gantt Chart Example



Gantt Chart Example



Case Studies

- Look at the NOTES for the Following Case Studies
  - Case Study 1: Successful project identification and formulation leading to a product launch.
  - Case Study 2: Challenges in identification and formulation and their impact on project outcomes.

Unit 1 - Topic 3
Market and Demand Analysis

Market and Demand Analysis

Definition of Market and Demand Analysis:

- Market and demand analysis is a crucial aspect of business strategy that involves **assessing market dynamics**, **understanding customer needs**, and **evaluating the potential demand** for a product or service.

Importance of Market and Demand Analysis:

- Informs business decisions, helps identify opportunities, minimizes risks, and guides product/service development.

Demand

- Demand is defined as that **want, need or desire** backed by **willingness and ability to buy** a particular commodity, in a given period of time.

Or

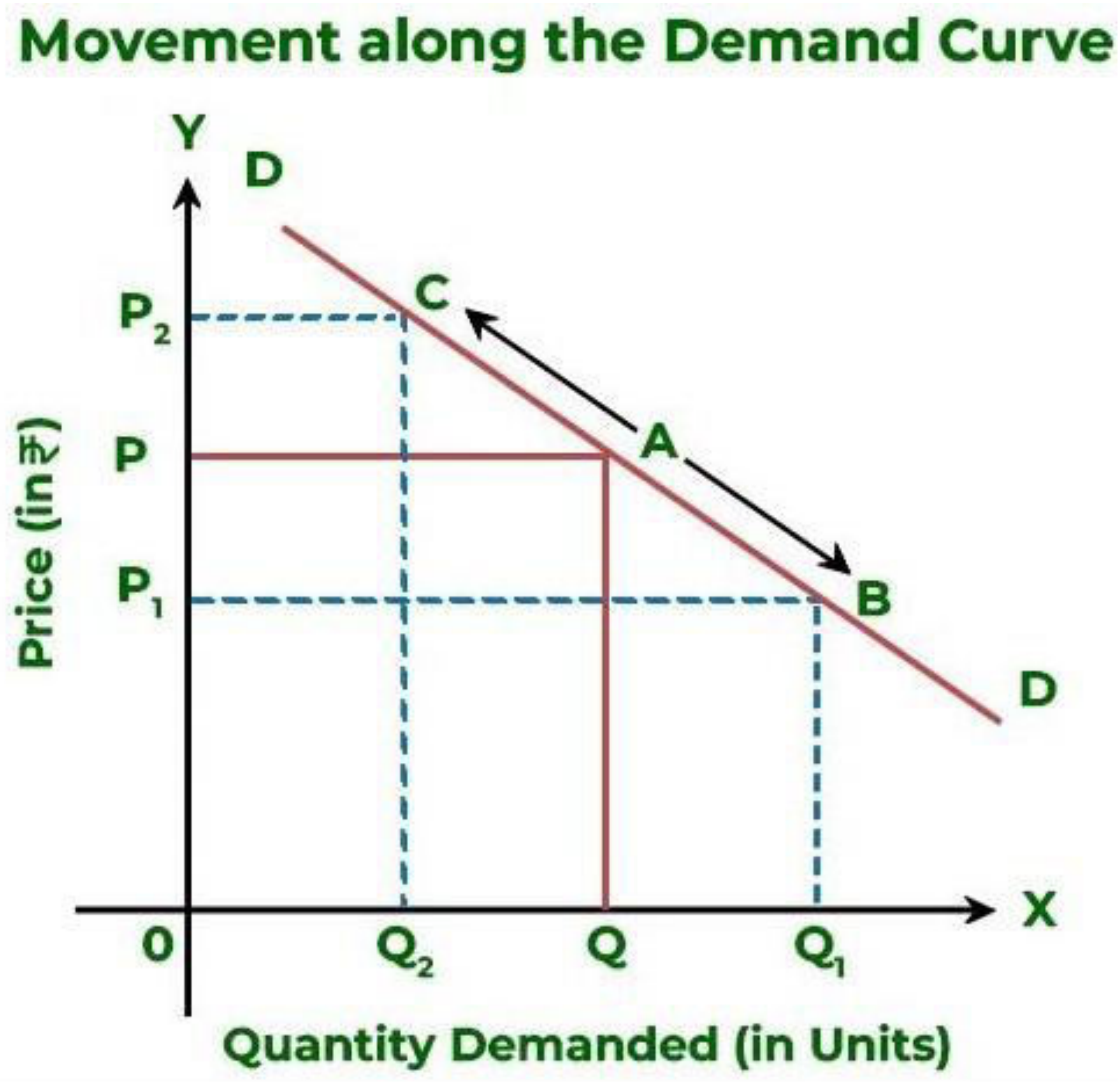
- demand is the **quantity of a commodity which consumers are willing to buy at a given price** for a particular unit of time.

1. Individual Demand
2. Market Demand
   - Primary Demand
     - Total Demand for all of the brands
   - Selective Demand
     - Demand for particular brand or product

Factors Influencing Demand

1. Tastes and Preferences of the Consumers.
2. Income of the People.
3. Changes in Price of the Related Goods.
4. Advertisement Expenditure.
5. The Number of Consumers in the Market.
6. Consumers Expectations with Regard to Future Prices

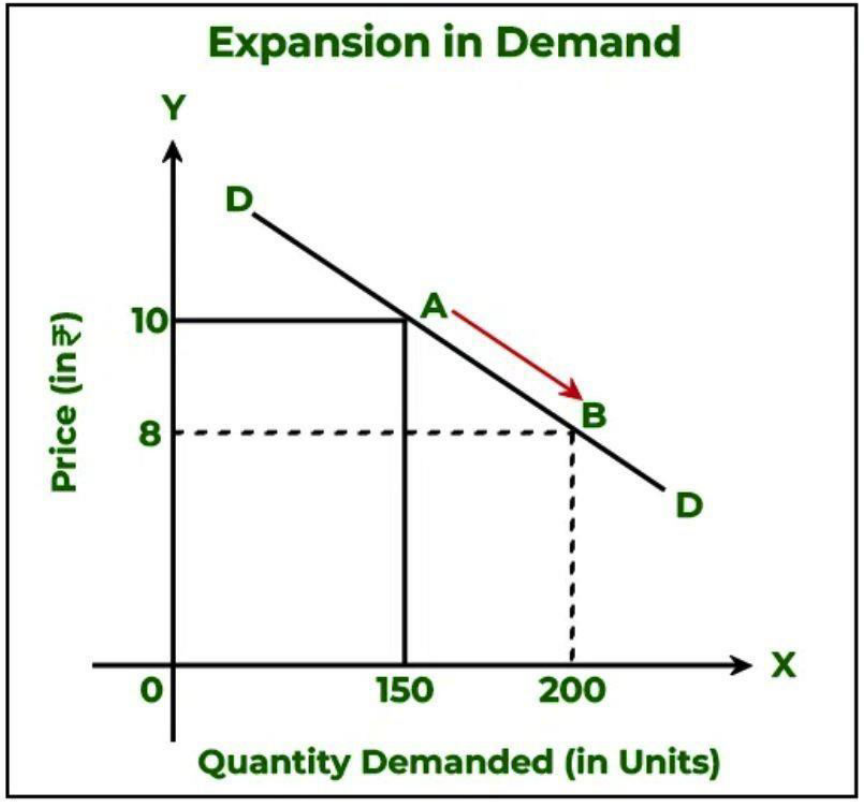
Demand Curve

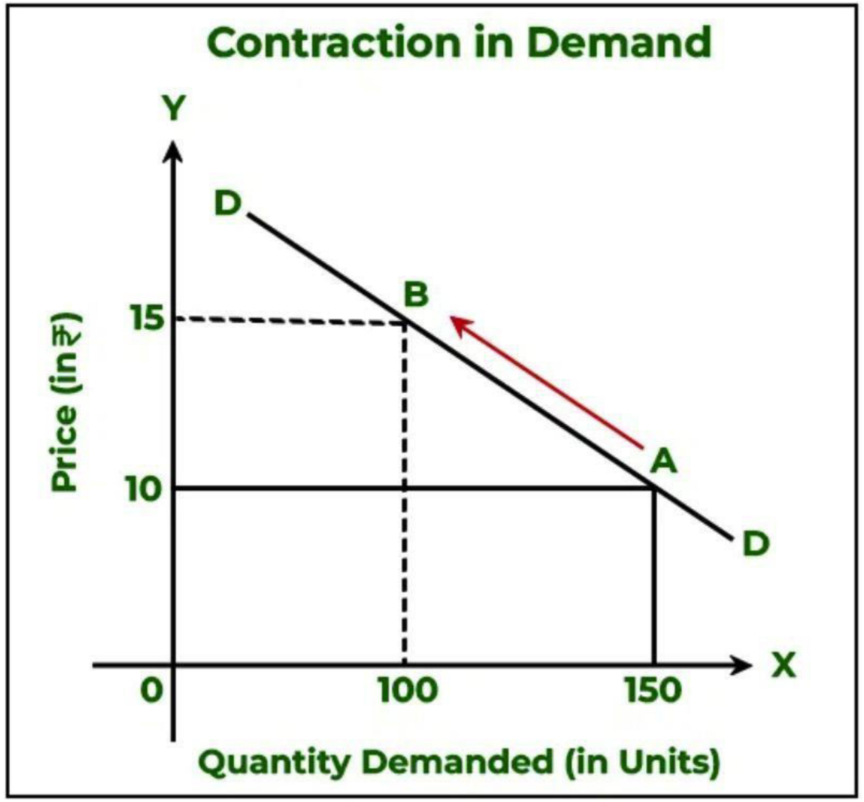


Change in Quantity Demand Curve

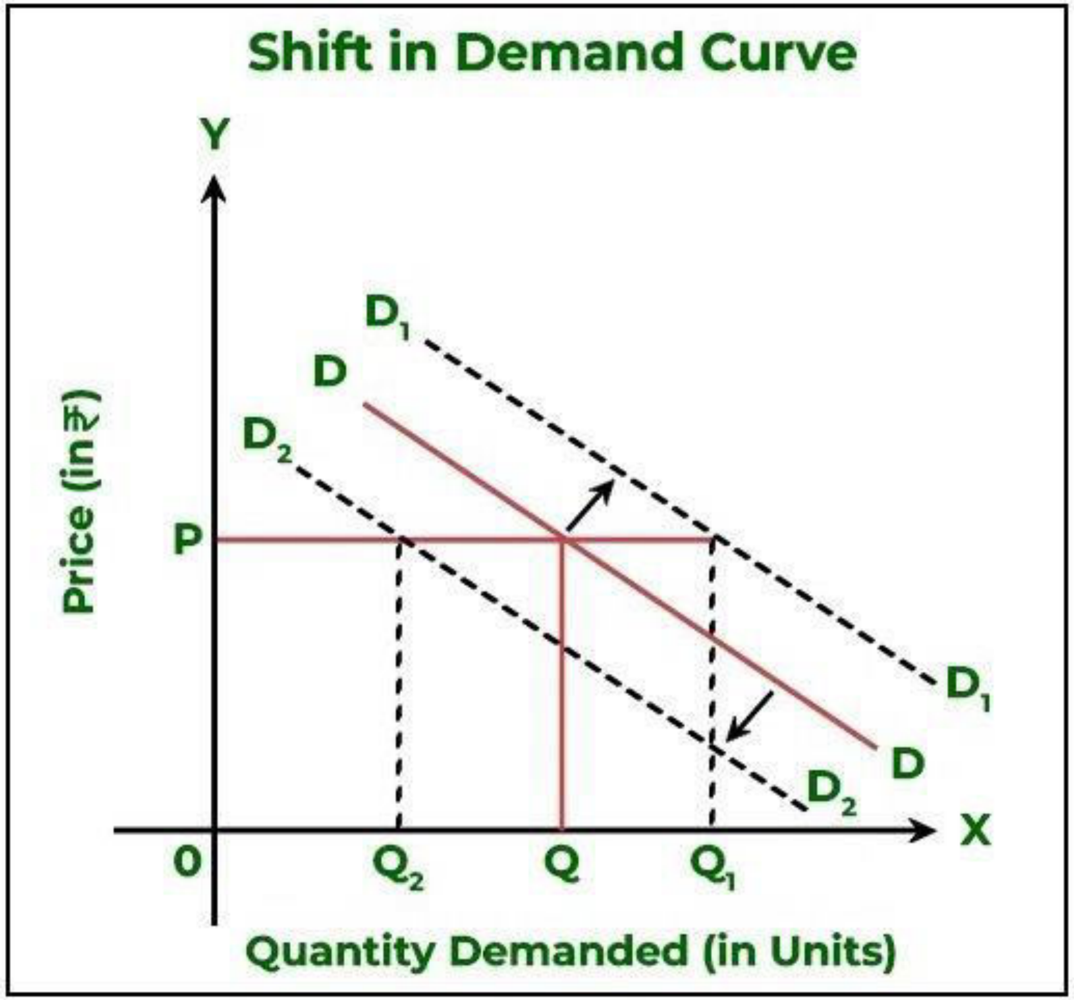
Price (₹)	Quantity (Units)
10	150
8	200

Price (₹)	Quantity (Units)
10	150
15	100





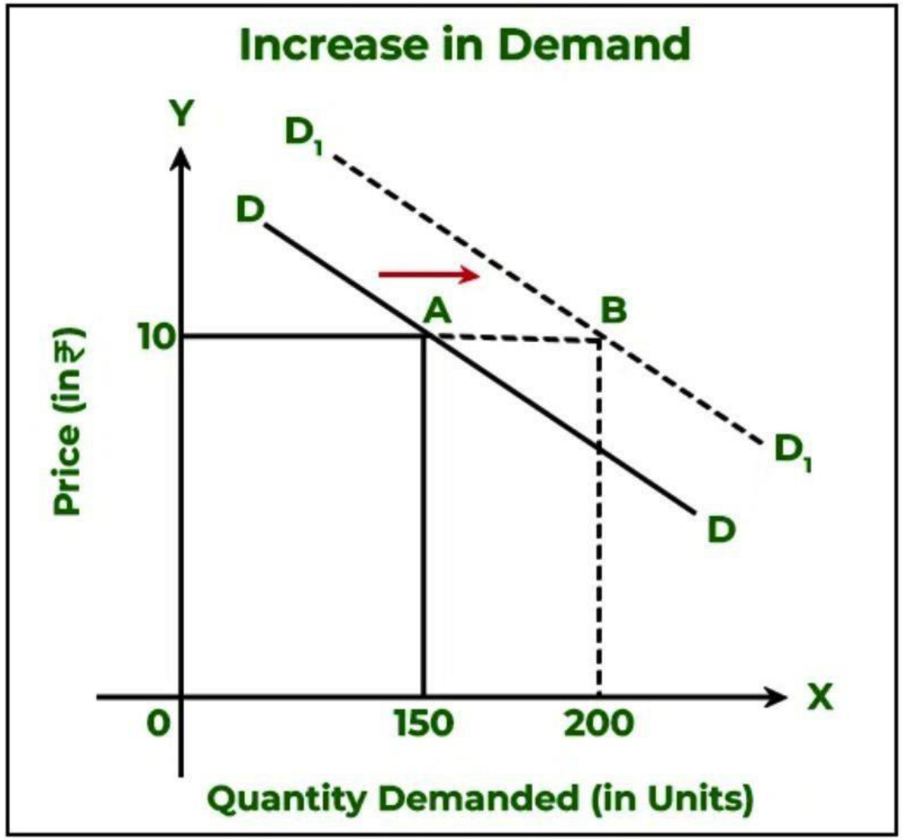
Shift in Demand Curve

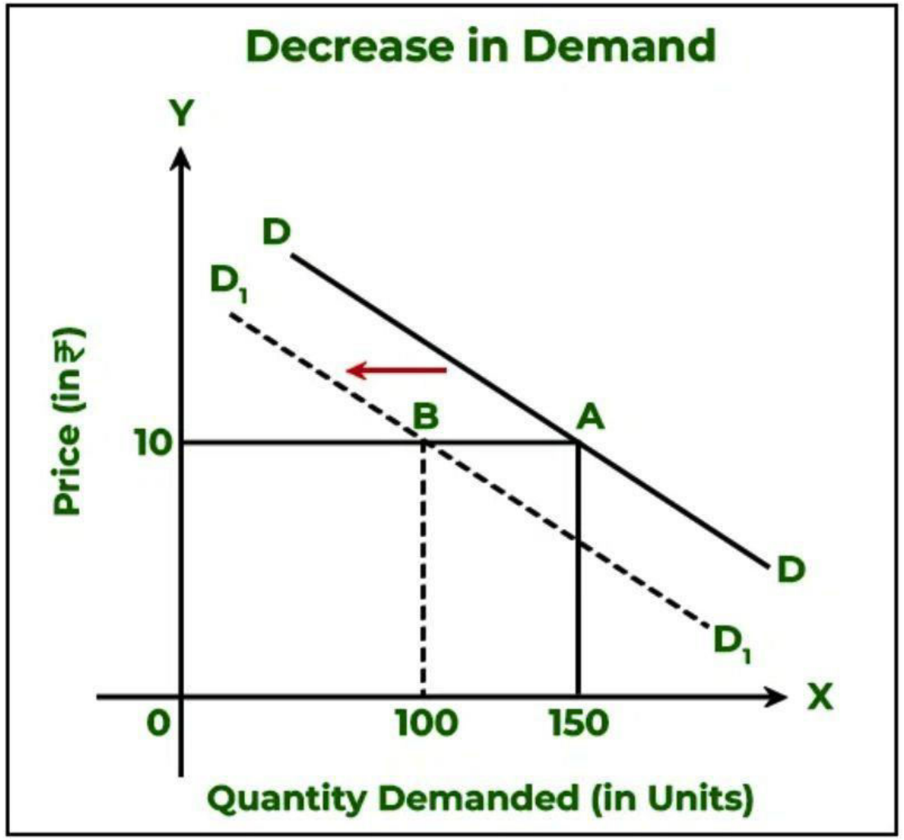


Shift in Demand Curve

Price (₹)	Quantity (Units)
10	150
10	200

Price (₹)	Quantity (Units)
10	150
10	100





Law of Demand

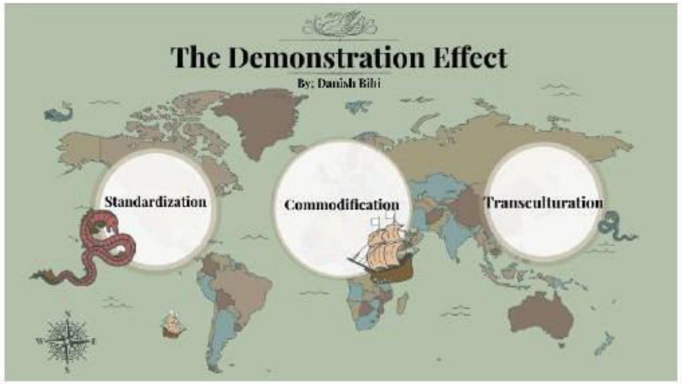
Law of Demand:

Recognizing the fact: Price of the product is the single most important variable of a product demand.

"Other things remaining constant when the price of a commodity rises, the demand for that commodity falls & vice versa."

"demand for a product is inversely proportional to its price."

Exceptions to Law of Demand



Identification of Market

Defining the Target Market:

- Target market refers to the specific group of consumers or businesses a product or service aims to serve.
- Demographics, psychographics, and geographic factors help define the target market.

Market Segmentation:

- Divides the broader market into smaller, more manageable segments based on common characteristics.
- Enables tailored marketing strategies for each segment.

Market Size and Potential:

- Estimating the total size of the market.
- Assessing growth trends and potential opportunities.

Understanding Demand Analysis

Understanding Demand:

- Demand is the quantity of a product or service that consumers are willing and able to purchase at a given price and time.
- Factors influencing demand include price, consumer preferences, and external economic factors.

Types of Demand:

- Elastic Demand: Quantity demanded is sensitive to price changes.
- Inelastic Demand: Quantity demanded remains relatively constant despite price changes.
- Cross Elasticity: Measures how demand for one product is affected by a change in the price of another.

Conducting Market Research

Market Research Methods:

- Surveys and Questionnaires: Gathers direct feedback from potential customers.
- Interviews: In-depth conversations to understand customer preferences.
- Observational Research: Observing consumer behaviour in real-world settings.

Analysing Competitors:

- Identifying key competitors in the market.
- Analysing competitor strengths, weaknesses, opportunities, and threats (SWOT analysis).

Unit 1 - Topic 4
Project Cost Estimation

Project Cost Estimation

Definition of Project Cost Estimate:

- Project cost estimation is the process of **forecasting the financial requirements of a project**. It involves **identifying and quantifying all costs associated with the project**, providing a basis for **budgeting and resource allocation.**

Importance of Accurate Cost Estimation:

- Critical for project planning, resource allocation, and decision-making.
- Helps prevent budget overruns and ensures financial feasibility.

Components of Project Cost

Direct Costs:

- Costs directly attributed to the project.
  - Examples: Labor, materials, equipment.

Indirect Costs (Overhead):

- Costs not directly tied to a specific project task.
  - Examples: Utilities, administrative expenses.

Fixed Costs:

- Costs that remain constant regardless of project volume.
  - Examples: Rent, insurance.

Variable Costs:

- Costs that fluctuate based on project activities.
  - Examples: Hourly wages, raw materials.

Methods of Cost Estimation

Analogous Estimation:

- **Uses historical data** from similar projects **as a reference**.
  - Example: **Estimating** the cost of a software development project **based on a similar past project.**

Parametric Estimation:

- Uses **statistical relationships and parameters** to estimate costs.
  - Example: Estimating construction costs per square foot based on historical data.

Bottom-Up Estimation:

- **Breaks down the project into detailed tasks**, estimating each individually.
  - Example: Estimating the cost of building a website by considering development, design, and testing separately.

Example Scenario: Construction Project

Direct Costs:

- Labor: Estimating the number of workers and their hourly rates.
- Materials: Calculating the quantity and cost of construction materials.
- Equipment: Assessing the cost of machinery and tools required.

Indirect Costs:

- Overhead: Considering administrative costs, utilities, and supervision.
- Contingency: Including a percentage for unexpected events or changes.

Importance of Contingency

Definition of Contingency:

- A provision for unforeseen events or changes in project scope.

Example:

- If the construction project encounters unexpected delays due to weather conditions, having a contingency budget can cover additional costs for overtime or rescheduling.

Summary of Cost Estimation

Key Points:

- Accurate cost estimation is essential for project success.
- Different methods provide flexibility in choosing the most suitable approach.

Next Step:

- Once the cost estimate is complete, it serves as the foundation for budgeting, resource allocation, and overall financial management throughout the project lifecycle.

Example for Software Project Cost Estimation

- Look at the NOTES

Unit 1 - Topic 4
Software Project Appraisal

3. Financial Appraisal

- This involves **evaluating the financial viability** of the software project.

Key Methods:

1. **Payback Period**
2. **Net Present Value (NPV)**
3. **Internal Rate of Return (IRR)**
4. **Benefit-Cost Ratio (BCR)**
5. **Profitability Index (PI)**
6. **Return on Investment (ROI)**

3.1 Payback Period

- Time required to recover the initial Investment.
- Formula:

$$Payback\ Period\ (t) = \frac{Inital\ Investment}{Annual\ Cash\ Inflow}$$

Example: "Calculate the payback period for a company investing $50,000, with annual revenue of $25,000?"

$$Payback\ Period\ (t) = \frac{\$50{,}000}{\$25{,}000} = 2\ Years$$

3.2 Net Present Value (NPV)

- Measures the **profitability of the project** by discounting future cash flows.
- Formula:

$$\text{NPV} = \sum \frac{Cash\ Inflow_t}{(1+r)^t} - Initial\ Investment$$

Example: Calculate the NPV for a company investing \$50,000 where they gain cashflow of \$10,000 over 3 years at 10% discount rate ?.

$$\text{NPV} = \sum \frac{\$10{,}000_3}{(1+0.10)^3} - \$50{,}000 = \$8{,}264$$

3.3 Internal Rate of Return (IRR)

- It represents the **discount rate** at which the **Net Present Value** (NPV) of all cash flows (both inflows and outflows) **equals zero**.
- Decision Rule:
  - **IRR >** Required Rate of Return (or Cost of Capital): The investment is profitable and should be considered.
  - IRR < **Required Rate of Return**: The investment is not viable.
  - IRR =Required Rate of Return: The investment breaks even.

Example: IRR = 15%, cost of capital = 10%, project is viable.

$$0 = CF_0 + \frac{CF_1}{(1+IRR)} + \frac{CF_2}{(1+IRR)^2} + \frac{CF_3}{(1+IRR)^3} + \ldots + \frac{CF_n}{(1+IRR)^n}$$

Or

$$0 = NPV = \sum_{n=0}^{N} \frac{CF_n}{(1+IRR)^n}$$

Where:

CF_0 = Initial Investment / Outlay

$CF_1, CF_2, CF_3 \ldots CF_n$ = Cash flows

n = Each Period

N = Holding Period

NPV = Net Present Value

IRR = Internal Rate of Return

The Project: A company buys new machinery for **$5,000** (Year 0 cash flow: -$5,000).

Expected Returns: It's expected to generate profits (cash inflows) of **$600, $1,100, $1,500, $ 2,000, $ 2,300 for 5 years**.

Year 0: - **$5,000**
Year 1: + **$600**
Year 2: + **$1,100**
Year 3: + **$1,500**
Year 4: + **$2,000**
Year 5: + **$2,300**

The Goal: Find the rate (IRR) that makes the present value of those payments equal exactly $5,000.

Note: (1 + r) ^ t is known as discount factor, In the case of 10% rate and one year, Discount factor = 1/(1 + 0.10) = 0.9091

The Formula: You're solving for ‘irr' in: (12%)

Year 0: - **$5,000 (DF=1)**
Year 1: + **$600 ($588.24)**
Year 2: + **$1,100 ($876.91)**
Year 3: + **$1,500 ($1,067.67)**
Year 4: + **$2,000 ($1,271.04)**
Year 5: + **$2,300 (($1,134.85)**

($61.29) < $0

3.4 Benefit-Cost Ratio (BCR)

- Ratio of benefits to costs. A BCR >1 indicates viability.
- Formula:

$$BCR = \frac{Total\ Benefits}{Total\ Costs}$$

Example: Calculate BCR for the company with Total benefits of \$75,000 and Total cost is \$50,000.

$$BCR = \frac{\$75{,}000}{\$50{,}000} = 1.5\ [Viable]$$

3.5 Profitability Index (PI)

- It **assess** the **feasibility or profitability of a project.**
- It **compares** the present value of benefits to the present value of costs.
- Similar to BCR but considers NPV.
- Formula:

$$PI = 1 + \frac{NPV}{Initial\ Investment}$$

Example: Financial Appraisal for a Software Project

Project: Development of an Online Learning Platform.

Initial Investment: $100,000

Projected Revenue: $50,000/year

Discount Rate: 8%

1. **Payback Period:**
2. **NPV:**
3. **IRR:**
4. ***BCR***

Conclusion:

Example: Financial Appraisal for a Software Project

Project: Development of an Online Learning Platform.

Initial Investment: \$100,000

Projected Revenue: \$50,000/year

Discount Rate: 8%

1. **Payback Period:** $PP = \frac{100{,}000}{50{,}000} = 2\ Years$
2. **NPV:** Using a discount rate of 8%, calculate future cash flows over 5 years:
   - $NPV = \sum \frac{50{,}000}{(1+0.08)t} - 100{,}000$
   - NPV ≈ \$108,660.
3. **IRR:** IRR ≈ 12%.
4. $BCR = \frac{250{,}000}{100{,}000} = 2.5$

Conclusion: The project is financially viable.

Example

Year	Project 1 (Cash flow)	Project 2 (Cash flow)	Project 3 (Cash flow)	Project 4 (Cash flow)
0	– 100,000	– 1,000,000	– 100,000	– 120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	20,000	200,000	30,000	30,000
4	20,000	200,000	20,000	25,000
5	100,000	350,000	20,000	50,000

Net Profit	60,000	150,000	30,000	45,000
Payback	5	5	4	4
ROI	12%	4%	10%	11%

NPV

Example :

Year	Cash flow	Discount factor (10 %)	Discounted cash flow
0	– 100,000	1	– 100, 000
1	10,000	0.9091	9,091
2	10,000	0.8264	8,264
3	10,000	0.7513	7,513
4	20,000	0.683	13,660
5	100,000	0.6209	62,090
		NPV	618

Software Project Appraisal

Definition:

- Software project appraisal involves **assessing and evaluating the feasibility, viability, and potential success** of a software project before its initiation.

Purpose:

- To make informed decisions regarding project investment, resource allocation, and overall project management.

Key Components of S/W Project Appraisal

1. Feasibility Analysis:
   - Technical Feasibility: **Assess the technical capabilities and requirements** of the project, including the availability of necessary technology and expertise.
   - Operational Feasibility: **Evaluate whether the proposed system will operate effectively** within the existing organizational environment.
   - Economic Feasibility: **Examine the cost-effectiveness and financial viability** of the project.

1. Risk Analysis:
   - **Identify potential risks and uncertainties** associated with the project.
   - **Evaluate the impact of risks** on project success and **develop mitigation strategies**.

Key Components of S/W Project Appraisal

3 Cost-Benefit Analysis:

- **Estimate the total cost of the project**, including development, maintenance, and operational costs.
- **Analyze the expected benefits**, both **tangible and intangible**, over the project's lifecycle.
- Perform a **cost-benefit analysis** to determine the project's overall return on investment (ROI).

4 Resource Analysis:

- **Evaluate the availability and allocation** of human resources, technology, and other essential assets.
- **Assess the skill sets** required and the availability of personnel with the necessary expertise.

Software Project Appraisal Techniques

1. Net Present Value (NPV):

- **Calculate the present value of all cash inflows and outflows** to **determine the project's profitability.**
- **Consider the time value of money to account** for the inflation or interest rate.

2. Return on Investment (ROI):

- **Express the net benefits of the project as a percentage** of the initial investment.
- Useful for comparing the relative profitability of different projects.

Software Project Appraisal Techniques

3. Payback period:

- Determine the time it takes for the project to recover its initial investment.
- Shorter payback periods are generally preferred.

4. Break-Even Analysis :

- Identify the point at which the project's total revenue equals its total cost.
- Useful for understanding when the project becomes financially self-sustaining.

Example for Software Project Appraisal

- Look at the NOTES

Unit 1 - Topic 5
PERT and CPM

PERT and CPM

INT422 Unit 1

Program Evaluation and Review Technique

Definition:

PERT, which stands for Program Evaluation and Review Technique, is a **project management method used for planning, scheduling, and coordinating tasks** within a project. It is particularly **useful** in situations **where the activities involved are uncertain, complex, or have interdependencies**.

Key Components and Features of PERT

1. Probabilistic Time Estimates:

- PERT incorporates three time estimates for each activity:
  - Optimistic Time (O): The shortest time an activity can be completed under **favorable conditions**.
  - Pessimistic Time (P): The longest time the activity might take under **unfavorable conditions**.
  - Most Likely Time (M): The best estimate of the time required under **normal conditions**.
- These estimates are used to calculate the expected duration (TE) using the formula $TE = (O + 4M + P) / 6$

Key Components and Features of PERT

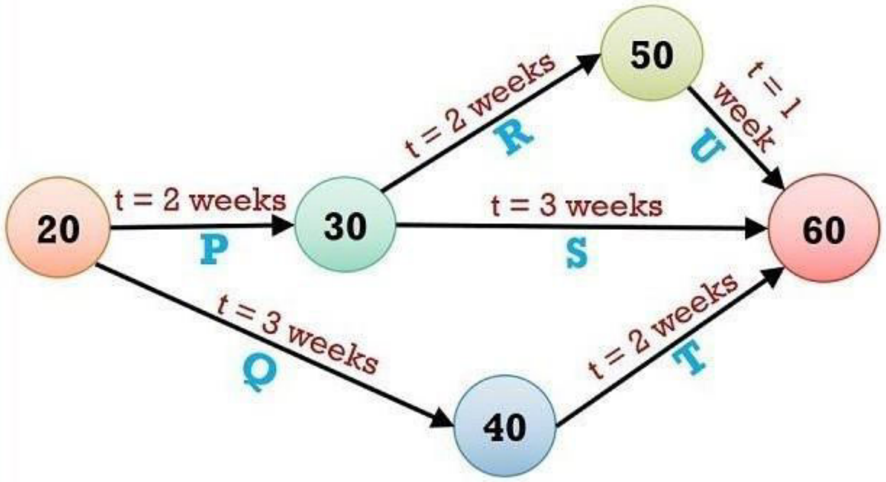
2. Network Diagrams:

- PERT uses **network diagrams** (PERT charts) to **represent** the **sequence of activities and their dependencies**. Nodes represent activities, and directed arrows show the flow and dependencies between them.

Key Components and Features of PERT

3. Critical Path Analysis:

The critical path is the **longest path through the network**, determining the **minimum project duration**. Activities on the critical path have no float or slack, meaning **any delay** in these activities directly impacts the project's completion time.



Key Components and Features of PERT

4. Probability Analysis (Optional):

PERT allows for the **calculation of the probability of completing the entire project within a specified time frame**. This involves using probability distributions to account for uncertainties in activity durations.

How PERT Works ?

1. Activity Identification:

   □Identify and list all the activities required for the project.

2. Time Estimation:

   □For each activity, determine optimistic, pessimistic, and most likely time estimates. Calculate the expected duration using the formula.

3. Network Construction:

   □Develop a network diagram that visually represents the sequence and dependencies of activities.

How PERT Works ?

4. Critical Path Analysis:

   - Identify the critical path by calculating the expected duration for each path in the network. The critical path is the sequence of activities with the longest duration.

5. Probability Analysis (Optional):

   - Optionally, perform probability analysis to **assess the likelihood of completing the entire project** within a specified time frame.

6. Float/Slack Calculation:

   - Determine the **float or slack for non-critical activities**, representing the flexibility in their start and finish times.

How PERT Works ?

7. Project Control and Monitoring:

- **Monitor** the progress of **activities**, especially those **on the critical path,** and **take corrective actions** to **prevent delays**. **Adjust project timelines** based on actual progress.

8. Periodic Review and Adjustment:

- Periodically **review the project's progress** and **adjust time estimates** or **schedules based on real-world developments or changes** in project requirements.

Critical Path Method (CPM)

- The Critical Path Method (CPM) is a **project management technique** used for **planning, scheduling, and managing complex projects**. **CPM** is particularly **effective** when the **activities within a project are well-defined and have known durations.**

Key Components and Features in CPM

1. Deterministic Time Estimates:

☐Unlike PERT, **CPM uses deterministic time estimates**, where a **single time estimate is assigned to each activity**. This estimate **represents the expected time for the activity to be completed**.

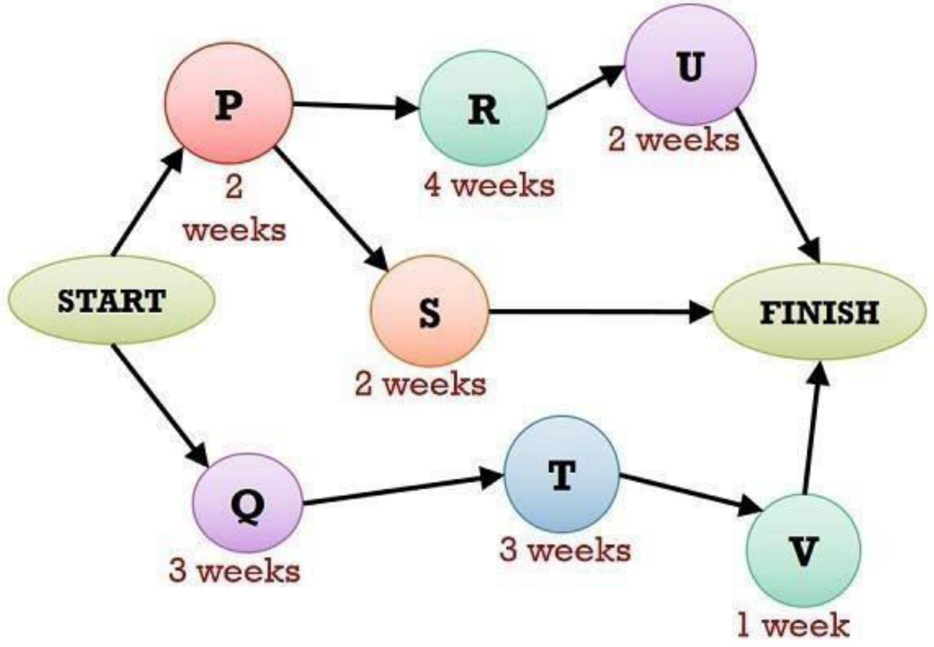
2. Network Diagrams:

☐Similar to PERT, CPM uses **network diagrams to represent the sequence of activities and their dependencies**. **Nodes represent activities**, and **directed arrows show the logical flow and dependencies between them**.

Key Components and Features in CPM

3. Critical Path Analysis:

- The critical path in CPM is the sequence of activities that **determines the minimum project duration**. Unlike PERT, **CPM focuses on deterministic estimates and identifies the longest path through the network**.



Key Components and Features in CPM

4. Float or Slack Calculation:

- CPM calculates the float or slack for each activity, representing the amount of time an activity can be delayed without delaying the project's overall completion. Activities on the critical path have zero float.

How CPM works ?

1. Activity Identification:
   - **Identify and list all the activities** required for the project.
2. Time Estimation:
   - **Assign a deterministic time estimate to each activity**. This is the expected time it will take for the activity to be completed.
3. Network Construction:
   - **Develop a network diagram** that visually **represents** the **sequence and dependencies of activities.**
4. Critical Path Analysis:
   - **Identify the critical path by calculating the total duration for each path in the network**. The critical path is the sequence of activities with the longest total duration.

How CPM works ?

5. Float/Slack Calculation:
   - ☐Determine the float or slack for each activity by subtracting its expected duration from the duration of the critical path. Activities on the critical path have zero float.
6. Project Control and Monitoring:
   - ☐Monitor the progress of activities, especially those on the critical path. Focus on managing and controlling activities to ensure they are completed on time.
7. Periodic Review and Adjustment:
   - ☐Periodically review the project's progress and adjust schedules based on actual achievements and changes in project requirements.

PERT vs CPM

Technique

PERT is a project management technique used to measure uncertain activities in a project and focuses on time planning and control.

CPM is a statistical project management technique that manages the well-defined activities of a project, such as time and cost management.

Focus

PERT focuses on minimizing the time required to complete a project by the designated deadline.

CPM focuses on a trade-off between cost and time with a significant emphasis on minimizing costs.

PERT vs CPM

Evolution

PERT evolved as a suitable technique for research and development projects.

CPM is best suited for non-research projects like construction.

Orientation

While PERT is event-oriented because it focuses on time.

CPM is often activity-oriented.

PERT vs CPM

Model

PERT uses a probabilistic model.

CPM is a deterministic model.

Time Estimates

PERT has three-time estimates and helps make high-precision time estimates.

CPM has only a onetime estimate and is appropriate for reasonable time estimation.

PERT vs CPM

Activities Type

While the PERT technique manages unpredictable activities.

The CPM can manage predictable activities effectively.

Job Nature

PERT is beneficial for the non-repetitive types of jobs.

CPM involves jobs with a repetitive nature.

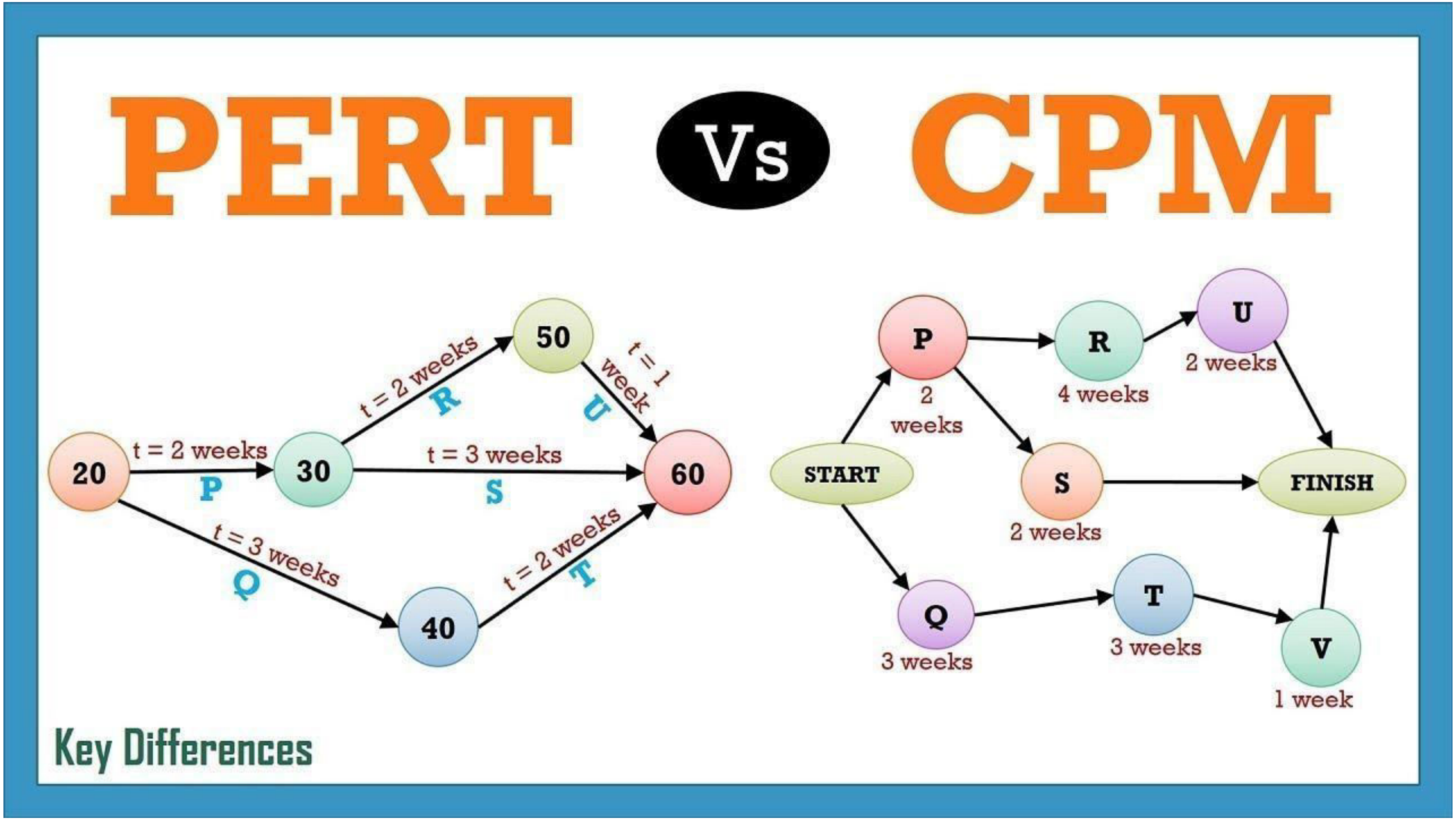
PERT vs CPM

Crashing Concept

In comparison, the crashing concept does not apply to PERT as there's no time certainty.

Crashing is a compression technique used in CPM to reduce the project duration and associated costs.

Key Components and Features in CPM

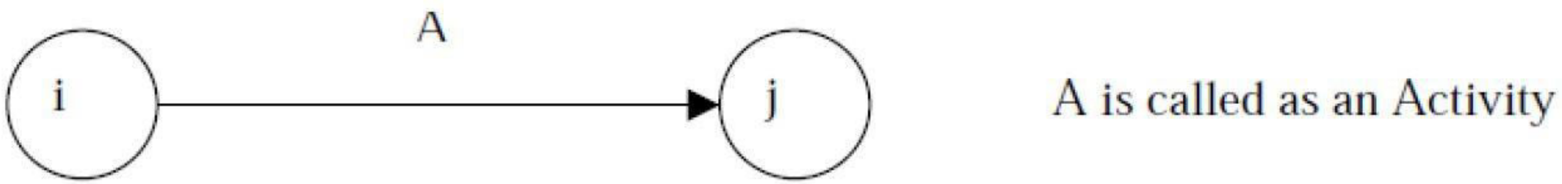


PERT / CPM Network Components

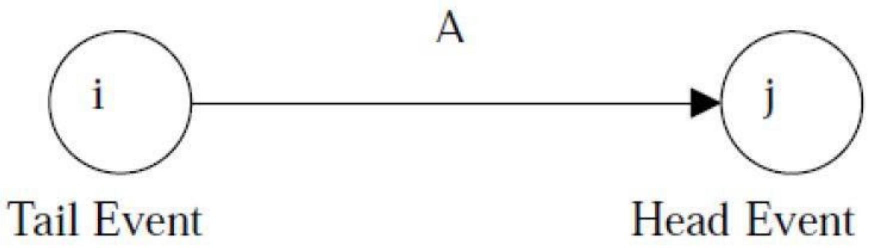
PERT / CPM networks contain two major components

1. Activities
2. Events

Activity: An **activity** represents an **action and consumption of resources** (time, money, energy) required to complete a portion of a project. **Activity** is represented by **an arrow**



***Event**:* An **event (or node) will always occur at the beginning and end of an activity**. The event has no resources and is represented by a circle. The i^{th} event and j^{th} event are the tail event and head event respectively



PERT / CPM Network Components

Merge and Burst Events

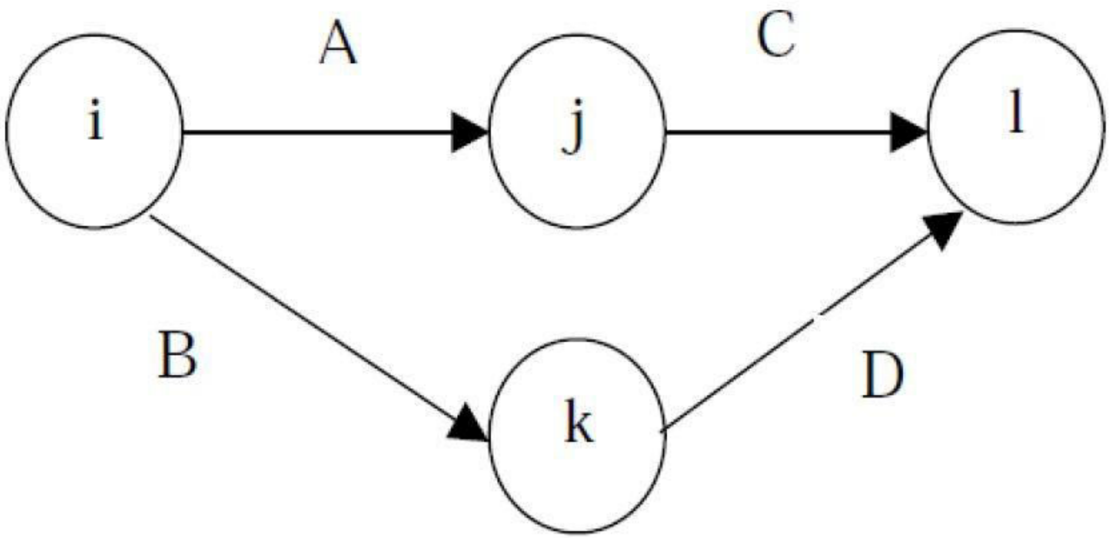
- **One or more activities can start and end simultaneously** at an event as shown below:

(a) Merge Event (b) Burst Event

PERT / CPM Network Components

Preceding and Succeeding Activities

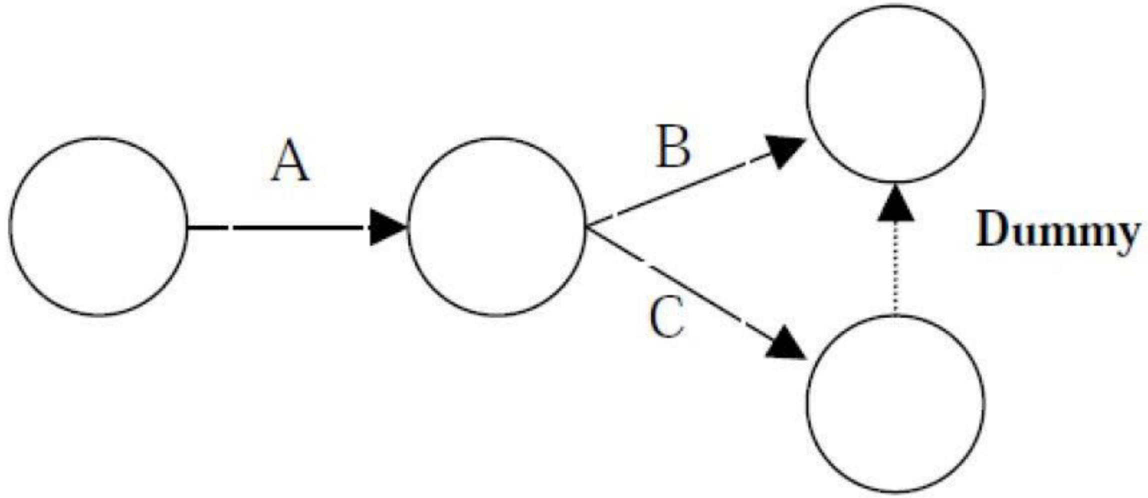
- **Activities performed before** given **events** are known as **preceding activities** and **activities performed after a given event are known as succeeding activities**. In the following figure, activities A and B precede activities C and D respectively.



PERT / CPM Network Components

Dummy Activity

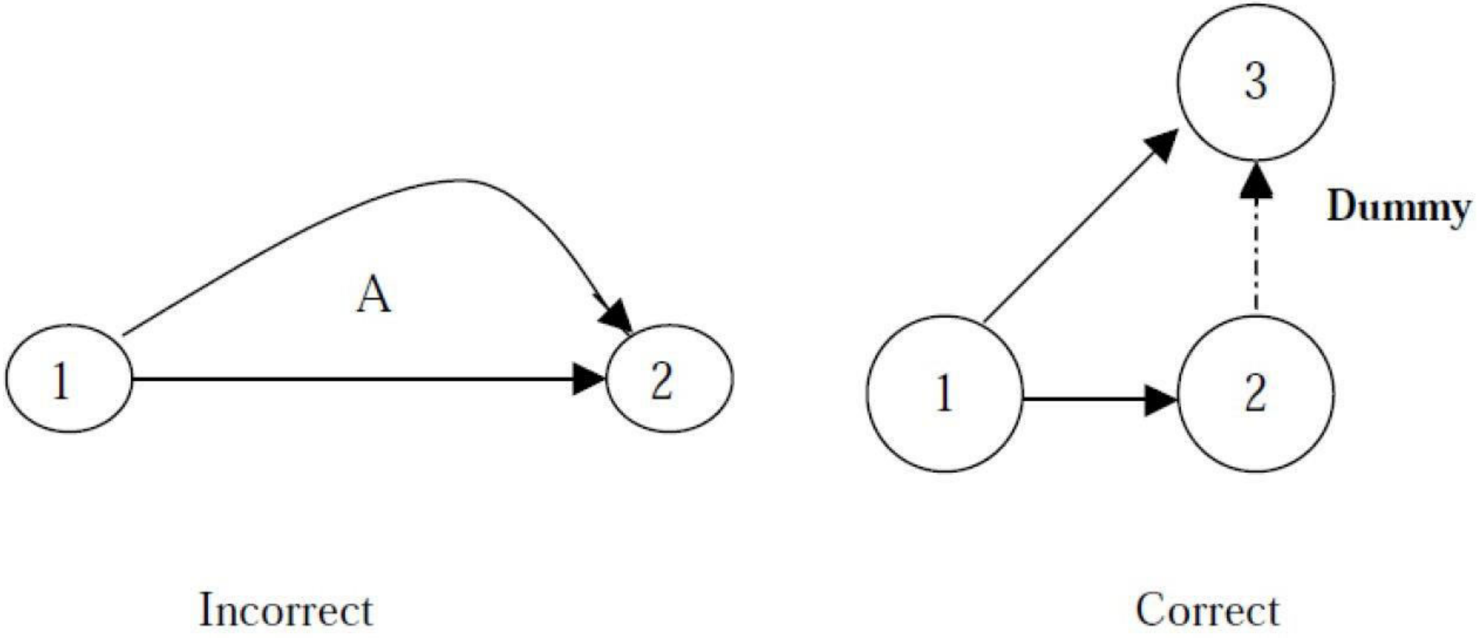
- An **imaginary activity which does not consume any resource and time is called a dummy activity**. **Dummy activities** are simply used to **represent** a **connection between events in order to maintain a logic in the network**. It is represented by a **dotted line** in a network, as shown be



PERT / CPM Network Components

ERRORS TO BE AVOIDED IN CONSTRUCTING A NETWORK

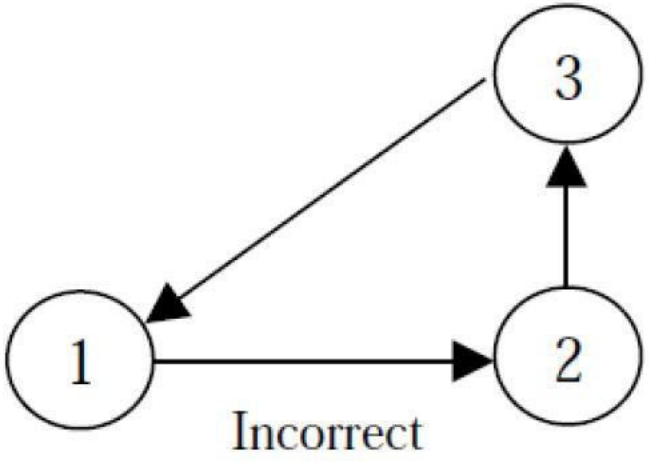
1. **Two activities starting from a tail event must not have a same end event.** To ensure this, it is **absolutely necessary to introduce a dummy activity**, as shown:



PERT / CPM Network Components

ERRORS TO BE AVOIDED IN CONSTRUCTING A NETWORK

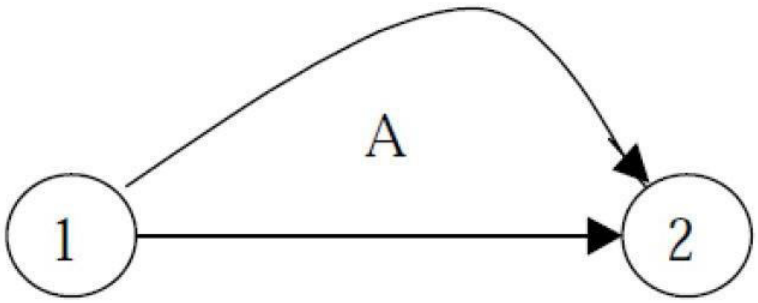
2. **Looping error should not be formed in a network,** as it represents performance of activities repeatedly in a cyclic manner, as shown below:

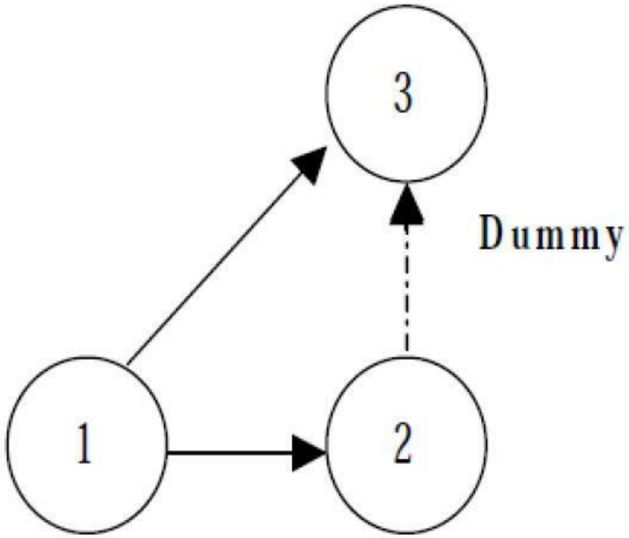


PERT / CPM Network Components

ERRORS TO BE AVOIDED IN CONSTRUCTING A NETWORK

3. **In a network, there should be only one start event and one ending event as shown below:**

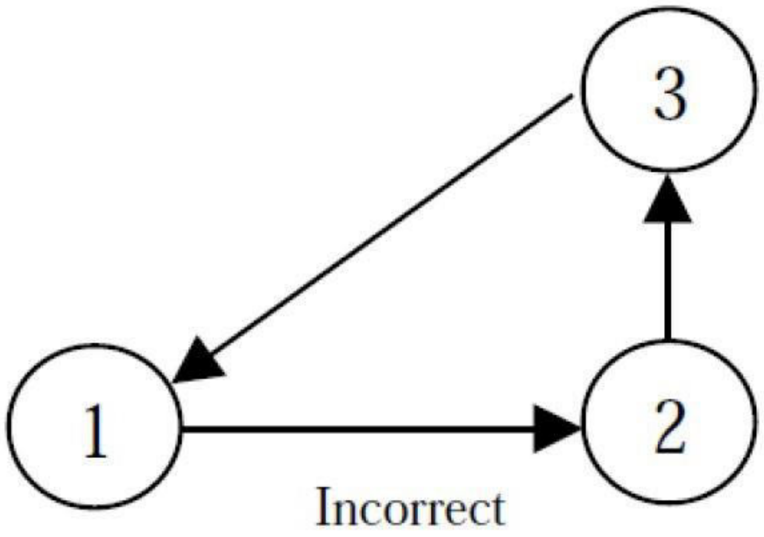




PERT / CPM Network Components

ERRORS TO BE AVOIDED IN CONSTRUCTING A NETWORK

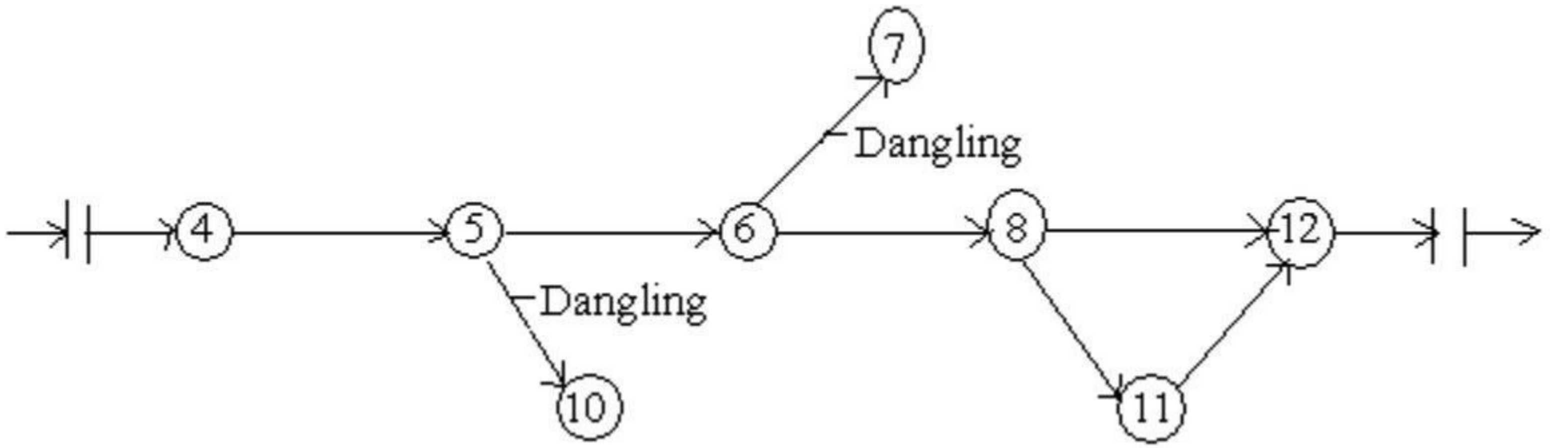
4. **The direction of arrows should flow from left to right avoiding mixing of direction as shown:**



PERT / CPM Network Components

ERRORS TO BE AVOIDED IN CONSTRUCTING A NETWORK

4. **To disconnect an activity before the completion of all activities in a network diagram is known as dangling.**



PERT / CPM Network Components

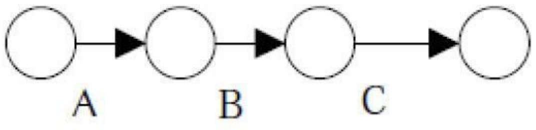
Rules for Constructing a Network Diagram

1. **No single activity can be represented more than once** in a network. The length of an arrow has no significance.
2. The **event numbered 1 is the start event** and an **event with highest number is the end event**. Before an activity can be undertaken, all activities preceding it must be completed. That is, the **activities must follow a logical sequence** (or – interrelationship) between activities.
3. In assigning numbers to events, **there should not be any duplication of event numbers** in a network.
4. **Dummy activities must be used only if it is necessary** to reduce the complexity of a network.
5. A **network should have only one start event and one end event**.

PERT / CPM Network Components

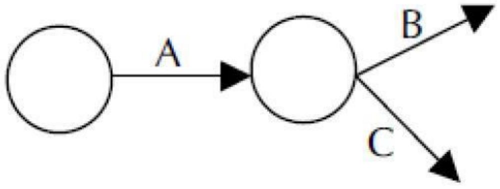
Rules for Constructing a Network Diagram

(a)



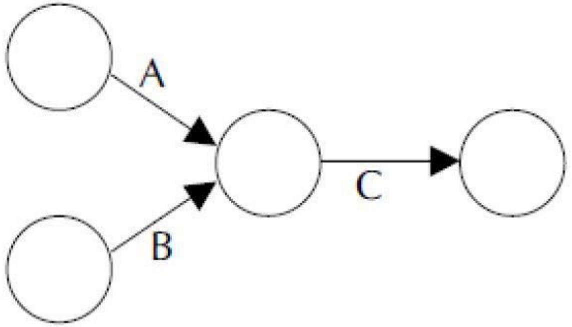
Activity B can be performed only after completing activity A, and activity C can be performed only after completing activity B.

(b)



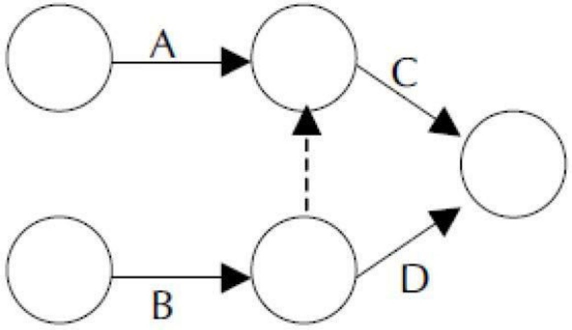
Activities B and C can start simultaneously only after completing A.

(c)



Activities A and B must be completed before start of activity C.

(d)



Activity C must start only after completing activities A and B. But activity D can start after completion of activity B.

PERT / CPM Network Components

PROCEDURE FOR NUMBERING EVENTS USING FULKERSON'S RULE

Step1: Number the start or initial event as 1.

Step2: From event 1, strike off all outgoing activities. This would have made one or more events as initial events (event which do not have incoming activities). Number that event as 2.

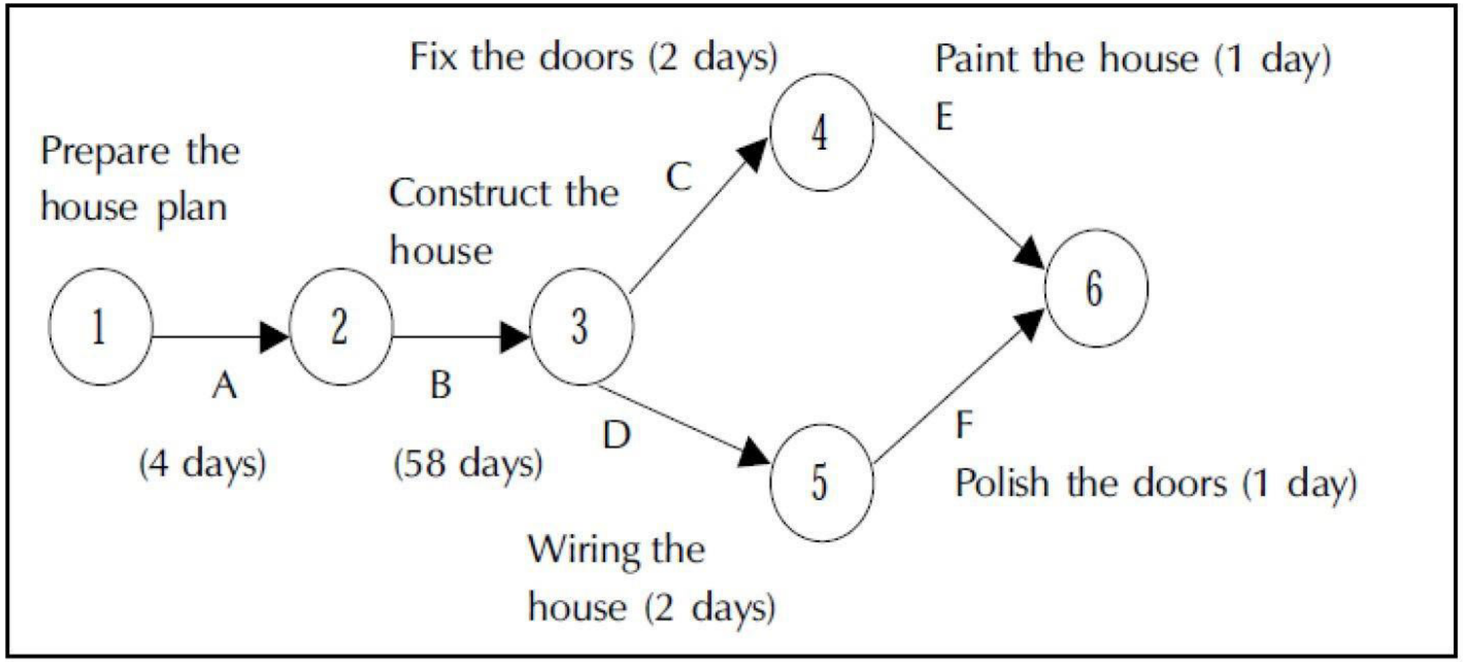
Step3: Repeat step 2 for event 2, event 3 and till the end event. The end event must have the highest number.

Example 1: Draw a network for a house construction project. The sequence of activities with their predecessors are given in following table:

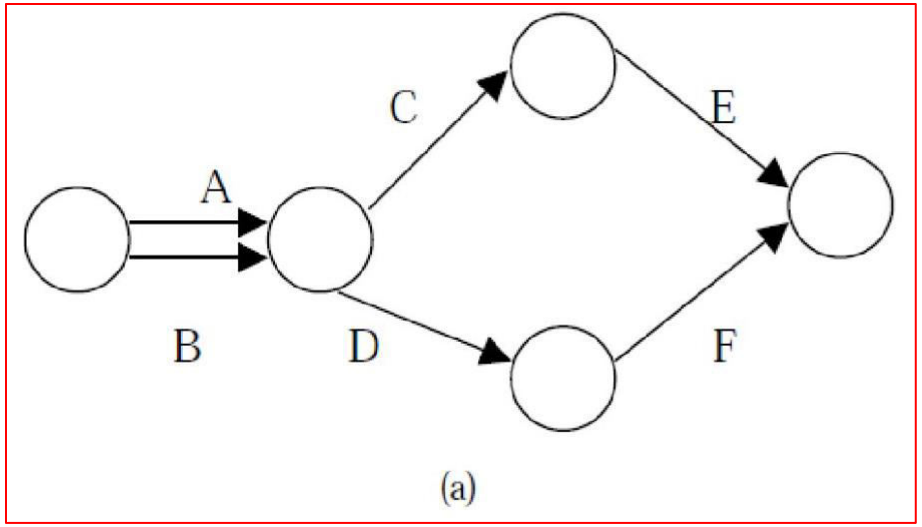
Name of the activity	Starting and finishing event	Description of activity	Predecessor	Time duration (days)
A	(1,2)	Prepare the house plan	--	4
B	(2,3)	Construct the house	A	58
C	(3,4)	Fix the door / windows	B	2
D	(3,5)	Wiring the house	B	2
E	(4,6)	Paint the house	C	1
F	(5,6)	Polish the doors / windows	D	1

Name of the activity	Starting and finishing event	Description of activity	Predecessor	Time duration (days)
A	(1,2)	Prepare the house plan	--	4
B	(2,3)	Construct the house	A	58
C	(3,4)	Fix the door / windows	B	2
D	(3,5)	Wiring the house	B	2
E	(4,6)	Paint the house	C	1
F	(5,6)	Polish the doors / windows	D	1

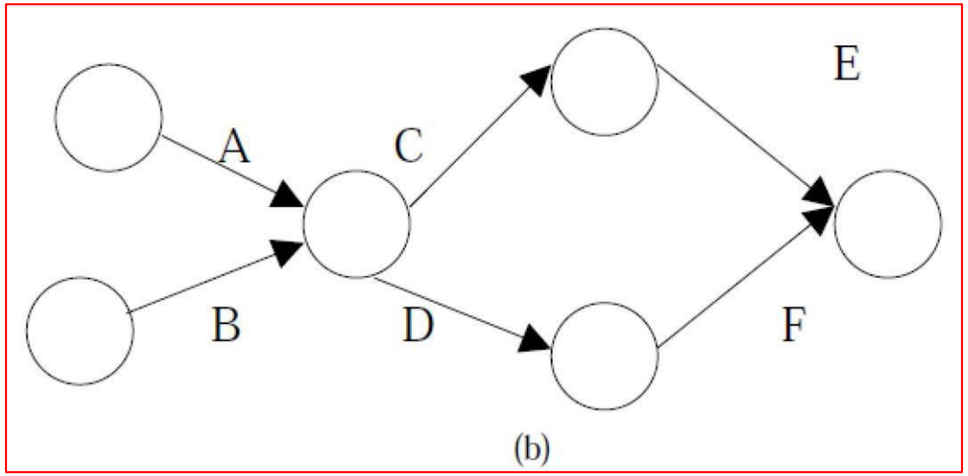
Solution:

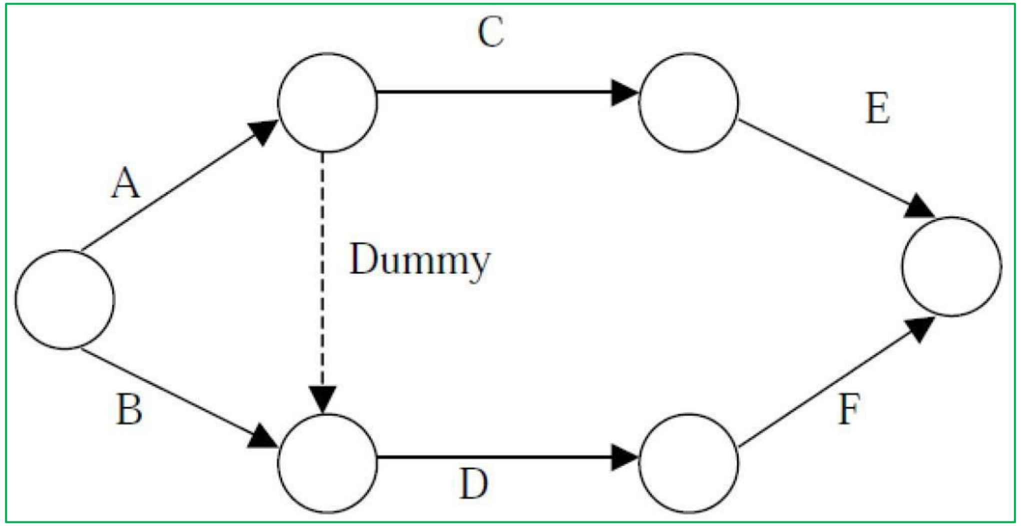


Example 2: Understand the need for Dummy activity and draw the table from the given Network Diagram

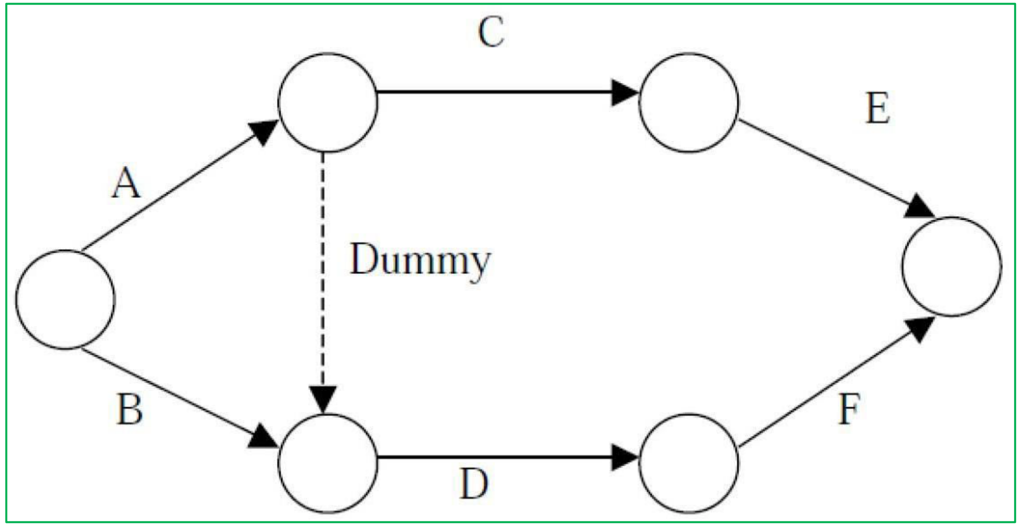


(a) and (b) are wrong.



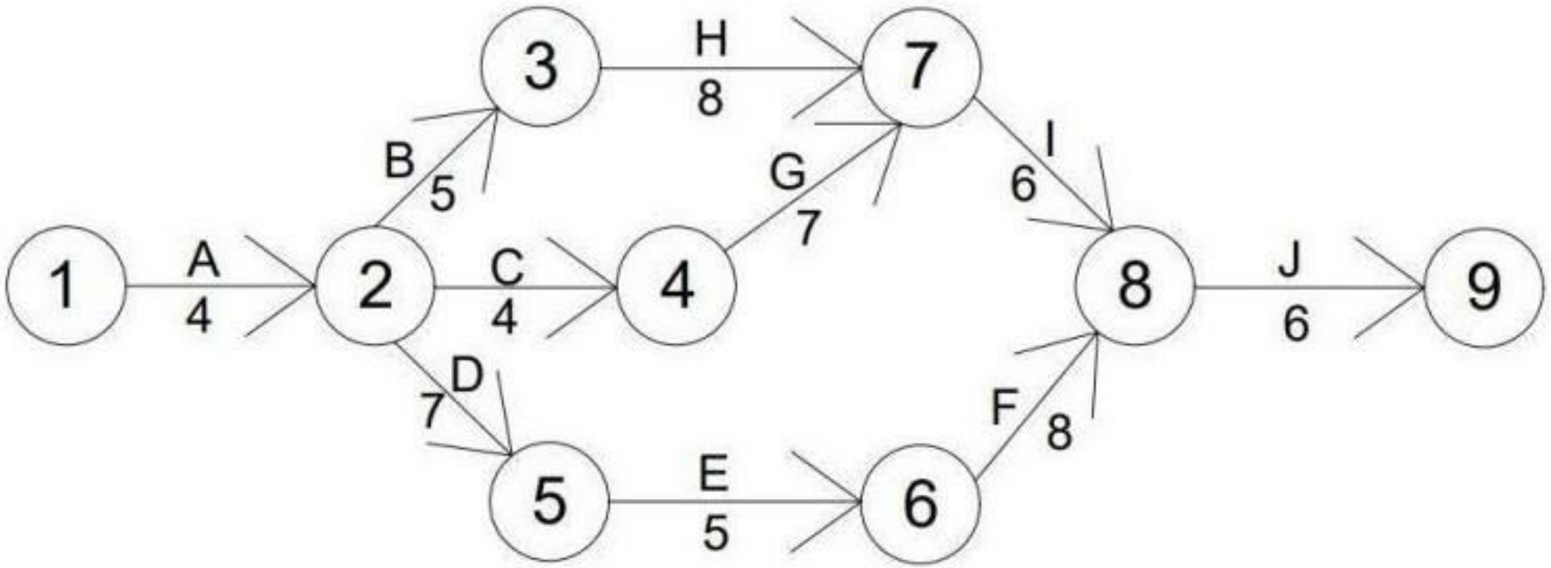


Example 2: Understand the need for Dummy activity and draw the table from the given Network Diagram



Name of the Activity	Starting and Finishing Event	Predecessor

Critical Path Method (CPM)



- This critical path method was developed in 1957 and
- Initially, it was suitable for both civil and mechanical engineering projects.
- This technique is useful **to determine** how best **to reduce the time required to perform production, maintenance and construction and also to minimize the direct and indirect cost** of the project.

Critical Path Method (CPM)

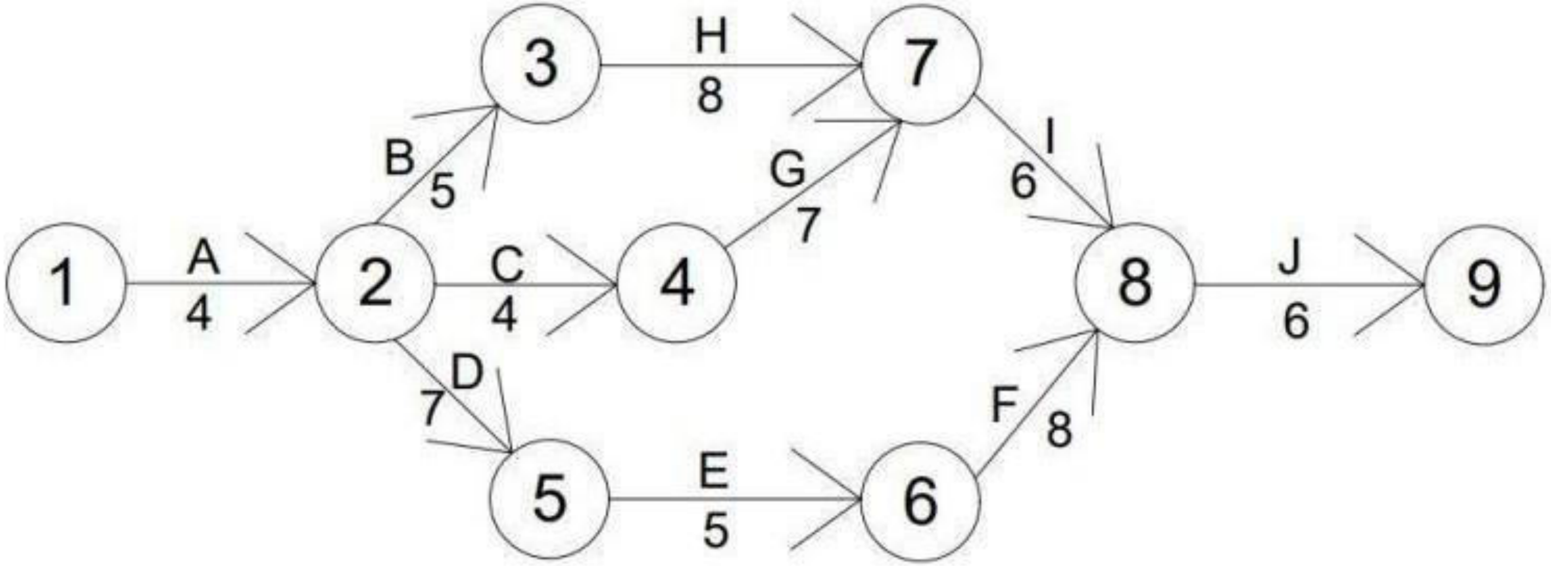
Using this critical path method C.P.M, a **planning engineer** will come to know the **sequence of the various activities** of the project.

- Critical path network can be prepared using the below steps :
  1. Preparation of network
  2. Estimation of the expected time to perform each activity.
  3. Determination of critical path schedule.
  4. Interpretation of the results.

Critical Path Method Terminologies

CPM terms:

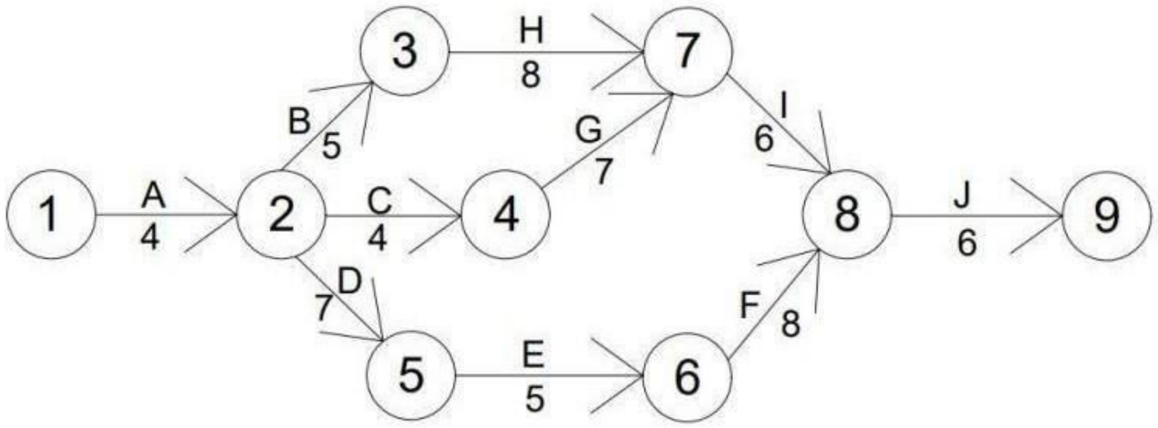
1. Activity
2. Dummy activity
3. Event
4. Network
5. Early start time (E.S.T)
6. Early finish time (E.F.T)
7. Late start time
8. Late finish time
9. Total float
10. Free float
11. Critical activities
12. Critical path



1. Activity

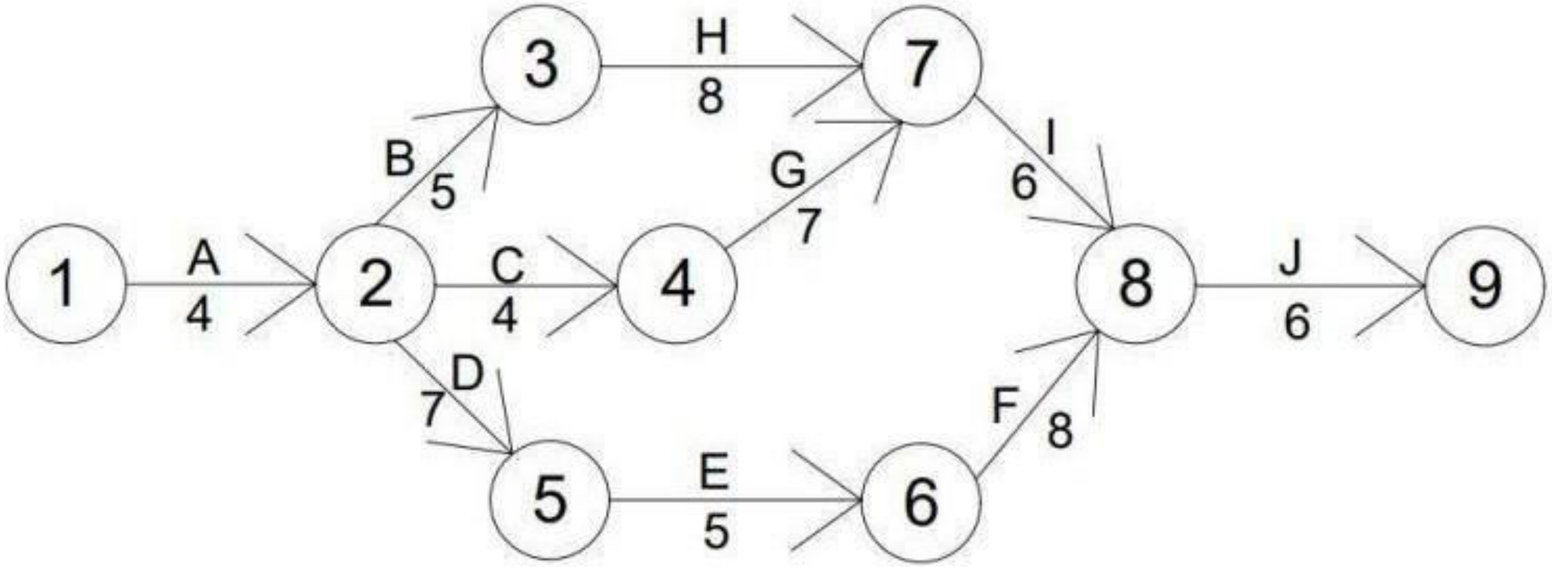
- **Activity** is part of the project.
- The network of the project is denoted by an **arrow**.
- The **start of the activity** is indicated by the **tail of the arrow** and the **end of the activity** is indicated by the **head of the arrow**.
- For a **given duration** of an **activity only one arrow** is utilised.
- The **arrows on the activity are not represented to scale**.
- Along with the arrows, the duration of the activity is shown.
- For example:

Between 2 and 4 arrows activity C is shown ,and its 4 units of time duration is shown beneath the arrow.



2.Dummy activity

- The dummy activity is the activity which does not ustilise any time or any resources for the completion of activity .

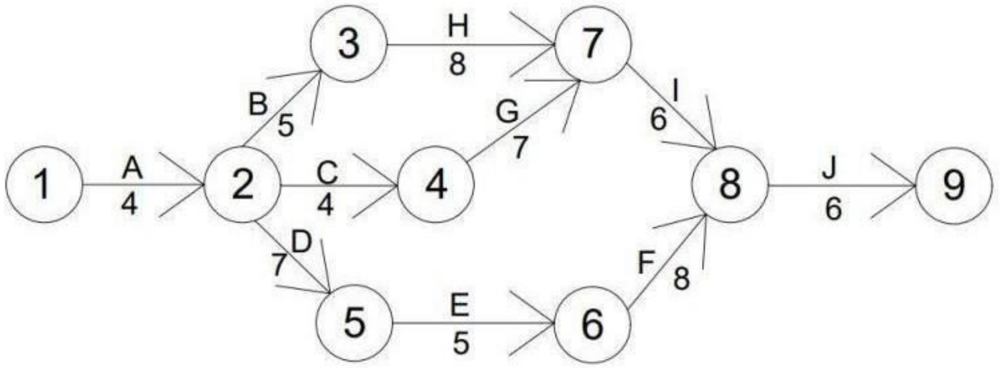


3.Event

- An **event** is the **stage or point** at which **all previous completed activities** merge and **jobs coming out of this point needs to be finished.**

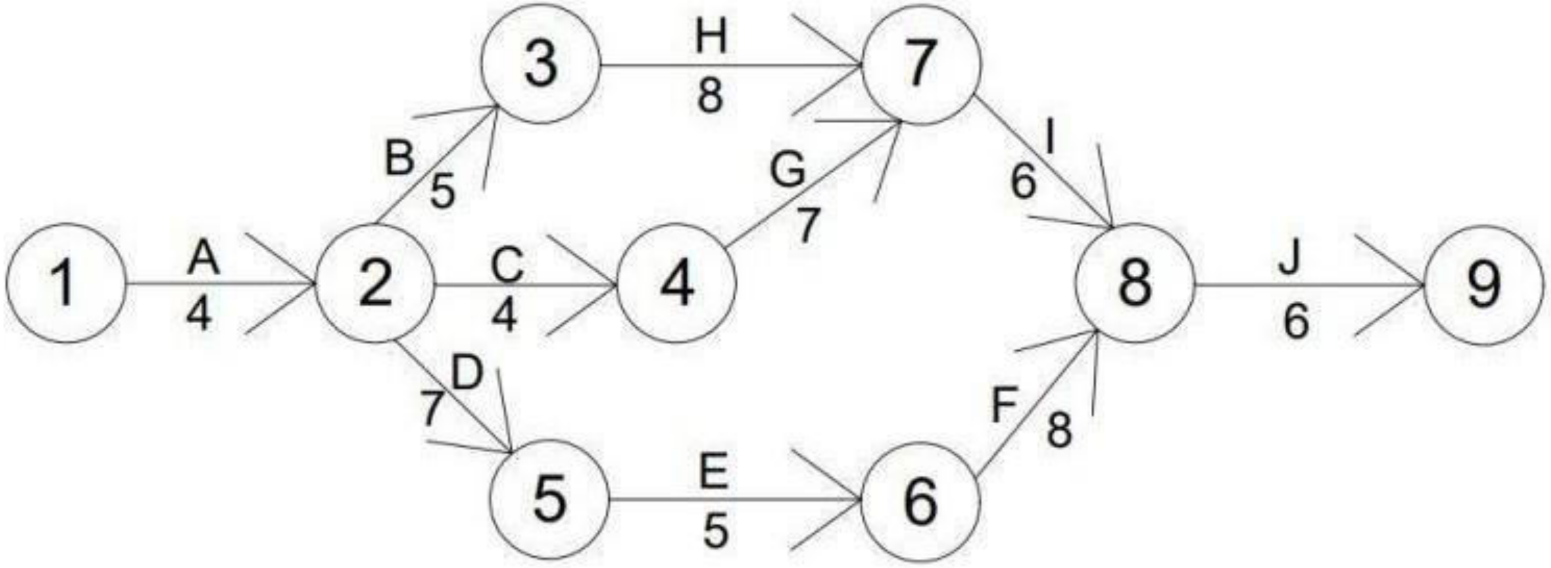
- At the junctions of arrows, they are represented by **circles** or **nodes**.

- In sequential order, they are **serially numbered**.



4. Network

- The representation of the **entire project activities in diagram**matically is termed as **network**.
- On this diagram, The various projects jobs which are expected to be completed are shown in the order.



Other CPM Terminologies from 5 to 12

5. Early start time

- An early start is the **earliest possible time** at which **an activity may start**.

6. Early finish time

- Early finish time is the **sum** of the **earliest start time of activity** and **the time required for its completion**.

7. Late start time

- **Without delaying** the project date, the **most possible time at which an activity may start** is termed as late start time.

8. Late finish time

- The **sum** of the **time required for completion** and the **late start time of activity** is termed as late finish time.

Other CPM Terminologies from 5 to 12

9. Total float

- ☐The difference between the estimated duration and maximum allowed time for an activity is termed as total float.
- ☐Without disturbing the project schedule it is the span of time by which the activity can start.
- ☐It is expressed by letter S.

10. Free float

- ☐The span of time without disturbing the start of succeeding activities by which the activity completion can be delayed is termed as Free float.
- ☐It is expressed by S.F.

11. Critical activities

- ☐The events which have no float are termed as critical activities.
- ☐All the critical activities need to be finished on schedule.

12. Critical path

- ☐Thejoining of the critical events in the network path is termed as critical path .

Basic rules for critical path method network

1. The **flow direction** is indicated by **arrows**, which is normally accepted from **left to right.**
2. The **length duration of the activity** is **not correspond to length of the arrow.** It is just geometrical configuration and does not have any importance.
3. The **activities merging** at it **must be completed before starting any new activity**.
4. The **indication of the logical conditions of dependency** in a network is **done** by an arrow.
5. The **construction of network** is done **logic**ally on the **principle of dependency**.
6. For **starting the project and also for closing the project,** only one event must be there in every network.
7. These **connection of any two events** may be **done directly** but **not with more than one activity.**
8. **Serial numbers** should be **given to events** and those **numbers should not done duplication** in a network.

Drawing a Network in critical path method

1. **List all the activities** required for completion of a project.
2. **Ascertain the dependence** of the activities on each other.
3. **Draw a network of the activities** roughly with a soft pencil so that it can be easily erased.
4. The **duration of each activity** can be **estimated** by keeping in mind about the availability of **equipment, manpower** and also the **likelihood of activities delays**.
5. The **estimate time** of each activity **needs to be noted**.
6. For **deciding the critical path**, the latest and earliest allowable start and finish times for every activity to determined.
7. In the **field** need to **finalise the network** for use.

Advantages of critical path method

1. It is useful to **ascertain the time schedule**
2. It helps to have **proper control of the project**.
3. It enables **better and detailed planning**.
4. It is very helpful **to communicate** the cost, project schedule, project planning ,performance ,time and project planning.
5. The **critical activities** which are **highlighted along the critical path should be attended with care until their completion**.

Despite the great planning cost of a project, **Critical path proves to be very useful for focusing the attention of project engineers** on the critical path activities and thus **avoids unnecessary expenditure**.

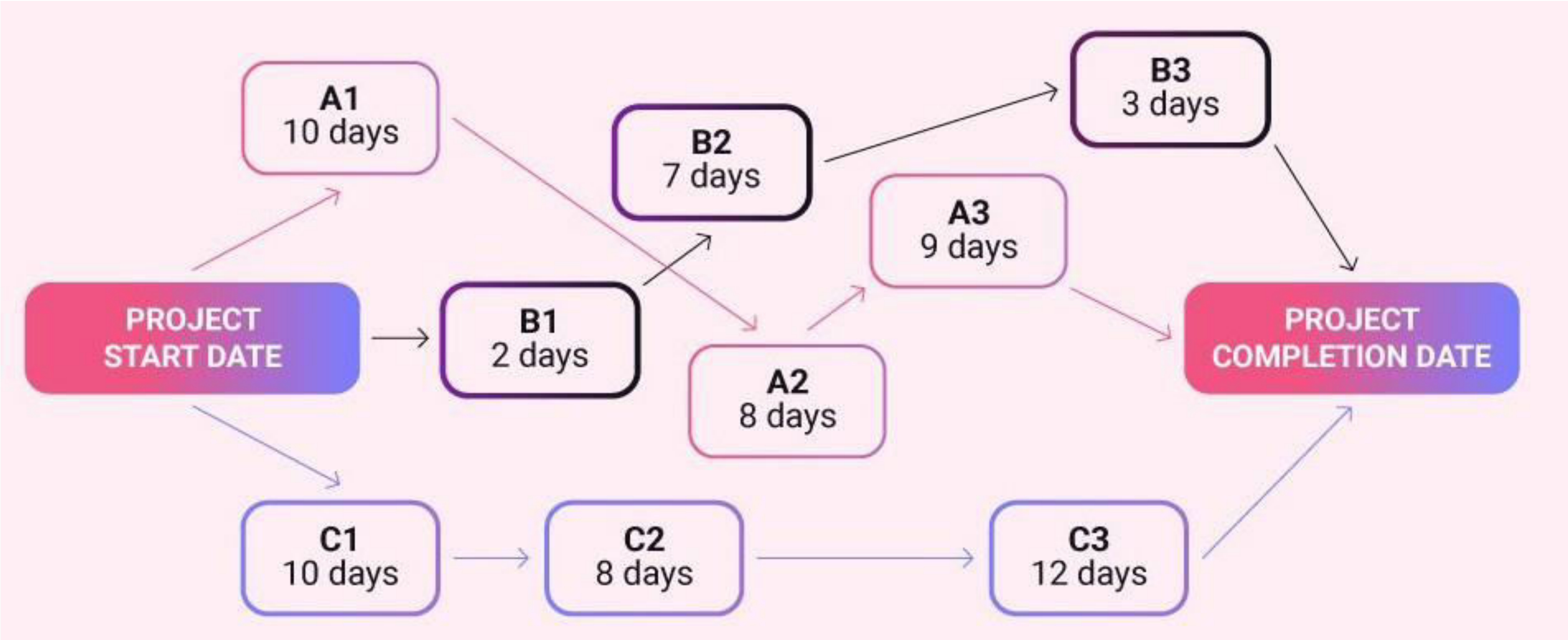
Expected length of the project , the critical path

1. **Critical path :**
   - ☐The **longest duration path** of a network.
   - ☐The **sum of the expected times of the total activities is maximum** in this path.
2. **Length of the project**
   - ☐The **sum of the expected times of all the activities along the critical path of the network** of a project is called the length of the project.
3. **The variance of the project**
   - ☐The variance of the project is the **total sum of the variance along the critical path the activities** of the network of a project.
4. **The standard deviation of the project**
   - ☐The standard deviation is the **square root of the variance** which is calculated along the network critical path of a project.

Critical Path Method Terminologies

1: Critical path

The critical path is the sequence of dependent activities with the longest overall duration.



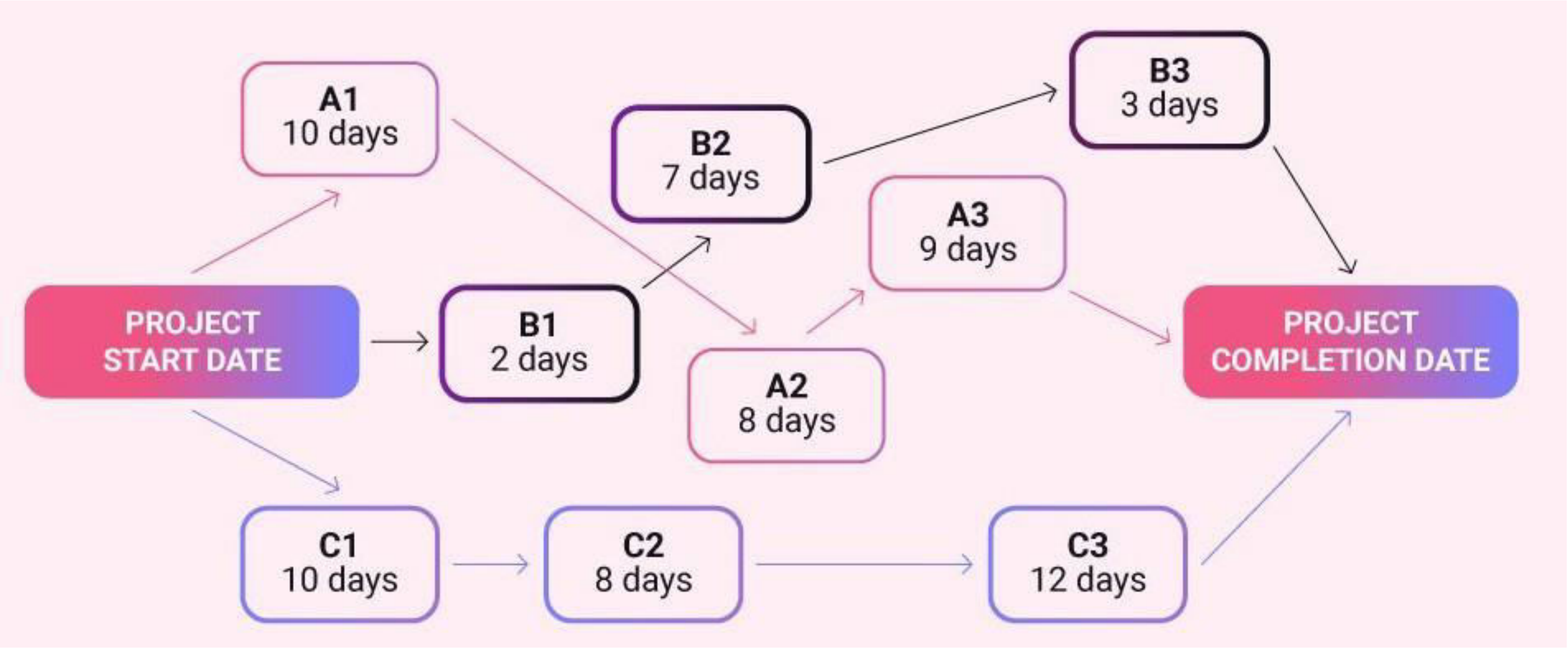
C1-C2-C3 string of activities constitutes the critical path, as they take 30 days to complete.

Critical Path Method Terminologies

2: Activity nodes

Activity nodes are rectangles used in CPM diagrams designed to hold all the crucial information needed for charting a project's critical path.

Each task required for project completion gets its own activity node. The illustration below features 9 activity nodes: **A1, A2, A3, B1, B2, B3, C1, C2, and C3**.



Critical Path Method Terminologies

3: Early start time (ES)

- Early start time denotes the **earliest possible point at which you can start working on a task**.
- The **ES for the first task(s) in a project is set to 0**.
- To **calculate the ES for all other tasks**, just **copy the EF (early finish time) of the preceding dependent task**.
  - **If a task is dependent on 2 or more** previous activities, **refer to the activity with the longer ES**.

Critical Path Method Terminologies

4: Duration (Dur)

Duration is self-explanatory — it denotes how long you expect a task will take to complete.

5: Early finish time (EF)

- **Early finish time denotes the earliest moment at which it is possible to finish a task**.
- To calculate the early finish time for a task, you take its ES and add the task duration (Dur).

$$EF = ES + Dur$$

- Since the first activity in a project generally has an ES of 0, the EF will be identical to the duration of the first activity.
- All dependent tasks that follow immediately after it will take the first task's EF as their ES, and add their duration to it to calculate their EF.

Critical Path in Network Analysis

Basic Scheduling Computations

The notations used are

- **(i, j)** = Activity with **tail event** i and **head event** j
- **E_i** = **Earliest occurrence time** of event i
- **L_j** = **Latest allowable occurrence time** of event j
- **D_{ij}** = **Estimated completion time** of activity (i, j)
- **$(Es)_{ij}$** = **Earliest starting time** of activity (i, j)
- **$(Ef)_{ij}$** = **Earliest finishing time** of activity (i, j)
- **$(Ls)_{ij}$** = **Latest starting time** of activity (i, j)
- **$(Lf)_{ij}$** = **Latest finishing time** of activity (i, j)

Critical Path in Network Analysis

1. Determination of Earliest time (E_j): Forward Pass computation

- **Step 1**

  The computation begins from the start node and move towards the end node. For easiness, the forward pass computation starts by assuming the earliest occurrence time of zero for the initial project event.

- **Step 2**

  i. Earliest starting time of activity (i, j) is the earliest event time of the tail end event i.e. $(Es)_{ij} = E_i$

  ii. Earliest finish time of activity (i, j) is the earliest starting time + the activity time i.e. $(Ef)_{ij} = (Es)_{ij} + D_{ij}$ or $(Ef)_{ij} = E_i + D_{ij}$

  iii. Earliest event time for event j is the maximum of the earliest finish times of all activities ending in to that event i.e. $E_j = \max [(Ef)_{ij}$ for all immediate predecessor of (i, j)] or $E_j = \max [E_i + D_{ij}]$

Critical Path in Network Analysis

2. Backward Pass computation (for latest allowable time)

- **Step 1**

  For ending event assume E = L. Remember that all E's have been computed by forward pass computations.

- **Step 2**

  Latest finish time for activity (i, j) is equal to the latest event time of event j i.e. $(Lf)_{ij} = L_j$

- **Step 3**

  Latest starting time of activity (i, j) = the latest completion time of (i, j) – the activity time or $(Ls)_{ij} = (Lf)_{ij} - D_{ij}$ or $(Ls)_{ij} = L_j - D_{ij}$

- **Step 4**

  Latest event time for event 'i' is the minimum of the latest start time of all activities originating from that event i.e. $L_i = \min [(Ls)_{ij}$ for all immediate successor of (i, j)] $= \min [(Lf)_{ij} - D_{ij}] = \min [L_j - D_{ij}]$

Critical Path in Network Analysis

3. Determination of floats and slack times

There are three kinds of floats

- **Total float** – The amount of time by which the completion of an activity could be delayed beyond the earliest expected completion time without affecting the overall project duration time.

  Mathematically

  $(Tf)_{ij}$ = (Latest start – Earliest start) for activity (i – j)

  $(Tf)_{ij} = (Ls)_{ij} - (Es)_{ij}$ or $(Tf)_{ij} = (L_j - D_{ij}) - E_i$

- **Free float** – The time by which the completion of an activity can be delayed beyond the earliest finish time without affecting the earliest start of a subsequent activity.

  Mathematically

  $(Ff)_{ij}$ = (Earliest time for event j – Earliest time for event i) – Activity time for (i, j)

  $(Ff)_{ij} = (E_j - E_i) - D_{ij}$

Critical Path in Network Analysis

3. Determination of floats and slack times

There are three kinds of floats

- **Independent float** – The amount of time by which the start of an activity can be delayed without effecting the earliest start time of any immediately following activities, assuming that the preceding activity has finished at its latest finish time.

  Mathematically

  $$(If)_{ij} = (E_j - L_i) - D_{ij}$$

  The negative independent float is always taken as zero.

- **Event slack** - It is defined as the difference between the latest event and earliest event times.

  Mathematically

  Head event slack = $L_j - E_j$, Tail event slack = $L_i - E_i$

Critical Path in Network Analysis

4. Determination of Critical Path

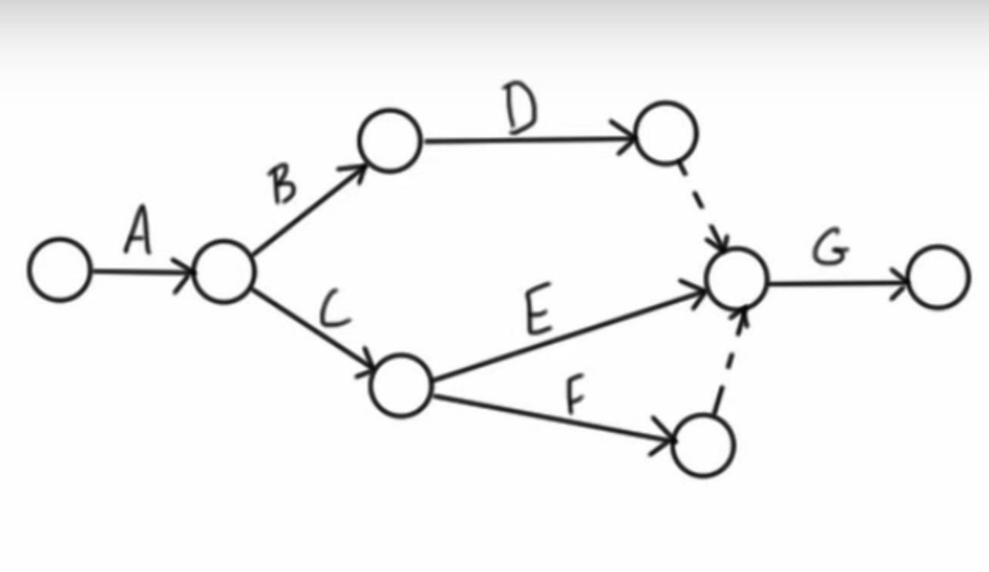
- **Critical event** – The events with zero slack times are called critical events. In other words the event i is said to be critical if $E_i = L_i$

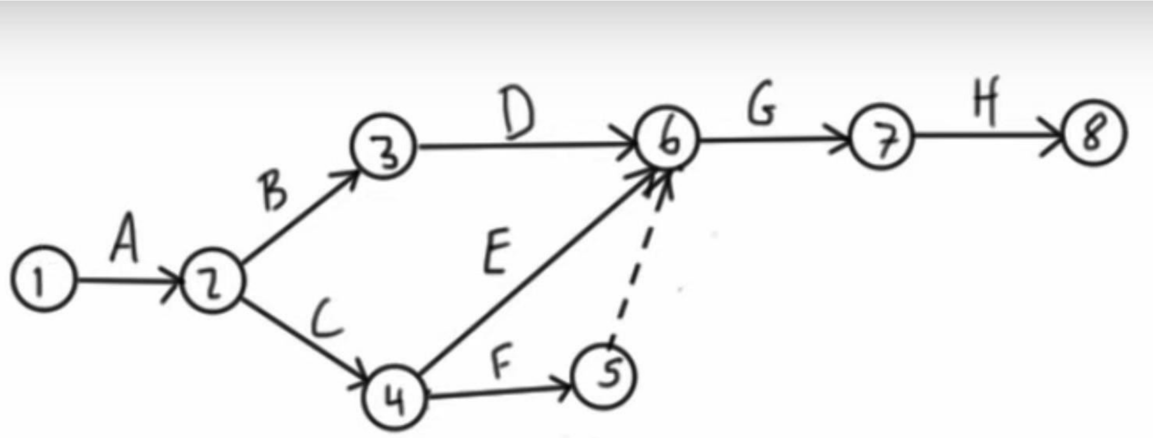
- **Critical activity** – The activities with zero total float are known as critical activities. In other words an activity is said to be critical if a delay in its start will cause a further delay in the completion date of the entire project.

- **Critical path** – The sequence of critical activities in a network is called critical path. The critical path is the longest path in the network from the starting event to ending event and defines the minimum time required to complete the project.

How to Draw Network Diagram

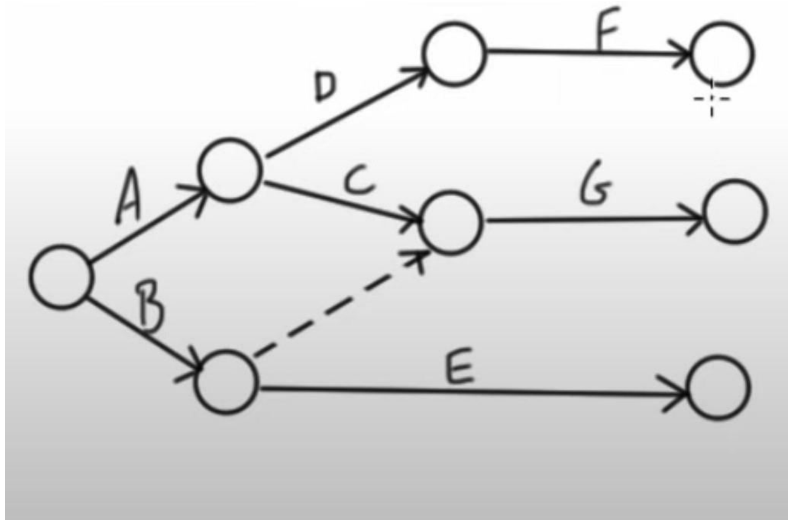
Activity	Predecessor
A	-
B	A
C	A
D	B
E	C
F	C
G	D,E,F
H	G

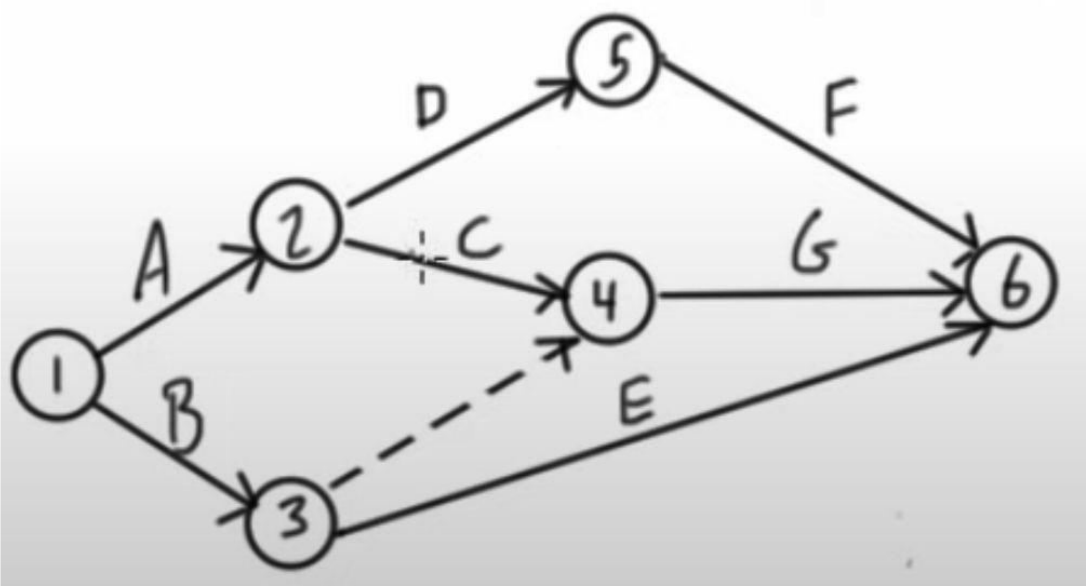




How to Draw Network Diagram

Activity	Predecessor
A	-
B	-
C	A
D	A
E	B
F	D
G	B,C

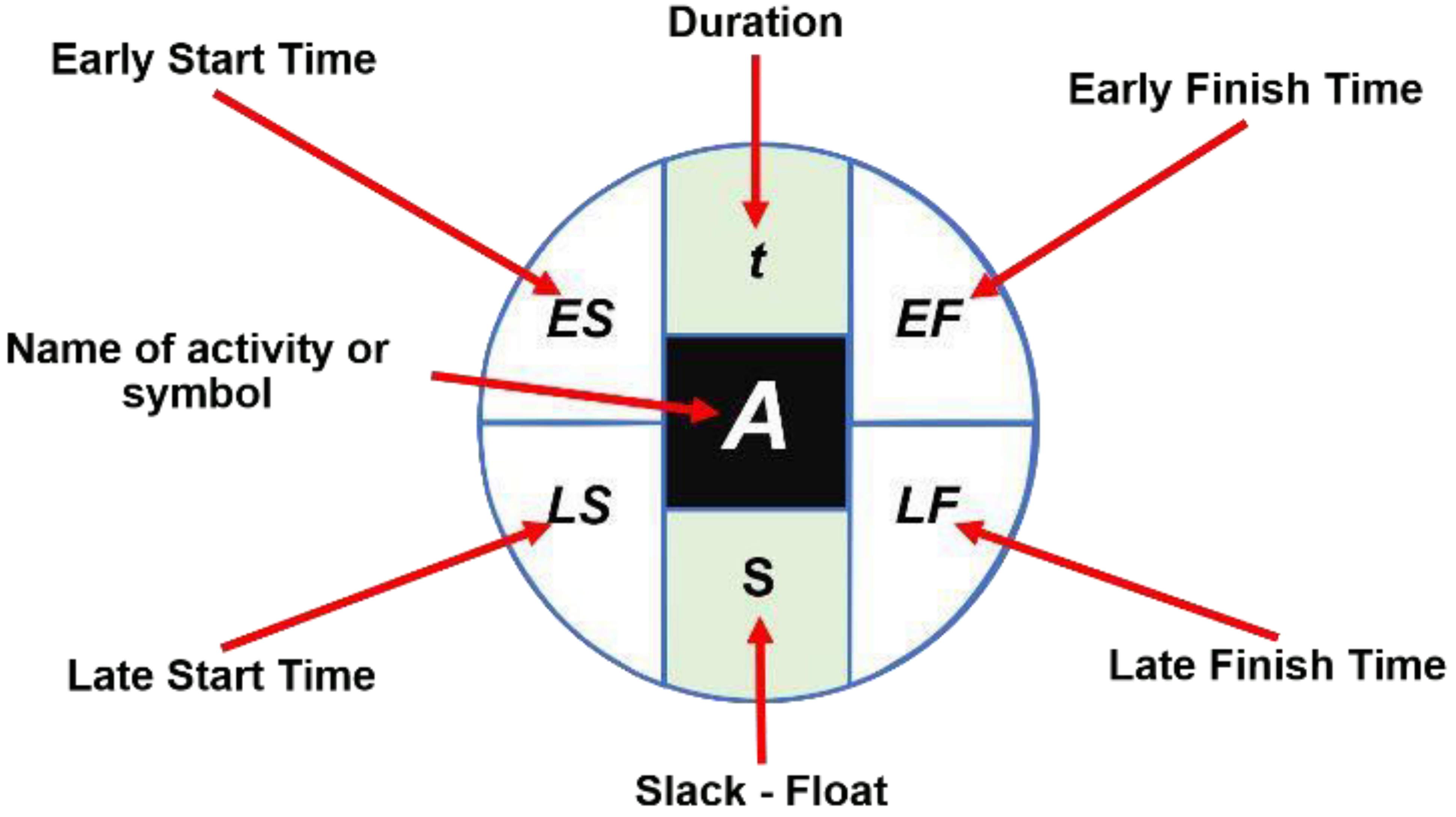


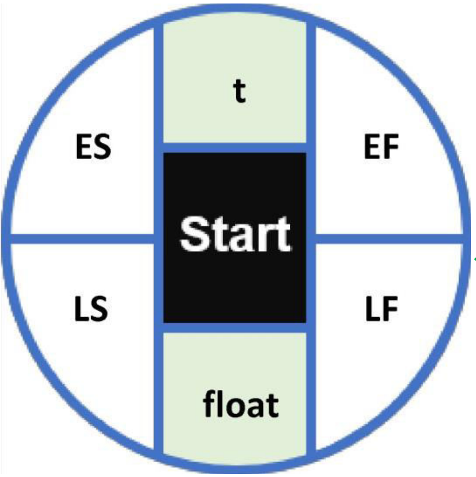


Critical Path in Network Analysis

- Step 1: Obtain the project data. Make a list of all the activities of the project along with their dependencies and their specific times.
- Step 2: Elaborate the network diagram. We have written a post that explains how to elaborate the project network diagram step by step.
- Step 3: Calculate the Early Start and Late Start Times. Determine the Early Start and Late Start Times for each activity.
- Step 4: Calculate the Early Finish and Late Finish Times. Determine the Early Finish and Late Finish Times for each activity.
- Step 5: Calculate the slack time. The critical path must be determined by finding out the slack for each activity of the process. The activities where there is no slack are the ones making up the critical path.

How to Calculate Early Start, Late Start, Early Finish, Late Finish and Slack (Float)

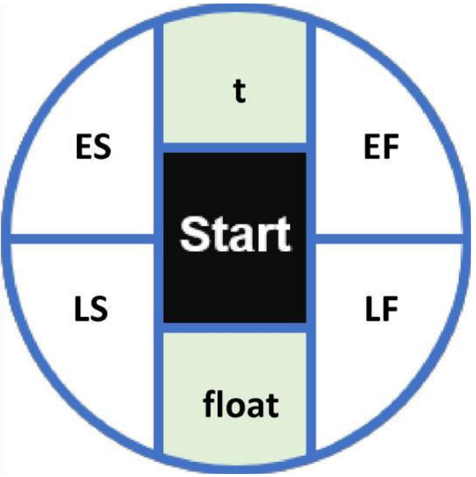




CPM

- **Duration (t):**
  - Indicates the time it takes to complete the activity.
- **Early finish Time (EF):**
  - This is the **earliest time that an activity can finish**. It is equal to the early start time plus its estimated duration (t):

    EF = ES + t
- **Early Start Time (ES):**
  - This is the **earliest time that an activity can be started** assuming all previous activities have been completed beforehand.
  - For **activities that have more than one precedent**, the **ES is the greatest of the late start time of their precedents**.



CPM

❖ Late Finish Time (LF):

- This is the latest time at which an activity can be completed without delaying the entire project. It is obtained by equaling the late start time of the activity that immediately follows.
- If **activities have more than one task immediately** following them, the **LF will be the least of the late start time of those activities**.

❖ Late Start Time (LS):

- It is the **latest time an activity can begin without delaying** the whole project. It is **equal to the Late Finish Time minus the expected duration** of that activity (t):

$$LS = LF - t$$

❖ Slack – Float (S):

- A period of time when an activity can be delayed without causing the entire project to be delayed. All activities contained in the critical path have zero slack.
- Here is how it is calculated mathematically:

$$S = LS - ES = LF - EF$$

CPM

- **Early Start (ES):** It is **equal to the Early Finish to the activity's precedent**. **If** it has **more than one precedent**, the **highest value** is **taken**.
- **Early Finish (EF):** It is equal to the Early Start of the activity plus its duration (t).

$$EF = ES + t$$

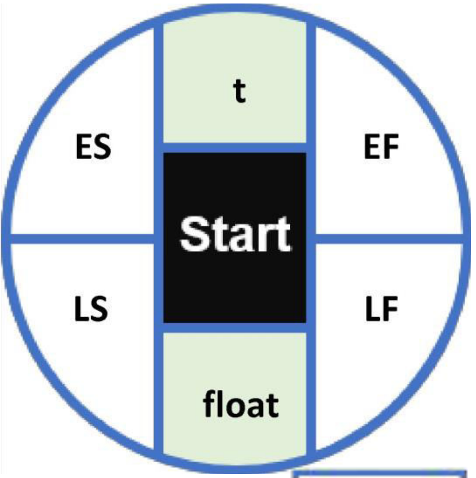
- **Late Start (LS):** It is **equal to the Late Finish minus its duration** (**t**).

$$LS = LF - t$$

- **Late Finish (LF):** It is **equal to the late start** of the activity that follows. **If** it has **more than one successor**, the **lowest** value is **taken**.
- **Slack (S):** It can be calculated in two ways.

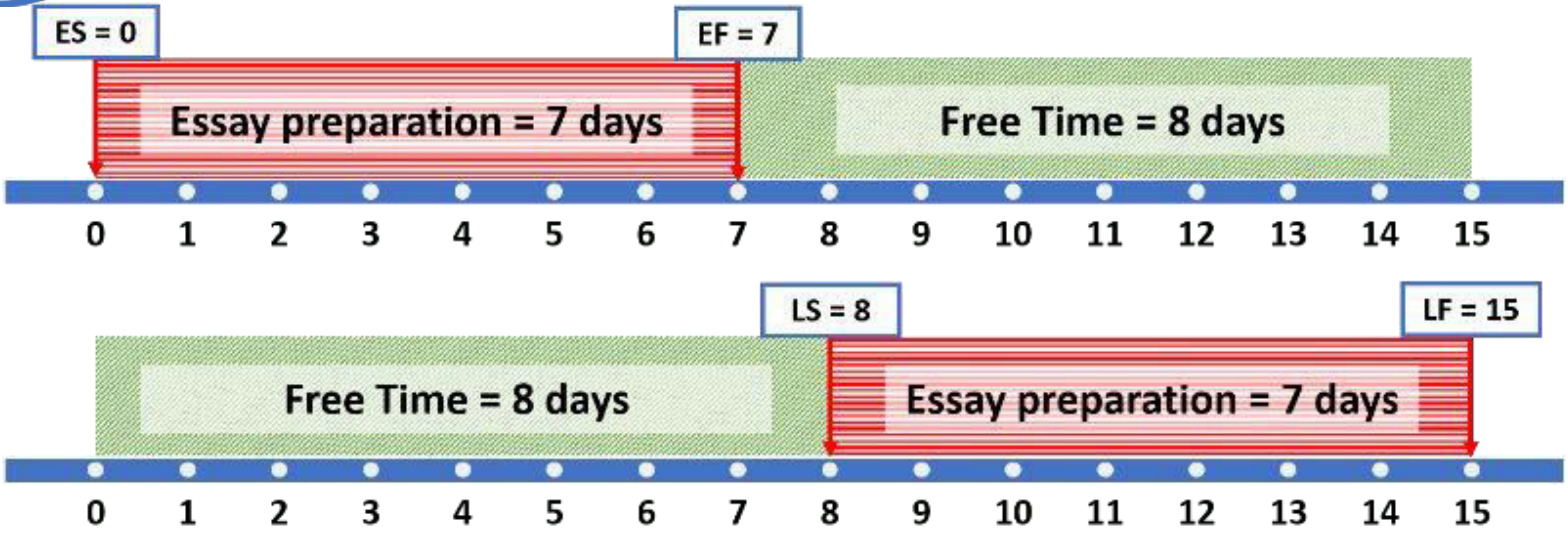
$$S = LS - ES = LF - EF$$

- Activities with zero clearance make up the critical path.

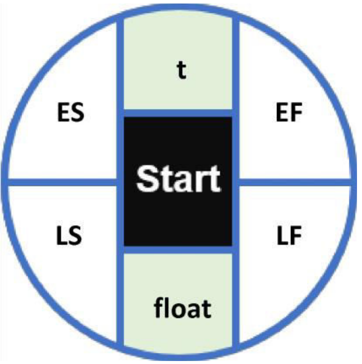


CPM

Imagine that your professor asks you to write an essay to be submitted in 15 days.

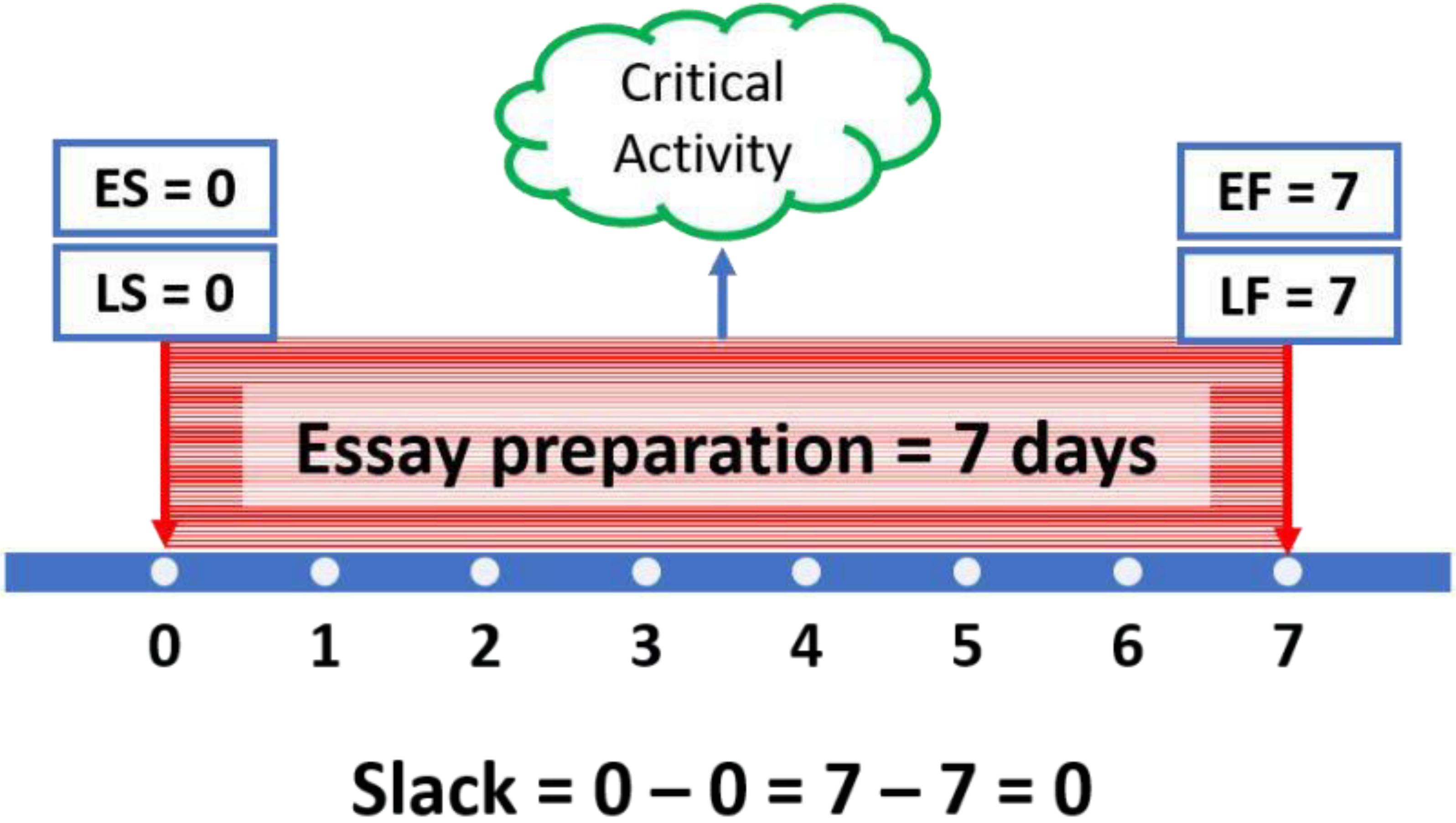


Slack = 8 – 0 = 15 – 7 = 8



CPM

If we change the scenario and our teacher gives us only the **one-week** deadline to submit the essay.



CPM Example

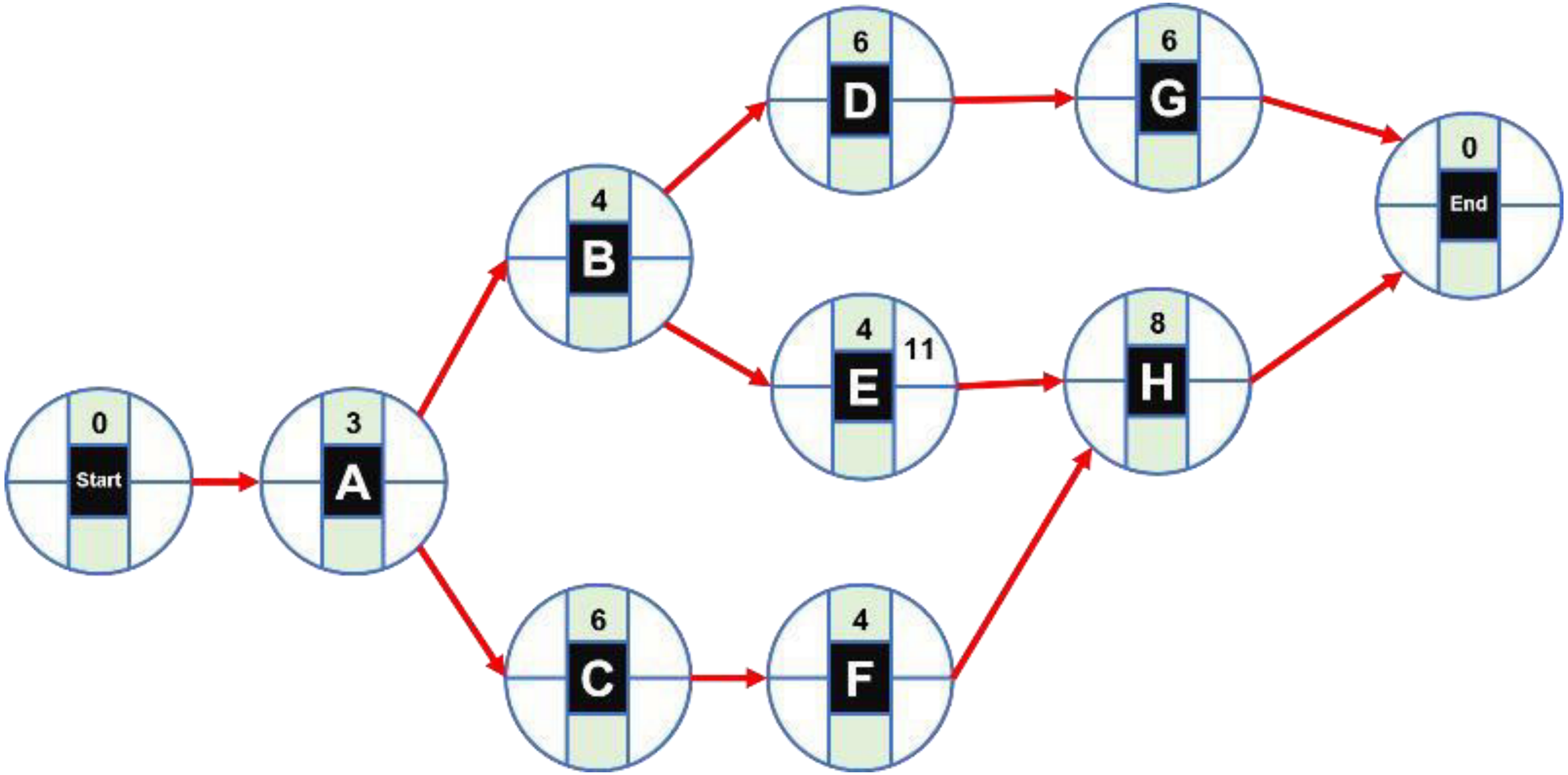
- Draw the activity-on-node (AON) project network associated with the following activities for Dave Carhart's consulting company project.

Activity	Immediate Predecessor(s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

1. How long should it take Dave and his team to complete this project?
2. What are the critical path activities?

CPM Example -Solution

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

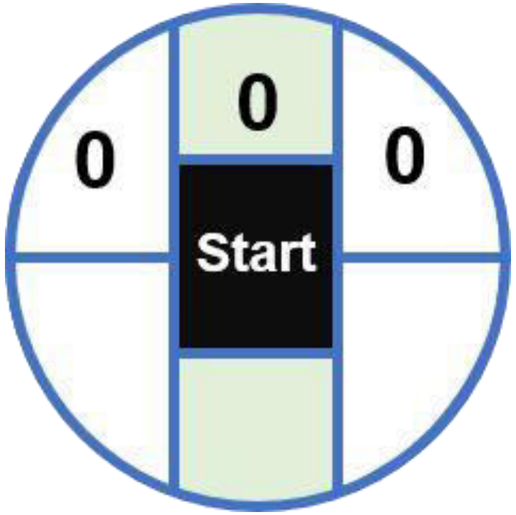


CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Start node:

This dummy node has all values equal to zero.



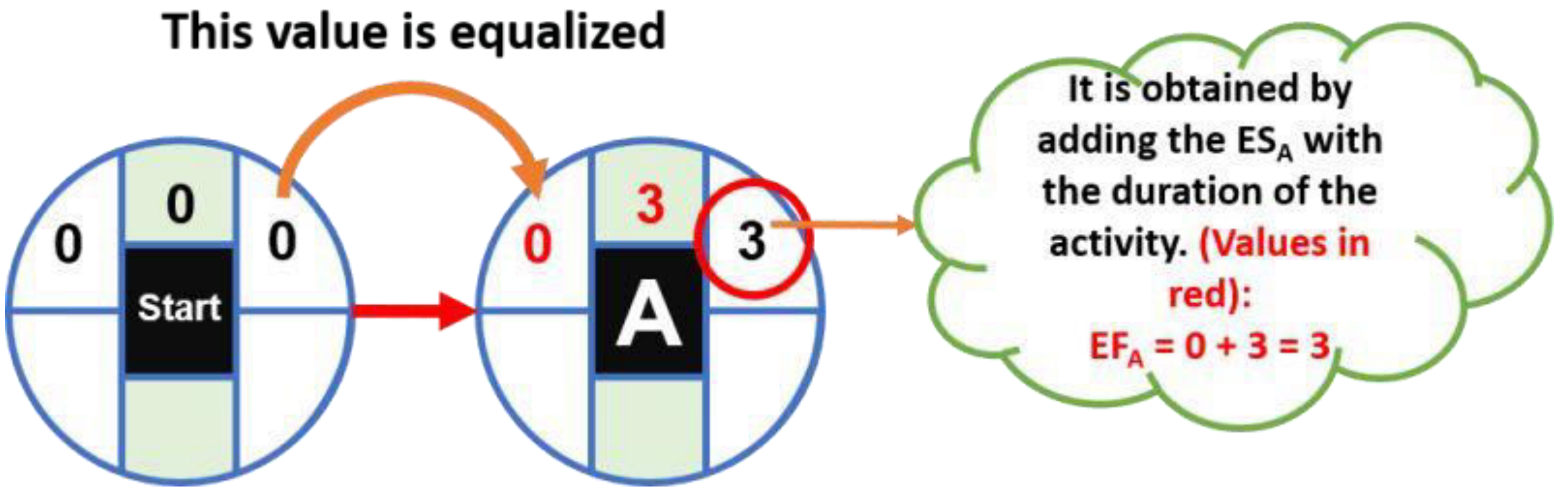
CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity A:

Since it is the first activity, its ES will be **equal** to the EF of the **starting node** (zero); the EF is calculated as follows:

$$EF = ES + activity\ time$$
$$EF_A = 0 + 3 = 3$$



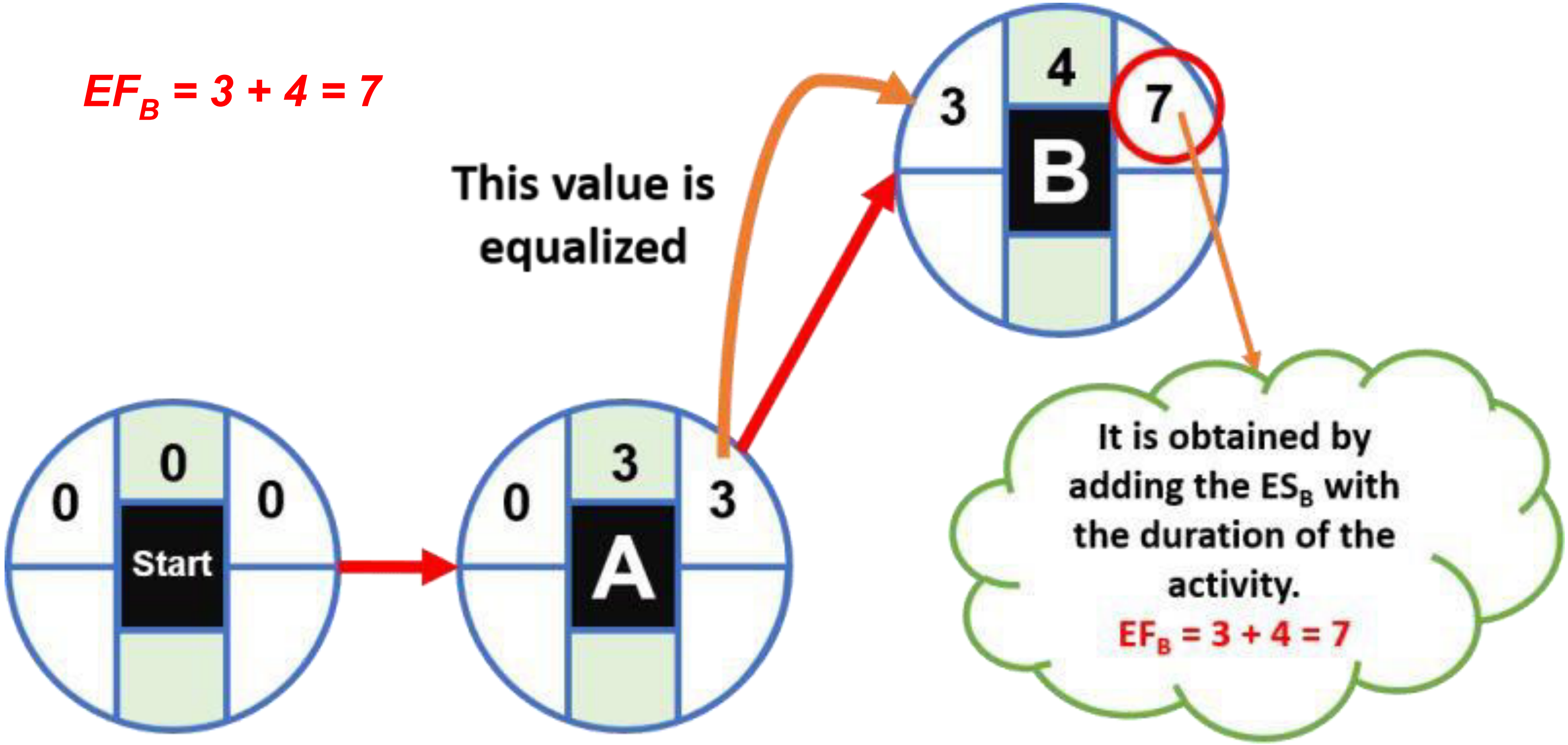
CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity B:

It has as precedent only activity A; therefore its ES will be equal to the EF of activity A. In the same way as the previous node, the EF of activity B is calculated by adding its ES + the corresponding time:

$EF_B = 3 + 4 = 7$

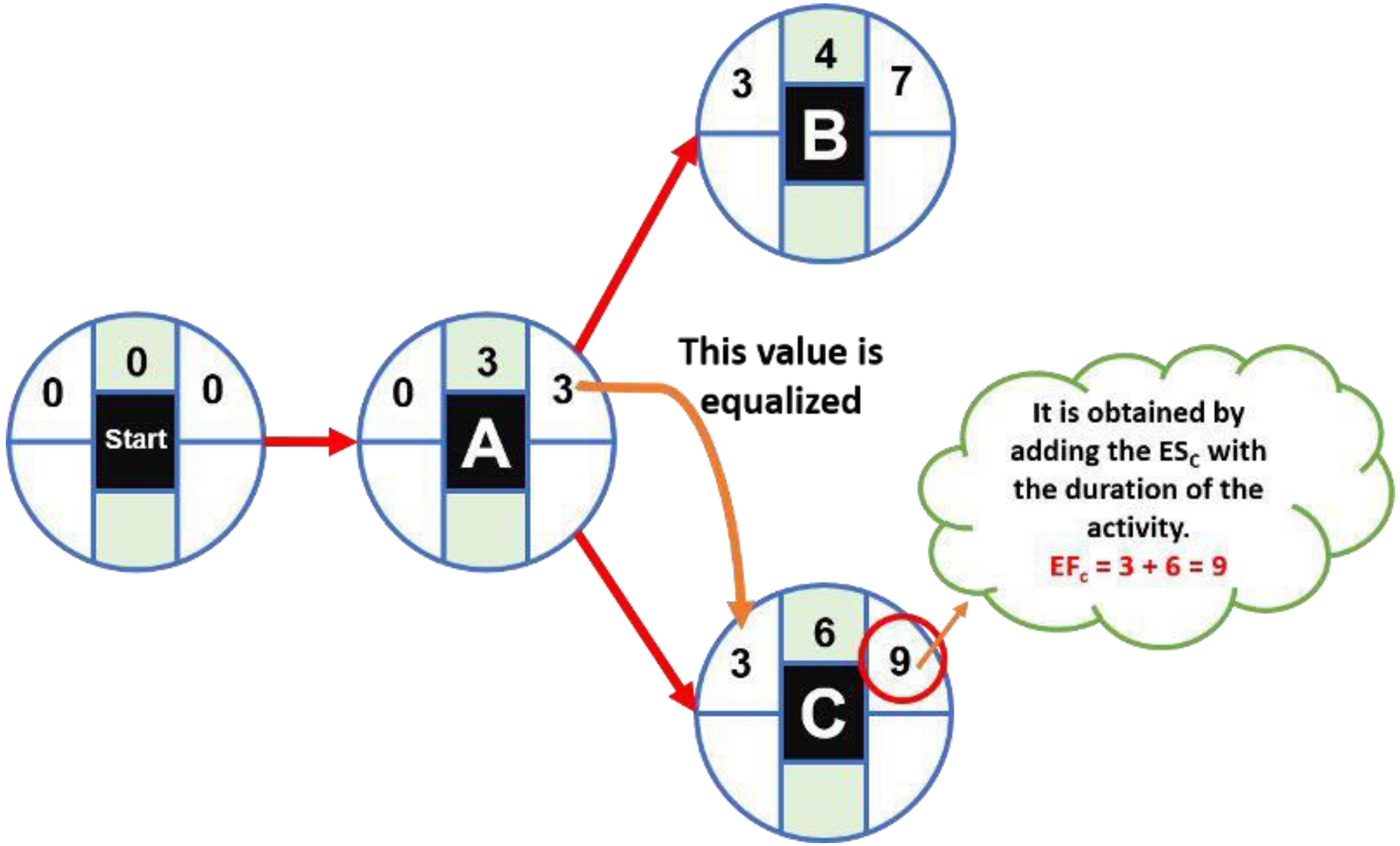


CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity C:

It has as precedent only activity A; therefore its ES will be equal to the EF of activity A. The EF of activity C is calculated by adding its ES + the corresponding time: $EF_C = 3 + 6 = 9$



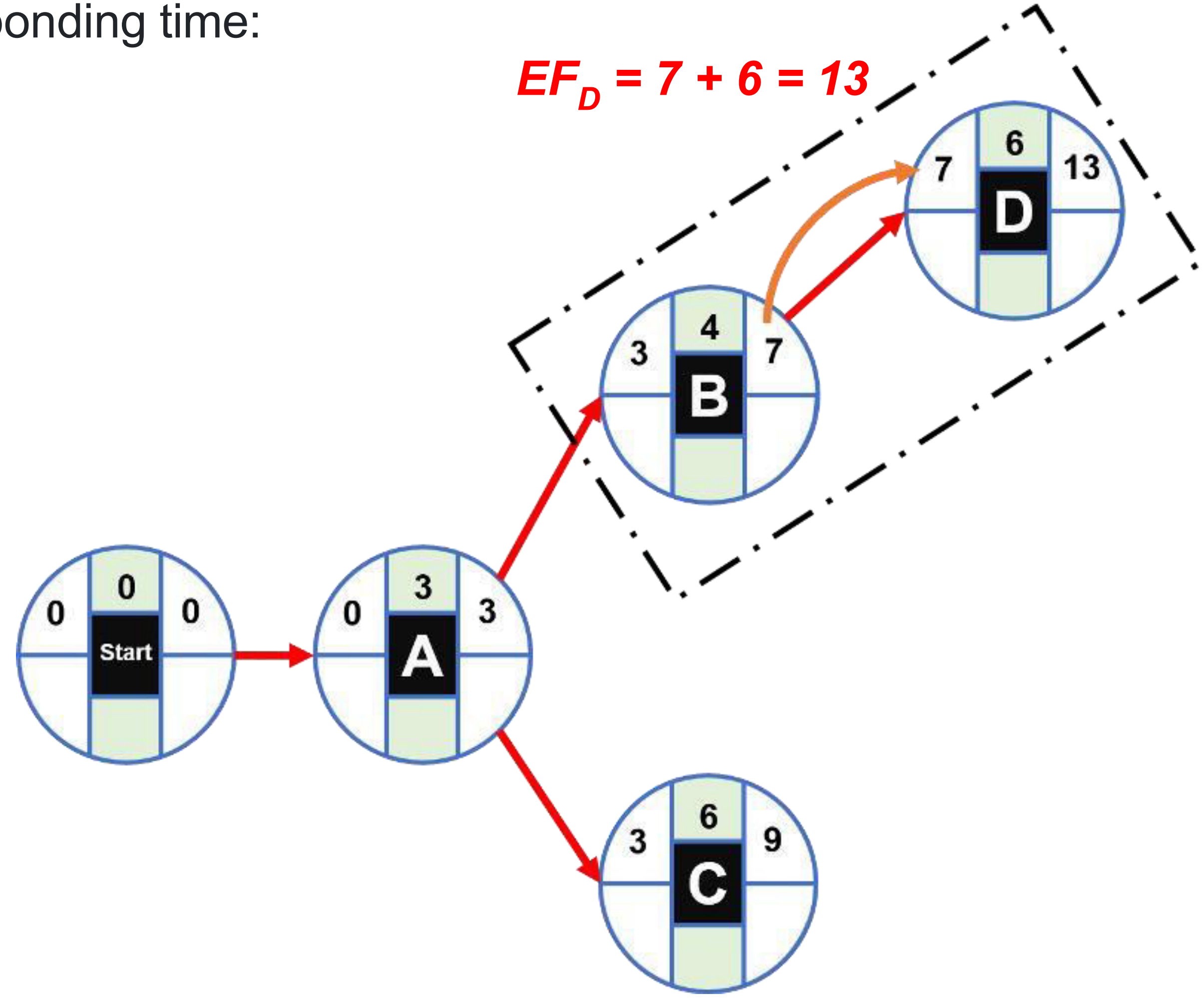
CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity D:

It has as a precedent only activity B; therefore its ES will be equal to the EF of activity B. The EF of activity D is calculated by adding its ES + the corresponding time:

$EF_D = 7 + 6 = 13$



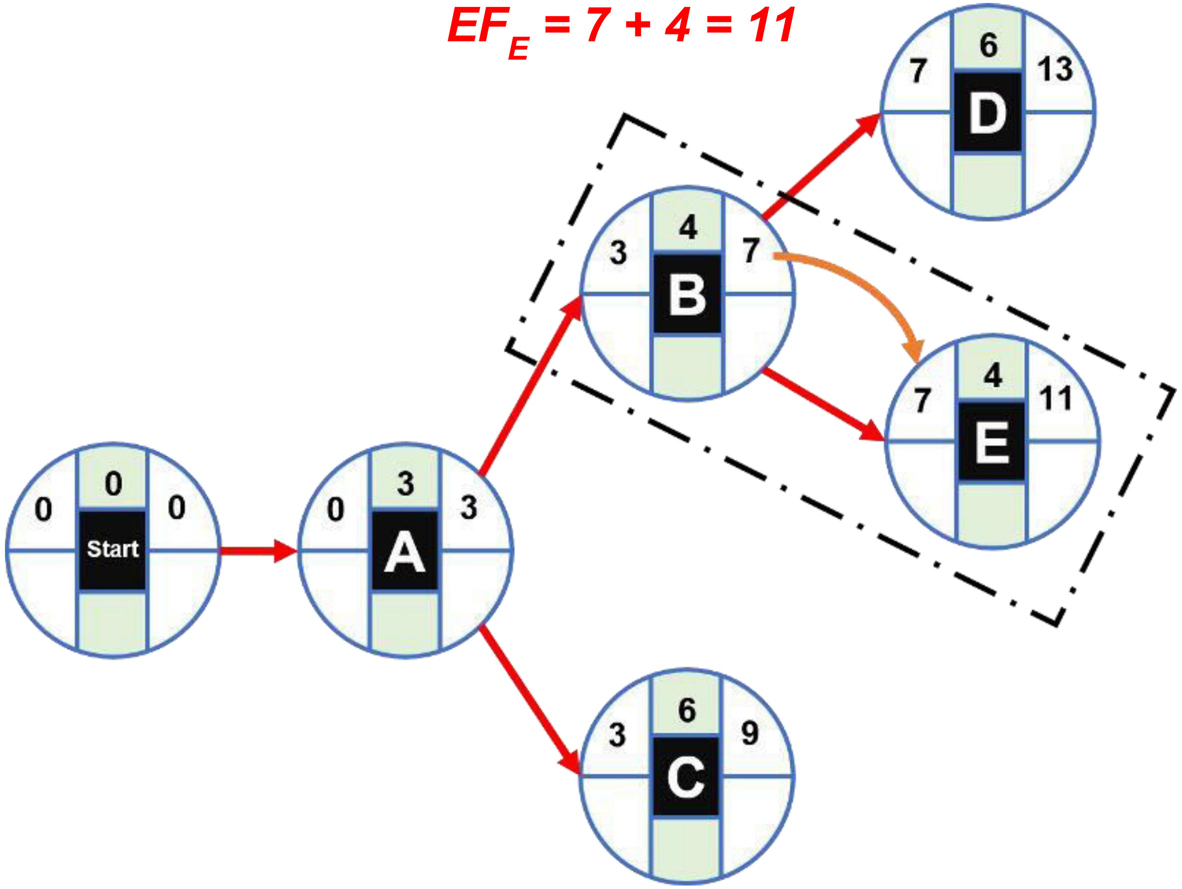
CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity E:

It has as a precedent only activity B; therefore its ES will be equal to the EF of activity B. The EF of activity E is calculated by adding its ES + the corresponding time:

$EF_E = 7 + 4 = 11$



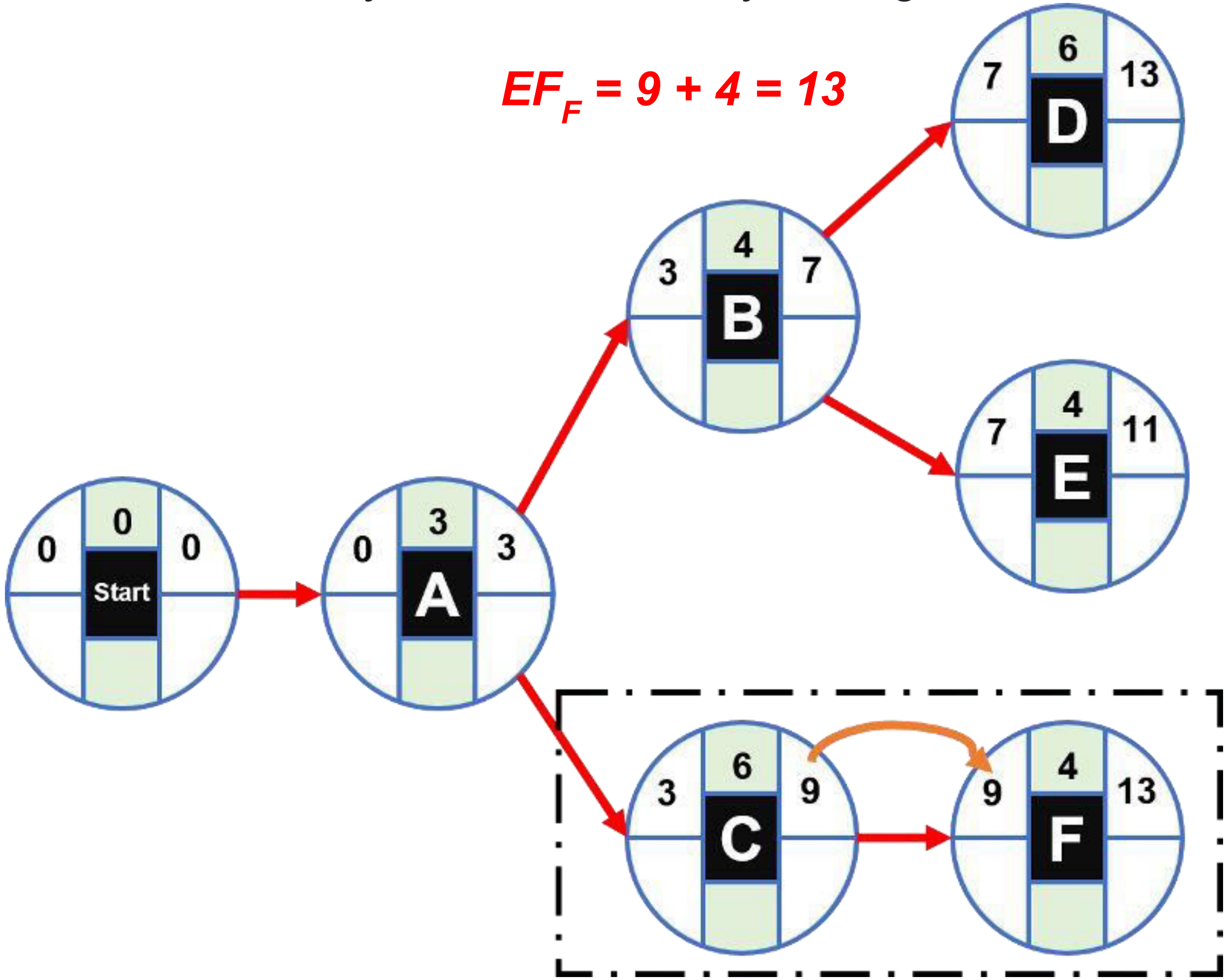
CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity F:

It has as precedent only activity C; therefore its ES will be equal to the EF of activity C. The EF of activity F is calculated by adding its ES + the corresponding time:

$EF_F = 9 + 4 = 13$



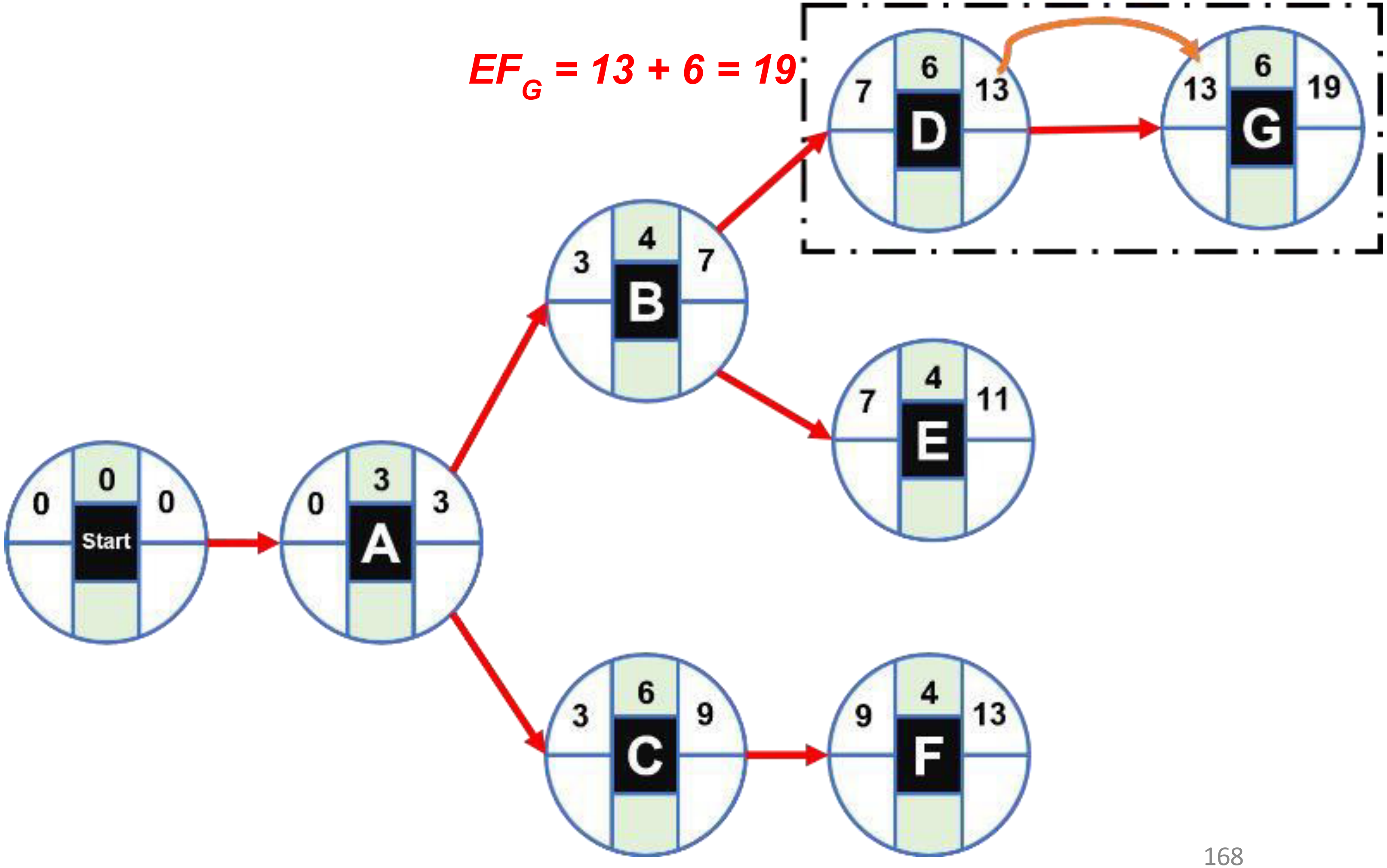
CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity G:

It has as precedent only activity D; therefore its ES will be equal to the EF of activity D. The EF of activity G is calculated by adding its ES + the corresponding time:

$EF_G = 13 + 6 = 19$



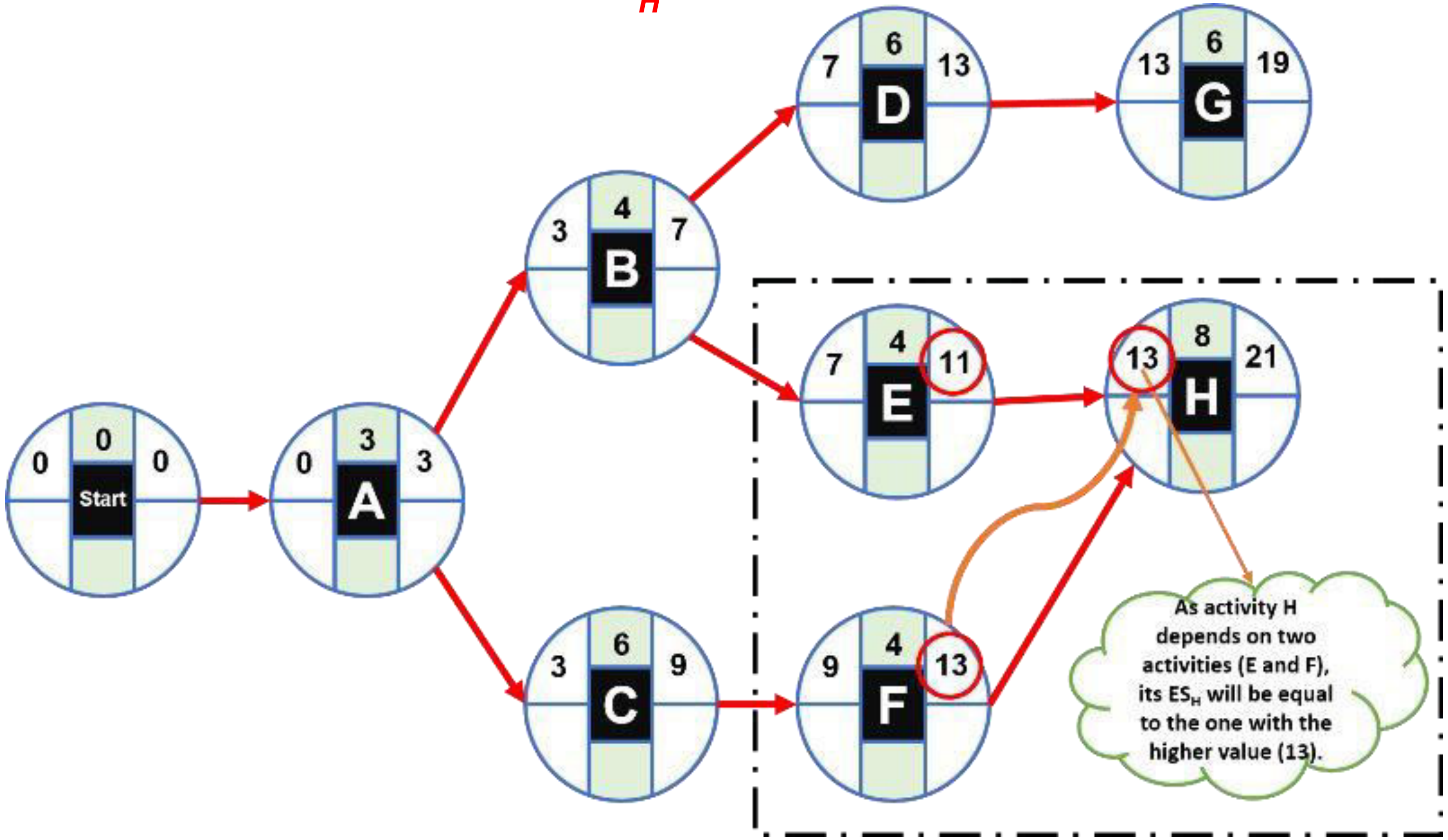
CPM Example –Solution (Forward Path)

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8

Activity H:

This activity has two precedents: E and F; therefore its ES will be equal to the highest EF of both activities. In this case, activity F has the highest value with 13. The EF of activity H is calculated by adding its ES + the corresponding time:

$$EF_H = 13 + 8 = 21$$

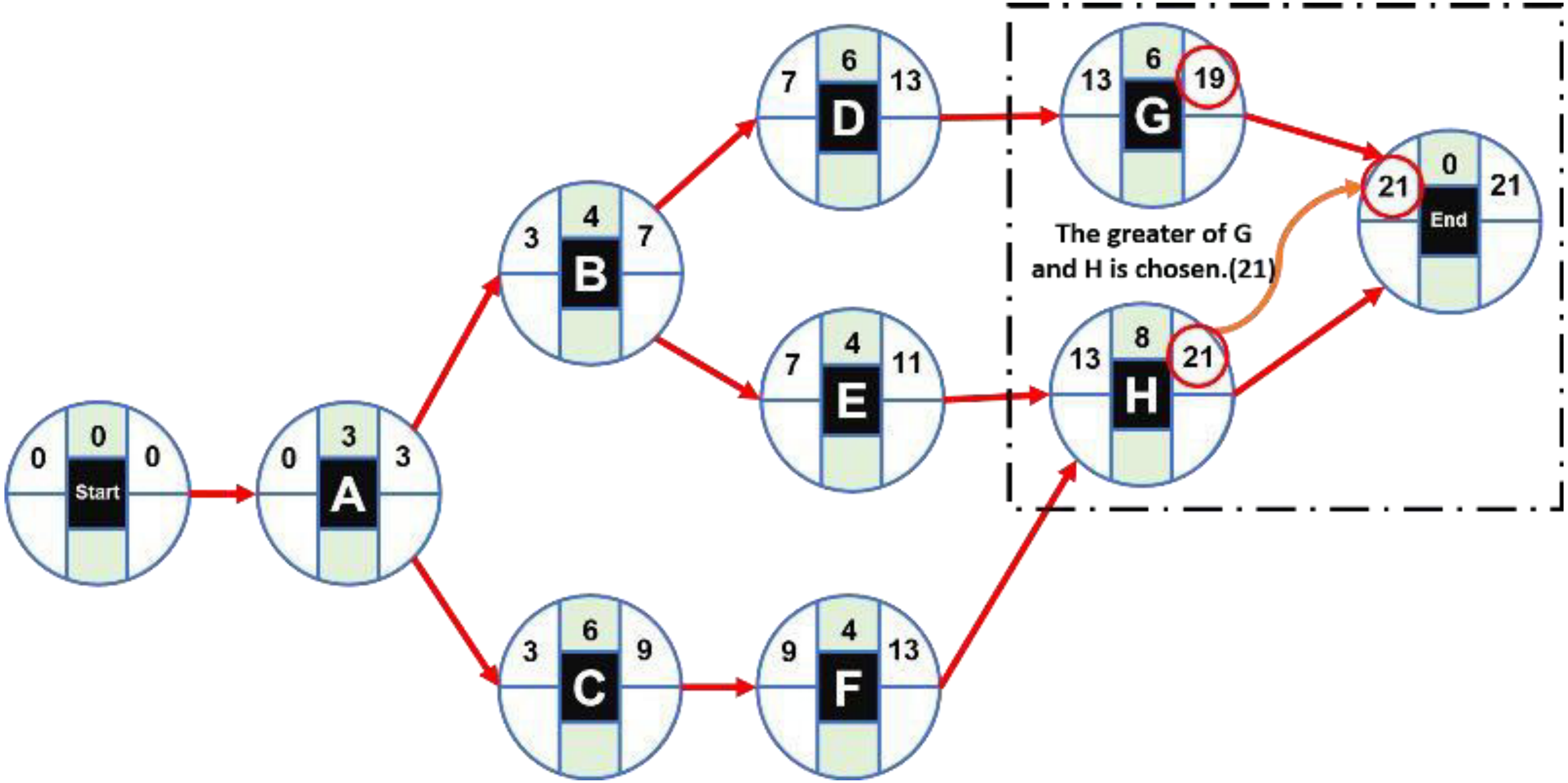


CPM Example –Solution (Forward Path)

End Node:

The end fictitious node is joined with the last activities G and H; and the highest value of the EF of both activities is placed as ES: 21. This value represents the total duration of the project. As this node has zero duration (because it is fictitious) its EF will be equal to 21 + 0 = 21.

Activity	Predecessor (s)	Time (Days)
A	–	3
B	A	4
C	A	6
D	B	6
E	B	4
F	C	4
G	D	6
H	E, F	8



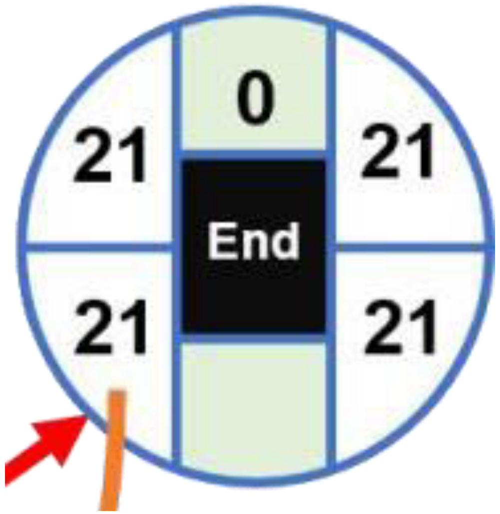
CPM Example –Solution (Backward Path)

To finalize the critical path calculation we will perform the backward traversal to calculate the LF and LS, starting from the final node; placing the values at the bottom of the node as follows:

End Node:

For the end node the LF value is equal to the project duration (21). The LS is calculated by subtracting the LF minus the duration (zero).

$$LS_{End} = 21 - 0 = 21$$

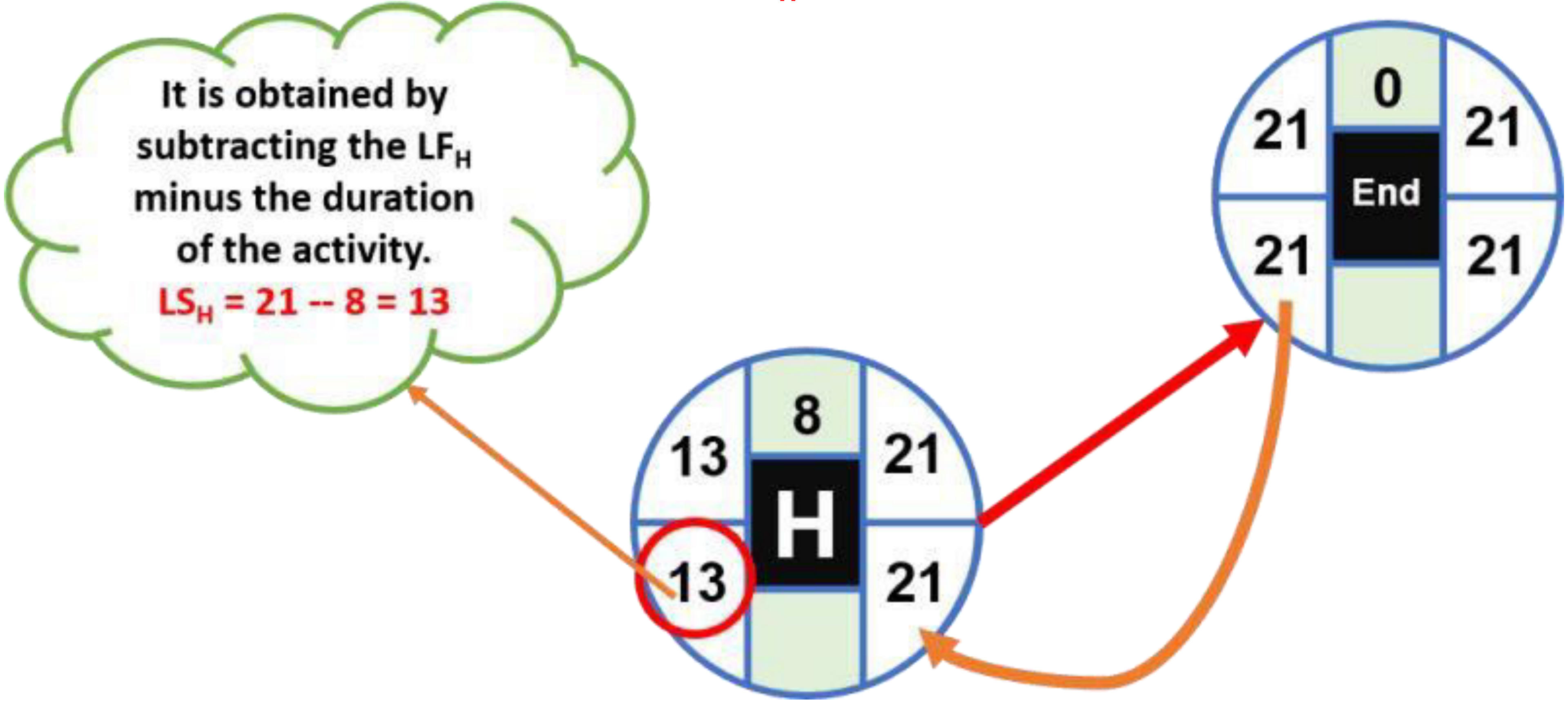


CPM Example –Solution (Backward Path)

Activity H:

Since the final node is the only successor to activity H, its LF will be equal to the LS of the final node (21). The LS of activity H is calculated by subtracting its LF minus its duration:

$$LS_H = 21 - 8 = 13$$

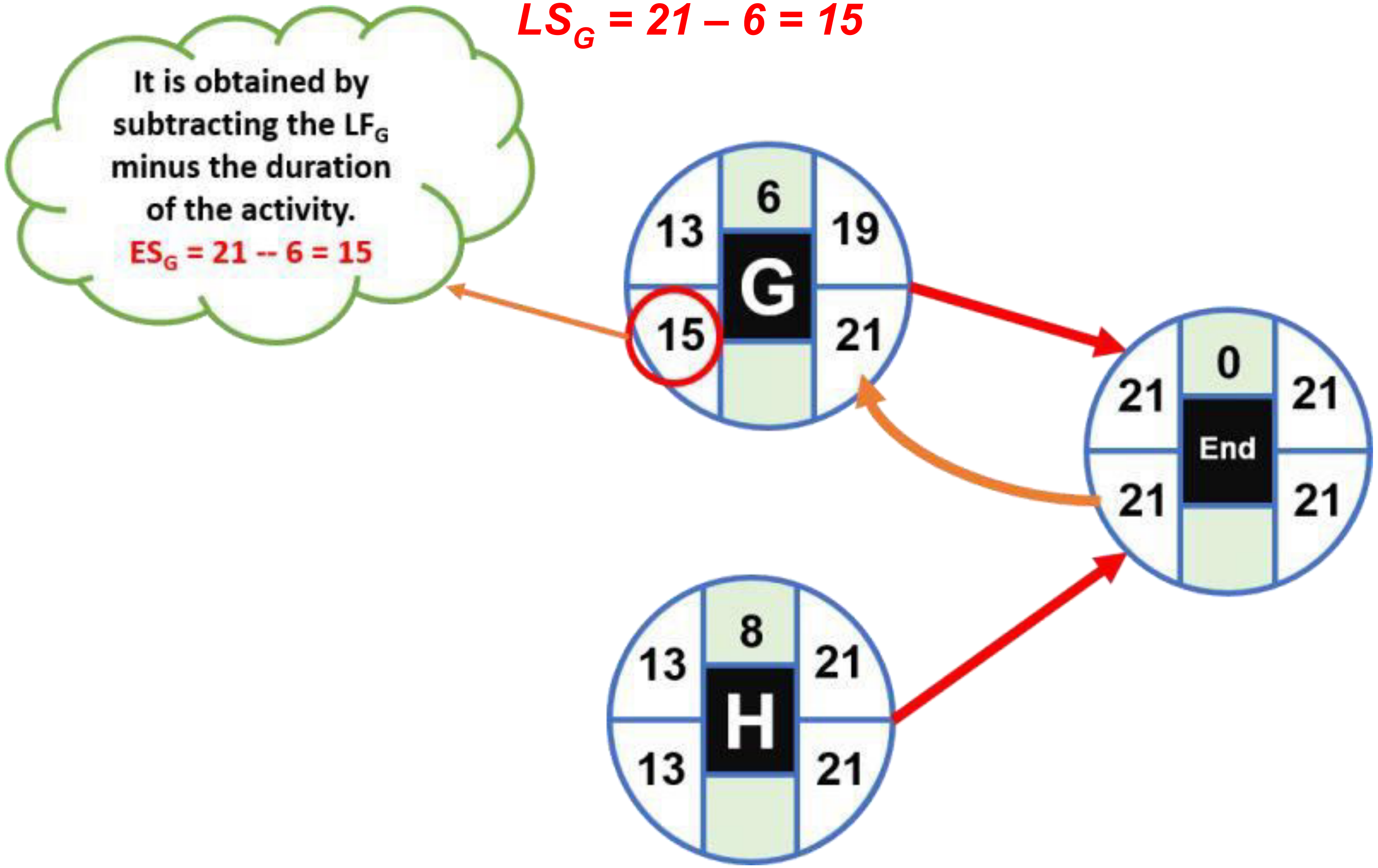


CPM Example –Solution (Backward Path)

Activity G:

Since the end node is the only successor of activity G, its LF will be equal to the LS of the end node (21). The LS of activity G is calculated by subtracting its LF minus its duration:

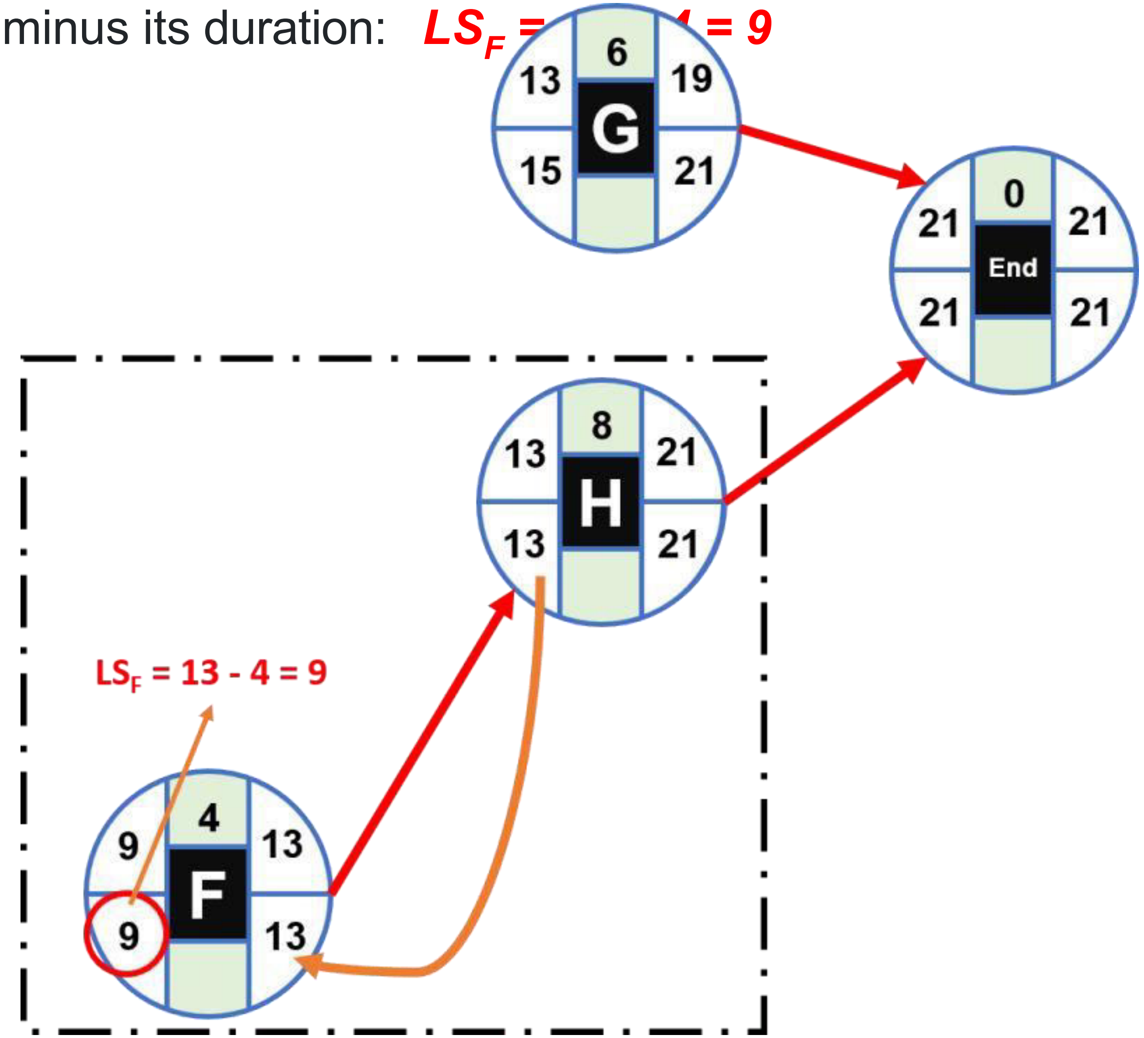
$$LS_G = 21 – 6 = 15$$



CPM Example –Solution (Backward Path)

Activity F:

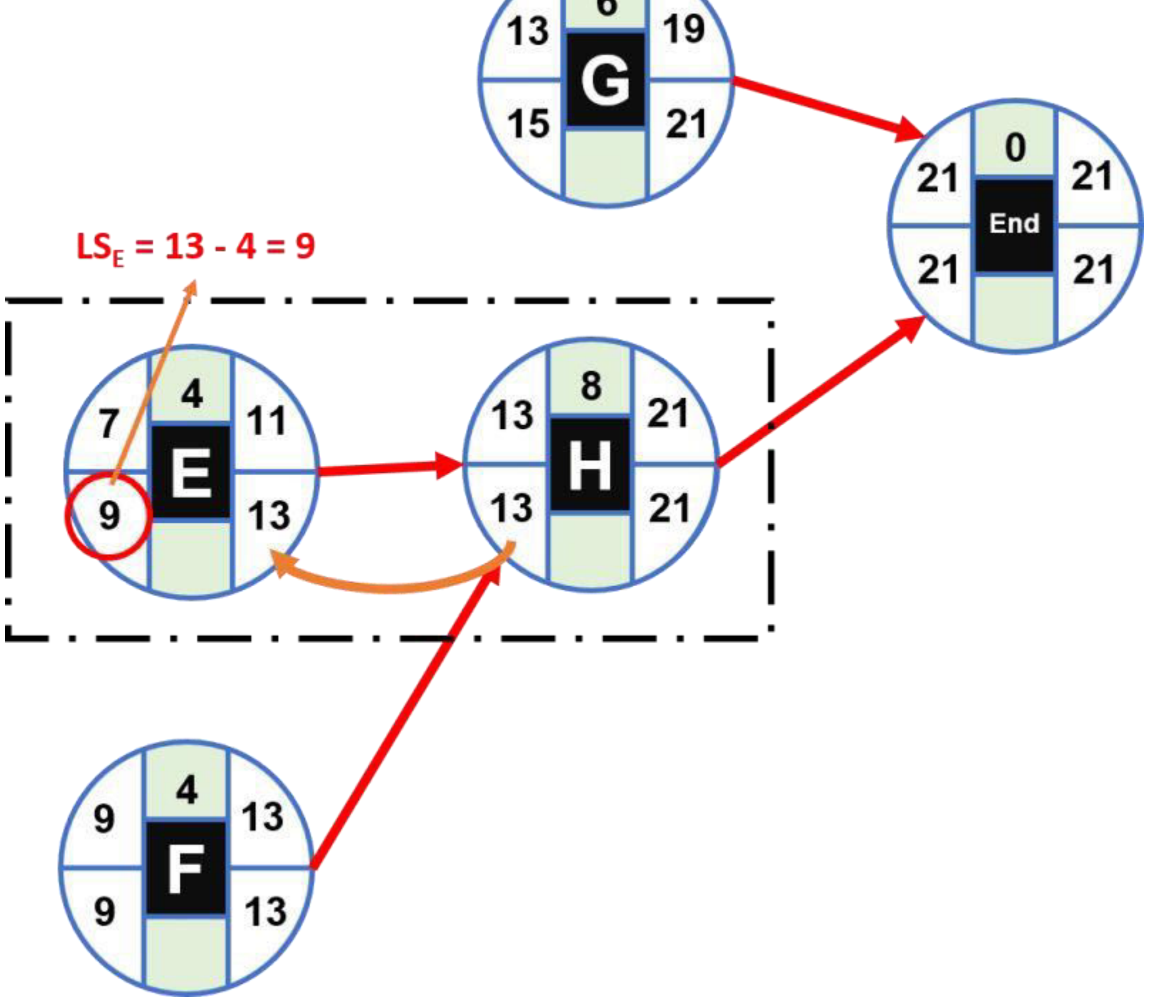
Since activity H is the only successor of activity F, its LF will be equal to the LS of activity H (13). The LS of activity F is calculated by subtracting its LF minus its duration: ***LS_F =*** [illegible] ***4 = 9***



CPM Example –Solution (Backward Path)

Activity E:

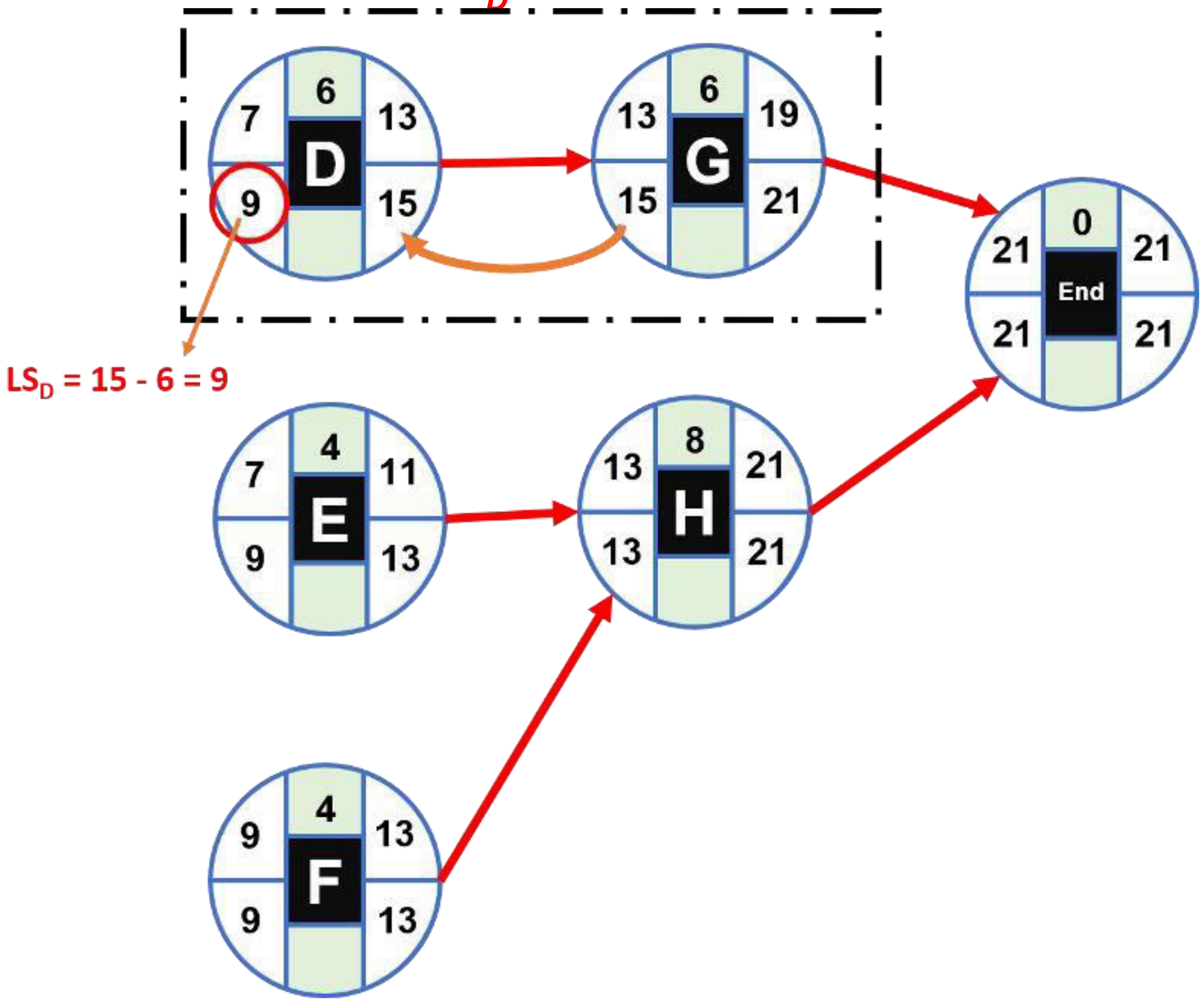
Since activity H is the only successor of activity E, its LF will be equal to the LS of activity H (13). The LS of activity E is calculated by subtracting its LF minus its duration: ***LS = 13 – 4 = 9***



CPM Example –Solution (Backward Path)

Activity D:

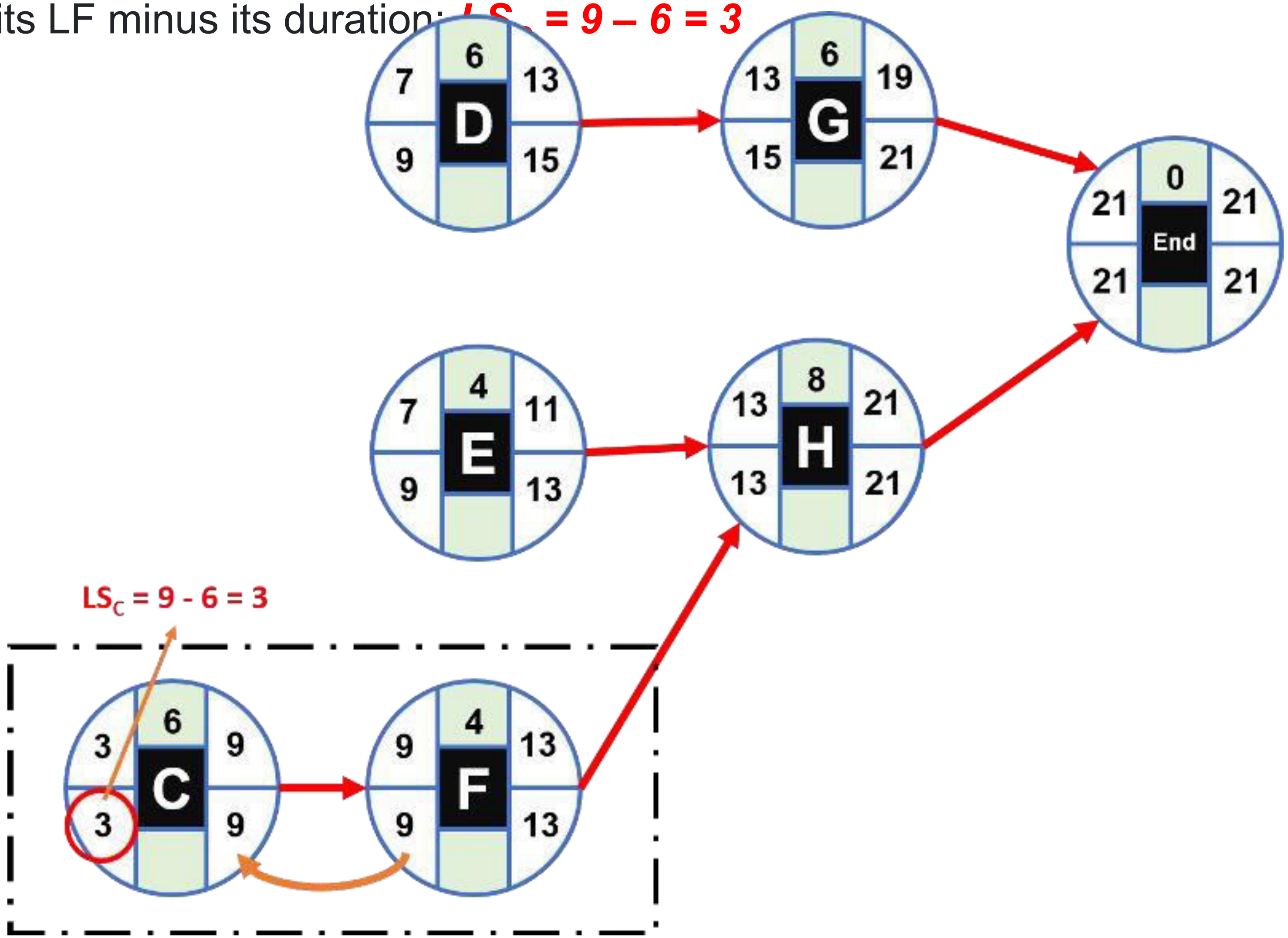
Since activity G is the only successor of activity D, its LF will be equal to the LS of activity G (15). The LS of activity D is calculated by subtracting its LF minus its duration: ***LS_D = 15 – 6 = 9***



CPM Example –Solution (Backward Path)

Activity C:

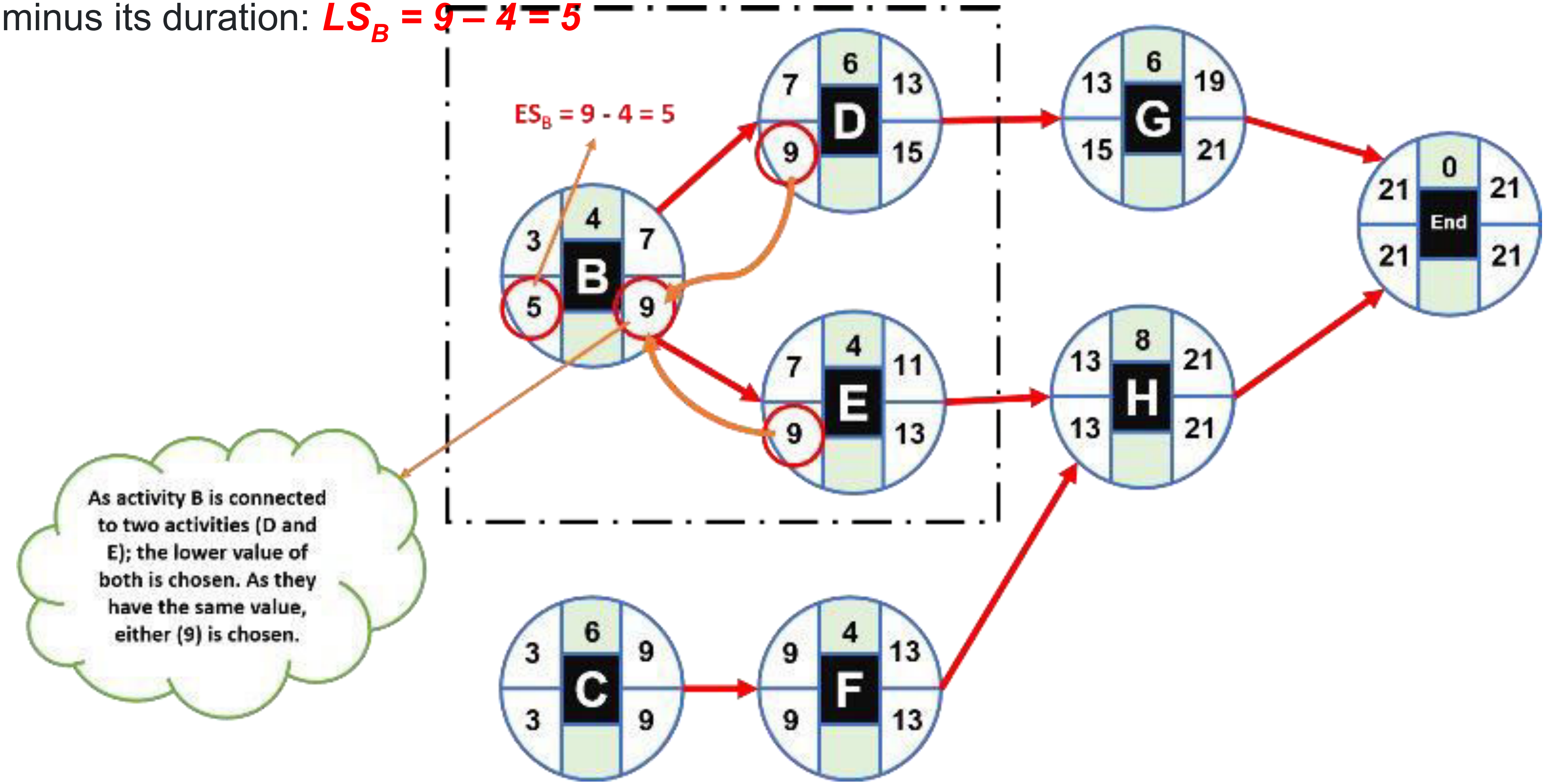
Since activity F is the only successor of activity C, its LF will be equal to the LS of activity F (9). The LS of activity C is calculated by subtracting its LF minus its duration: $LS_C = 9 – 6 = 3$



CPM Example –Solution (Backward Path)

Activity B:

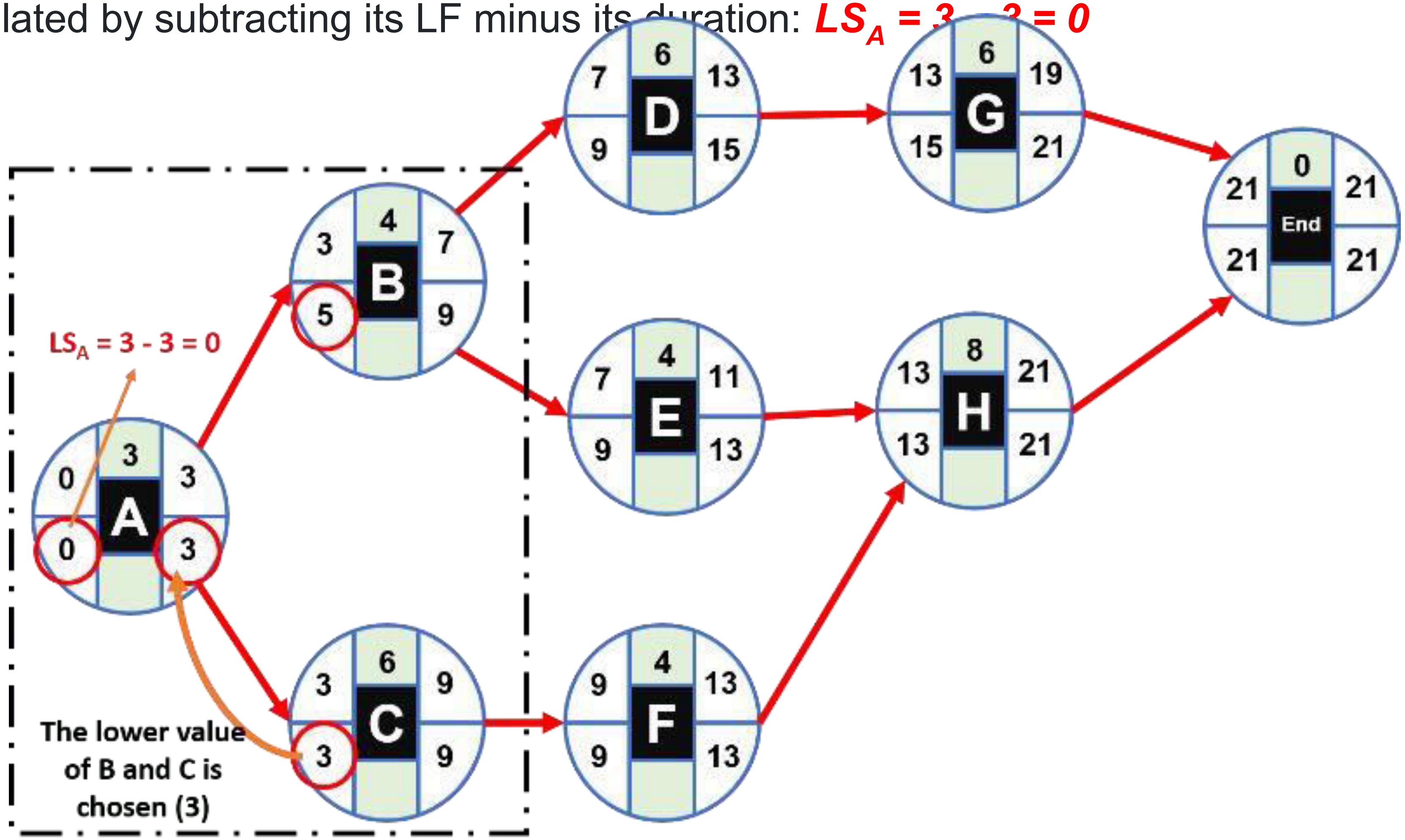
How activity B has as successors activities D and E, its LF will be equal to the smaller value of the LS of both. In this case, since both have a value of 9; that value will be the LF of activity B. The LS of activity B is calculated by subtracting its LF minus its duration: $LS_B = 9 - 4 = 5$



CPM Example –Solution (Backward Path)

Activity A:

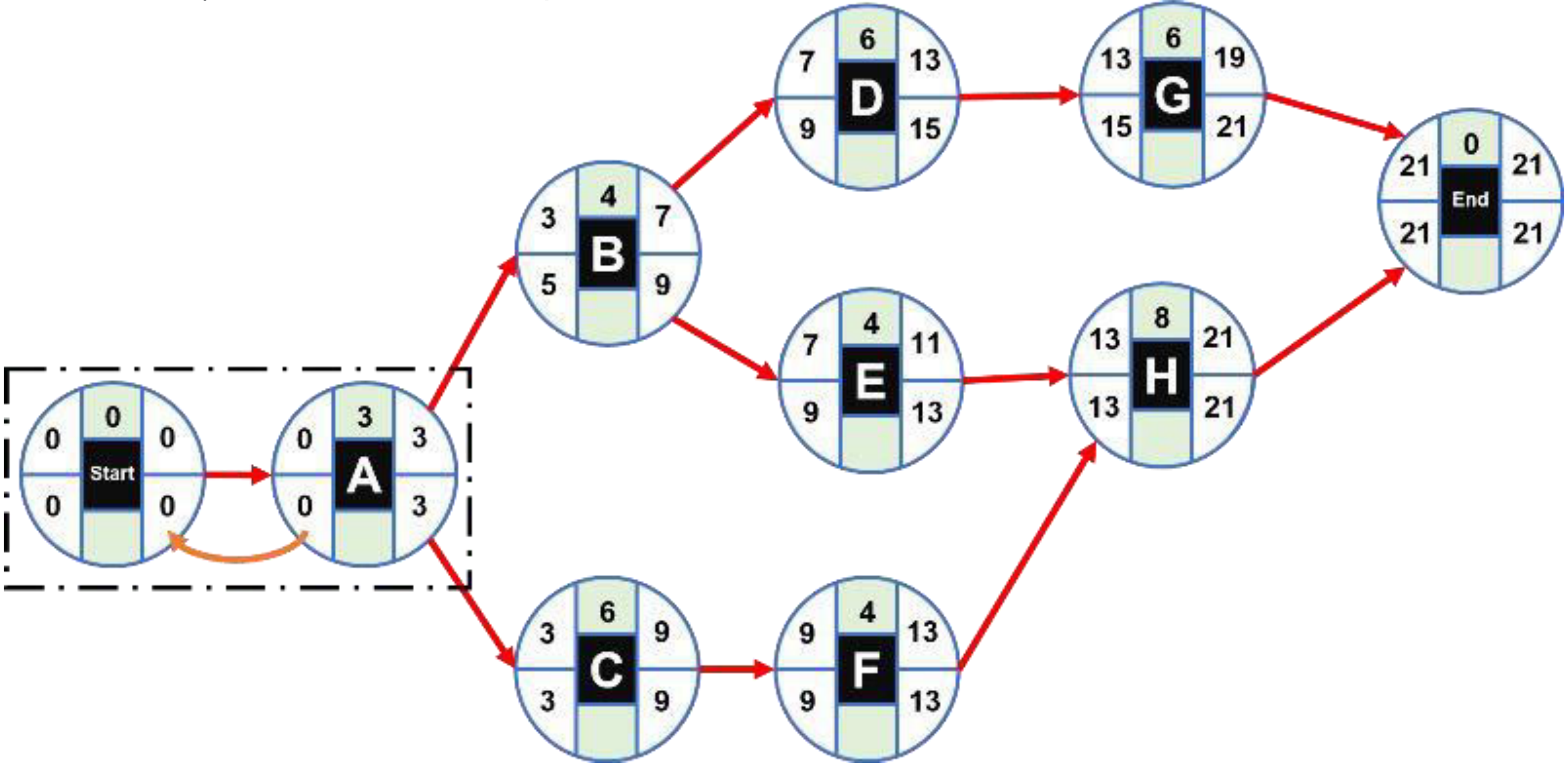
How activity A has as successors activities B and C, its LF will be equal to the smaller value of the LS of both. In this case, the lowest value is that of activity C (3); therefore, that value will be the LF of activity A. The LS of activity A is calculated by subtracting its LF minus its duration: ***LS_A = 3 – 3 = 0***



CPM Example –Solution (Backward Path)

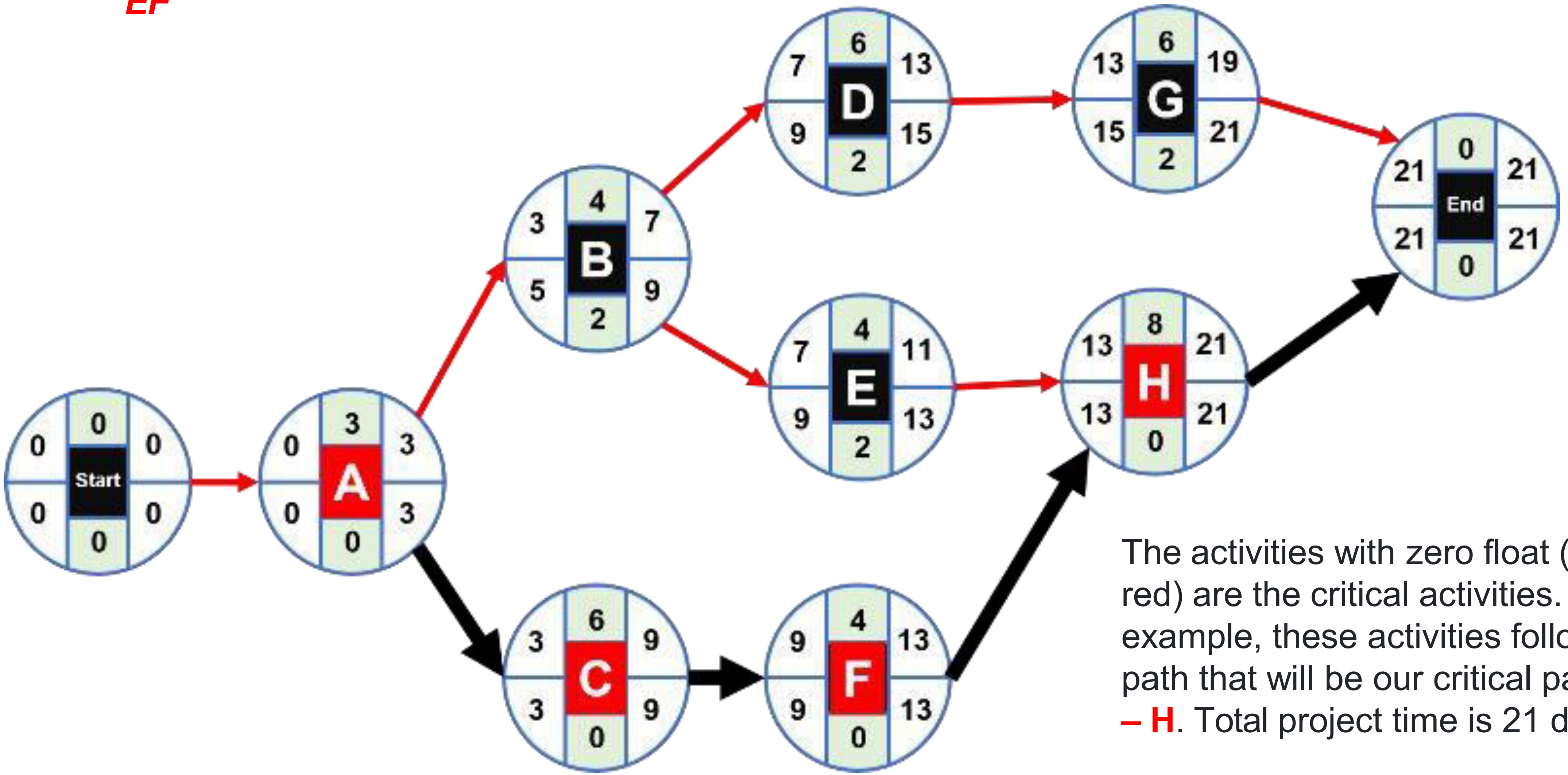
Initial Node:

Using the same analysis as above we complete the values of the initial node with zero.



CPM Example –Solution (Slack)

Finally, we calculate the slack for each node with the following formula: ***Slack = LS – ES = LF – EF***



The activities with zero float (marked in red) are the critical activities. In this example, these activities follow a single path that will be our critical path: **A – C – F – H**. Total project time is 21 days.

Example 2

- Shirley Hopkins is developing a program in leadership training for middle-level managers. Shirley has listed a number of activities that must be completed before a training program of this nature can be conducted. The activities, immediate predecessors, and times appear in the accompanying table.

Example 2

Activity	Immediate Predecessor	Time (days)
A	–	2
B	–	5
C	–	1
D	B	10
E	A,D	3
F	C	6
G	E,F	8

Provide the following:

1. AON diagram for these precedencies
2. What is the critical path?
3. What is the total project completion date?
4. What is the slack time for each individual activity?

Example 2 - Solution

Each activity is broken down by its precedents and descendants:

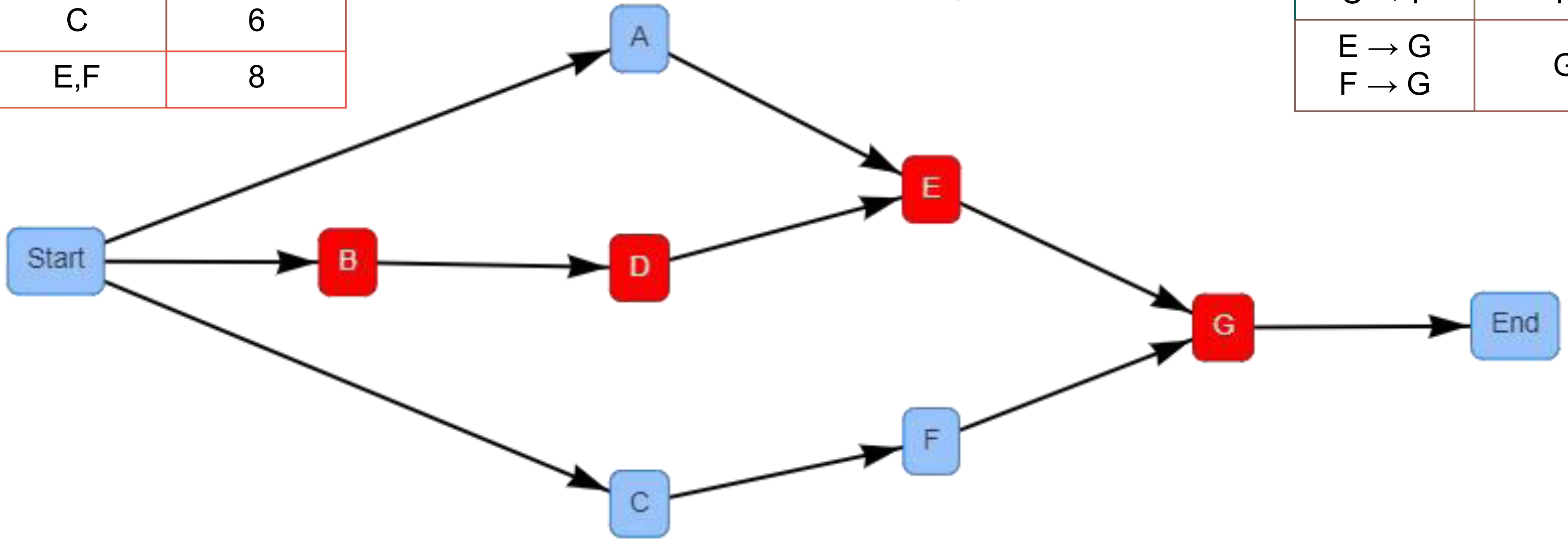
Predecessor	Activity	Successor
Start → A	A	A → E
Start → B	B	B → D
Start → C	C	C → F
B → D	D	D → E
A → E D → E	E	E → G
C → F	F	F → G
E → G F → G	G	G → End

Example 2 - Solution

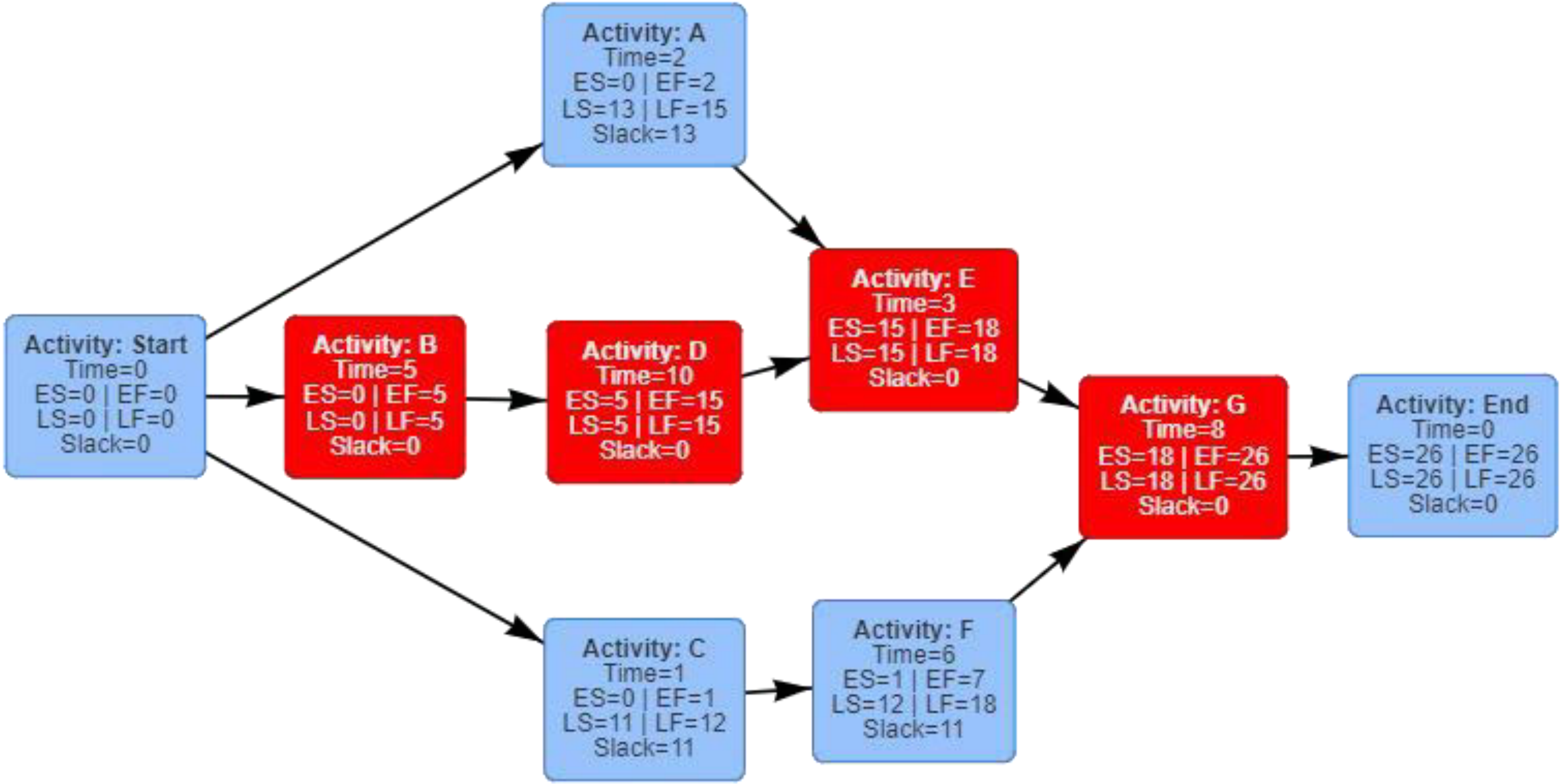
Activity	Immediate Predecessor	Time (days)
A	–	2
B	–	5
C	–	1
D	B	10
E	A,D	3
F	C	6
G	E,F	8

Predecessor	Activity	Successor
Start → A	A	A → E
Start → B	B	B → D
Start → C	C	C → F
B → D	D	D → E
A → E D → E	E	E → G
C → F	F	F → G
E → G F → G	G	G → End

CPM Network Diagram



Example 2 - Solution



Example 2 - Solution

Activity	Time	Early Start (ES)	Early Finish (EF)	Late Start (LS)	Late Finish (LF)	Slack (S)
A	2	0	2	13	15	13
B	5	0	5	0	5	0
C	1	0	1	11	12	11
D	10	5	15	5	15	0
E	3	15	18	15	18	0
F	6	1	7	12	18	11
G	8	18	26	18	26	0

The critical path is: **B → D → E → G**

The total project time is: **26 hours**